

Sixteen defender-centric problem formulations

SkyHAD / 3D mathematical specification / English edition

Purpose and scope

This document formalizes cooperative Red asset defense against a specified opposing Blue policy, across two particle interfaces and three interfaces each for fixed-wing and quadrotor agents. Each interface has a Survival and a Damage task: exactly 16 themes, with two pages per theme. The first page of each theme defines state, action and dynamics; the second defines interactions, observations and the defender Dec-POMDP objective.

The implementation is authoritative. In particular, particle position updates use the old velocity; native actor rows include all living entities; automatic firing is separate from navigation; and a frozen opposing policy is an experimental choice, not an automatic property of the native environment. The local reference paper's Chapter II was used only as a structural guide. Its prose and alternative kinematic or health equations are not reproduced.

Shared notation

$\mathcal{R}, \mathcal{B}, \mathcal{P}$ denote Red defenders, Blue opponents and assets. p, v, q, ω, h, D are position, velocity, body-to-world quaternion, body rate, health and cumulative raw asset damage. Subscripts A, D designate Attack and Disturb roles. λ is the scene scale; Δ is one external physical step; H is the caller's sampling limit. All distances use metres and time uses seconds. $\text{sat}_L(z)$ clips Euclidean norm, whereas clip clips components/scalars.

Reading the formulations

The task is cooperative within Red after freezing π_B ; the full two-team interaction is a POSG when both teams learn. A rigid-body state is a constrained 13-vector, not a position-only particle. Native `state()` is a privileged feature packing and may omit policy memory or task history. Complete simulation snapshots and a mathematical Markov state contain more information.

Survival optimizes a default terminal ± 10 payoff and protects every asset. Damage minimizes raw cumulative asset damage while asset health stays finite and unchanged. Red team death does not create a natural world terminal event. A sampling limit does not grant a native victory.

General grouping/SMDP extension

Grouping assigns persistent responsibilities and currently uses acceleration executors; it does not multiply the 16 themes. Add assignments, reserve, clocks and policy memory to state. For a macro transition lasting d physical steps,

$$\bar{R}_t = \sum_{j=0}^{d-1} \gamma^j R_{t+j}, \quad y_t = \bar{R}_t + \gamma^d b_t V(s_{t+d}).$$

The present grouping Damage reward is the undiscounted sum, corresponding to $\gamma = 1$. At Survival horizon, its configured horizon rule can assign a result, unlike native sampling truncation. Built-in opponents are reactive, balanced and concentrated, with a rush acceleration executor. A frozen external learned policy is a separate opponent specification.

Contents / exact page map

No.	Model	Control / task	Pages
1	Particle	Acceleration / Survival	2–3
2	Particle	Acceleration / Damage	4–5
3	Particle	Position / Survival	6–7
4	Particle	Position / Damage	8–9
5	Fixed wing	Acceleration / Survival	10–11
6	Fixed wing	Acceleration / Damage	12–13
7	Fixed wing	Actuator / Survival	14–15
8	Fixed wing	Actuator / Damage	16–17
9	Fixed wing	Position / Survival	18–19
10	Fixed wing	Position / Damage	20–21
11	Quadrotor	Acceleration / Survival	22–23
12	Quadrotor	Acceleration / Damage	24–25
13	Quadrotor	Actuator / Survival	26–27
14	Quadrotor	Actuator / Damage	28–29
15	Quadrotor	Position / Survival	30–31
16	Quadrotor	Position / Damage	32–33

Reward customization in Survival

Each formulation uses its native default payoff. An explicit four-vector w replaces Survival scalar reward by $w^T \ell_i$, where $\ell_i = (r_i^{\text{hit}}, r_i^{\text{target}}, r_i^{\text{enemy}}, 10\epsilon_i z)$, where $\epsilon_i = +1$ for Red and -1 for Blue. All geometry below is post-step. For a living Attack agent,

$$r_i^{\text{hit}} = \sum_{j \in \mathcal{U}: c_j \neq c_i, h_j > 0} \max(0, 1 - d_{ij}/r_1),$$

$$r_i^{\text{target}} = - \min_{k: h_k > 0} d_{ik}/L,$$

$$r_i^{\text{enemy}} = \begin{cases} - \min_{j \in \mathcal{B}: h_j > 0} d_{ij}/L & i \in \mathcal{R}, \\ \min_{j \in \mathcal{R}: h_j > 0} d_{ij}/L - 1 & i \in \mathcal{B}. \end{cases}$$

An empty comparison set gives zero, including the Blue enemy slot; dead agents have zero first three slots. “Hit” is a geometric proxy, not actual fire damage. Scout/Disturb keep their role scalar in slot 1 and zero slots 2–3. Let δ_i be nearest XY-wall distance and $B_i = .05e^{-\delta_i/\lambda} \mathbf{1}[\delta_i < 100\lambda]$. Scout slot 1 is $c_s |\text{sector}_i| - B_i$; Disturb slot 1 is $\mathbf{1}[J_i, h_i > 0](.1c_d \sum_{j \in \mathcal{U}} \text{sign}_{ij} G_{ij} - B_i)$, with enemy sign $+1$, friendly sign -1 . Defaults $c_s = c_d = 0$. The episode slot already contains the team payoff; adding it again doubles the terminal reward. Damage rejects these weights.

Primary parameter/model sources

[1] BYU MAGICC, [published Aerosonde parameter set](#). Physical data only; the project authors its force/controller code independently.

[2] T. Lee, M. Leok, N. H. McClamroch, [Control of Complex Maneuvers for a Quadrotor UAV using Geometric Methods on SE\(3\)](#), v2, 2010, Sections II and VII. Axis conventions are rotated to ENU/FLU.

[3] M. Faessler, A. Franchi, D. Scaramuzza, [Differential Flatness of Quadrotor Dynamics Subject to Rotor Drag for Accurate Tracking of High-Speed Trajectories](#), RAL, 2018. This project's ideal model omits those drag terms.

Theme 1 / Particle / Acceleration / Survival

Physical state, navigation action and discrete transition

A. Scenario and complete state

Red defenders $\mathcal{R}$ protect stationary assets $\mathcal{P}$ against Blue $\mathcal{B}$. Set $\mathcal{U} = \mathcal{R} \cup \mathcal{B}$, $\mathcal{E} = \mathcal{U} \cup \mathcal{P}$, $N = |\mathcal{U}|$, $K = |\mathcal{P}|$, $M = N + K$. Episode-fixed roles are Attack, Scout or Disturb.

$$s_t = (\{x_i, h_i, r_i, c_i, \text{id}_i\}_{i \in \mathcal{U}}, \\ \{p_k, h_k, D_k\}_{k \in \mathcal{P}}, t, \zeta_t).$$

$x_i = (p_i, v_i) \in \mathcal{W} \times \mathbb{R}^3$ is a six-value kinematic state; no quaternion, inertia, wing or rotor is attached to a particle. Participation is $\ell_i = \mathbf{1}[h_i > 0]$. Reset UAV health is 1, target health $h_k^0 = 2$ by default; $v_k = 0$, $D_k = 0$ initially. D_k accumulates raw asset damage.

Configuration C fixes roster, bounds, controls and task. ζ_t contains persistent assignments and opponent memory, if used. Reactive controllers have no hidden integrators. Last command, functional bits, pre-step positions and step damage are overwritten transition records; retain them for exact reward/event replay. RNG state is required for exact replay, not actor input.

B. Implemented kinematic map

For a living, noncolliding particle, $\Delta = 1$ s,

$$\begin{aligned} p_i^* &= p_i + \Delta v_i, & p_i' &= \Pi_{\mathcal{W}}(p_i^*), \\ w_i &= v_i + \Delta \alpha_i, & \tilde{v}_i &= \mathcal{S}_{35,120}(w_i; v_i). \end{aligned}$$

For $\|w\| \geq 10^{-3}$,

$$\mathcal{S}_{m,V}(w; v) = \frac{w}{\|w\|} \text{clip}(\|w\|, m, V).$$

For $\|w\| < 10^{-3}$, use $mv/\|v\|$ when $\|v\| \geq 10^{-3}$ and me_1 otherwise. The cancellation fallback retains heading. After speed normalization, an outward component is set to zero on each clipped axis: upper wall $v_a > 0$ or lower wall $v_a < 0$.

The final 3D turn map limits the angle between old and candidate velocities to $\theta_{\max} = \pi$ per step, so the default map is identity. For a smaller cap, rotate the old heading toward the candidate about $v_i \times \tilde{v}_i$, preserving candidate speed. A deterministic orthogonal axis resolves the antiparallel case; near-zero products $\|v\|_i \|\tilde{v}_i\| \leq 10^{-3}$ have angle zero.

Position uses old velocity, with no $\frac{1}{2}\alpha\Delta^2$ term. Boundary projection can reduce the final speed below 35 m/s. Agents killed by collision do not move; agents killed by combat complete this step's motion, and their velocity is zeroed on the next inactive update.

Interpretation

This is a kinematic benchmark. Its action controls future velocity and thus subsequent interception geometry, rather than instantaneous translation. Flight decisions and automatic functional decisions are distinct.

C. Acceleration intent

Each living agent chooses either $a_i \in [-1, 1]^3$ or an index into

$$\mathcal{V} = \{0\} \cup \{z/\|z\| : z \in \{-1, 0, 1\}^3 \setminus \{0\}\},$$

giving 27 discrete primitives, zero first. The physical intent is $\alpha_i = 40a_i$ m/s². Continuous inputs have component bounds, so diagonal intents can have norm $\sqrt{3}40$; discrete nonzero intents have norm 40. 40 is a control scale, not a clamp on actual rigid-body acceleration. The chosen vector directly enters the particle map as $\alpha_i = 40a_i$. No inner rigid-body controller or actuator state is introduced. Zero input preserves unconstrained velocity; it does not command a stop.

Episode geometry and initial law

The default scale is $\lambda = 1$, with

$$\mathcal{W} = \lambda([-2500, 2500]^2 \times [0, 2500]) \text{ m}.$$

Assets are sampled independently from $\lambda([-2300, -1900] \times [-1200, 1200] \times [500, 1500])$, or supplied as a fixed layout. Red chooses a target uniformly and samples an area-uniform XY annulus: $\rho^2 \sim U((800\lambda)^2, (2200\lambda)^2)$, $\vartheta \sim U(0, 2\pi)$; XY is clipped to $\mathcal{W}$. Blue samples $x \sim U(0, 2500\lambda)$ and $y \sim U(-2500\lambda, 2500\lambda)$. UAV altitude is sampled in $[200, 1500]\lambda$; target altitude is separate.

Random velocity has a Gaussian unit direction and speed $\|(u_1, u_2, u_3)\|/\sqrt{3}$, where $u_a \sim U(35, 120)$. Initial positions and velocities are generated by the environment-local RNG. The default collision distance is $d_c = 20\lambda = 20$ m. The full/zero-damage radii are $(r_0, r_1) = (300, 600)\lambda = (300, 600)$ m. The default firing threshold is $r_f = r_0$; configuration may select $0 < r_f \leq r_0$.

2D compatibility

Only particle themes support $d = 2$. The three coordinate slots remain, $p_z = 100\lambda$, $v_z = \alpha_z = 0$. Acceleration uses a 2-vector or nine planar primitives; position remains a 3-vector with prescribed altitude. Radii become $(200, 400)\lambda$. Planar resets reject too-close openings, and the heading cap is applied on the circle. These are compatibility settings of this theme, not additional themes.

Task dependence

Navigation and observations are identical between Survival and Damage. Asset health evolution, completion, and defender reward differ as specified on the next page. Neither task changes the particle into a rigid-body vehicle.

Theme 1 / Particle / Acceleration / Survival

Task interaction, exact information structure and defender learning objective

E. Collision and automatic interaction

Predict living UAV paths: particles use $p_i + \tau \Delta v_i$, rigid models use synchronized RK4 endpoints clipped to $\mathcal{W}$. On each matched segment, let $r = p_i - p_j$, $d = (q_i - p_i) - (q_j - p_j)$:

$$\tau_* = \begin{cases} 0 & \|d\|^2 \leq 10^{-12}, \\ \text{clip}(-r^T d / \|d\|^2, 0, 1) & \text{otherwise,} \end{cases} \quad d_* = \|r + \tau_* d\|.$$

Collision sets both UAV HP to zero when $\min d_* \leq d_c$; targets are excluded. Pairs use i -major, $j < i$ order. Dead outer agents are skipped at entry and dead inner agents are skipped. An outer agent killed during its inner sweep continues that sweep. After collision, surviving Attack agents automatically fire at step-start positions:

$$I_i = \begin{cases} \mathbf{1}[\min_{k \in \mathcal{P}: h_k > 0} \|p_i - p_k\| < r_f], & i \in \mathcal{B}, \\ \mathbf{1}[\min_{j \in \mathcal{B}_A: h_j > 0} \|p_i - p_j\| < r_f], & i \in \mathcal{R}. \end{cases}$$

An empty minimum is $+\infty$; dead sources have $I_i = 0$. A denotes Attack. Fire is a separate rule, not a navigation action. With unit attack intensity,

$$\eta(d) = \begin{cases} 1 & d < r_0, \\ (r_1 - d)/(r_1 - r_0) & r_0 \leq d < r_1, \\ 0 & d \geq r_1, \end{cases}$$

$$L_k = \sum_{b \in \mathcal{B}_A: h_b > 0} I_b \eta(\|p_b - p_k\|),$$

$$L_i^{\text{atk}} = \sum_{j \in \mathcal{U}_A: c_j \neq c_i, h_j > 0} I_j \eta(\|p_j - p_i\|).$$

One source can affect multiple enemies; friendly explosions are excluded. All losses use a common post-collision, pre-motion snapshot and add before HP clipping.

Disturb and Scout

For source j , $d_{ji} = \|p_i - p_j\|$ and $\theta_{ji} = \angle(v_j, p_i - p_j)$; angle is zero if either norm is below 10^{-3} . Let

$$G_{ji} = \mathbf{1}[10^{-3} < d_{ji} < 600\lambda, \theta_{ji} < \pi/6] \\ \times e^{-(d_{ji}/\lambda)^2/2} / \sqrt{2\pi},$$

$$S_c(j) = \sum_{i \in \mathcal{U}: c_i = c, h_i > 0} G_{ji}.$$

Living Red Disturb uses $J_j = \mathbf{1}[S_R(j) < S_B(j)]$, Blue uses $J_j = \mathbf{1}[S_R(j) \geq S_B(j)]$. Dead sources contribute zero.

$$L_i^{\text{dis}} = .1 \sum_{j \in \mathcal{U}_D: h_j > 0} J_j G_{ji},$$

$$\tilde{h}_i = \max(0, h_i - L_i^{\text{atk}} - L_i^{\text{dis}}).$$

Set $h'_i = 0$ if $r_i = A, I_i = 1$, otherwise $h'_i = \tilde{h}_i$. Disturb damages either side, not assets. Scouts use $d < 2000\lambda$, $\theta < \pi/3$; this sector does not restrict native actor rows. Collision victims do not move. Combat victims complete their motion; inactive velocity is zeroed next step.

G. Survival payoff and completion

Assets obey $h'_k = \max(0, h_k - L_k)$, $D'_k = D_k + L_k$.

$$z(s') = \begin{cases} -1 & \exists k : h'_k < 10^{-3}, \\ +1 & \forall b \in \mathcal{B}_A : h'_b < 10^{-3}, \\ 0 & \text{otherwise.} \end{cases}$$

Target loss has priority. Natural completion is $d = \mathbf{1}[z \neq 0]$; default Red reward $R_t = 10z$, Blue $-R_t$. Red elimination alone does not end the world. Individual $h'_i \leq 0$ retires a Survival actor after its final transition. Horizon H truncates without a win bonus; use global bootstrap $b_t = 1 - d$, including individual death and truncation. Actor action-loss masks use pre-step participation. Explicit reward weights replace the default payoff as defined on page 1.

F. Exact observations and CTDE information

For each living entity $j \neq i$, the actor row is

$$e_{ij} = [(p_j - p_i)/L, (v_j - v_i)/(2V_{\max}), \\ h_j, 1, \mathbf{1}_{c_j=R}, \mathbf{1}_{c_j=B}, \mathbf{1}_{j \in \mathcal{P}}], \\ L = \sqrt{\sum_a (b_a^+ - b_a^-)^2}.$$

Here $V_{\max}$ is the preset maximum speed, not the reference speed. Rows have 11 values, in fixed roster order: Red Scout/Disturb/Attack, Blue Scout/Disturb/Attack, then targets, excluding self. Dead rows are zero. Every living entity is included; Attack, Scout and Disturb share this packing. Side flags do not encode role. Own health is absent from the self block.

The actor input is only an $(M-1) \times 11$ entity table ($V_{\max} = 120$); it has no own-state block. With deterministic encoding, $Z_i(o | s) = \delta_{\text{encode}_i(s)}(o)$. Other UAV attitudes/rates, opponent memory and time are omitted; hence local information is partial despite noiseless sensing.

Privileged $g(s) = \text{state}()$ packs 11 values per entity,

$$[(p - b^-)/(b^+ - b^-), v/(2V_{\max}), h, \ell, R, B, P],$$

with dead blocks zero. Rigid models append 13 raw state values per UAV (dead zero), then $[t, \sum_k D_k, \mathbf{1}_{\text{Damage}}]$: size $11M + 13N + 3$. Particle size is $11M$, omitting time and damage. g is a CTDE feature, not the complete Markov state.

H. Defender Dec-POMDP and MARL objective

For a specified frozen opposing policy π_B , define

$$\mathcal{M}_R = (\mathcal{R}, \mathcal{S}, \{\mathcal{A}_i\}, T^{\pi_B}, \\ \{\mathcal{O}_i\}, Z, R, \gamma, \rho_0, H),$$

$$T^{\pi_B}(s' | s, a_R) = \int \delta_{F_C(s, a_R, a_B)}(s') \\ \times \pi_B(da_B | o_B, \zeta_t).$$

F_C is the stated collision, simultaneous interaction and controlled-motion map, including any memory evolution; ρ_0 is the reset law. Native execution accepts both teams' actions. Freezing a rule or learned opponent is an experiment choice; with both teams learning, the model is a POSG.

$$\tau_i = (o_{i,0}, a_{i,0}, \dots, o_{i,t}),$$

$$\pi_R(a_R | \tau_R) = \prod_{i \in \mathcal{R}} \pi_i(a_i | \tau_i),$$

$$\pi_R^* \in \arg \max_{\pi_R} \mathbb{E} \left[\sum_{t=0}^{T-1} \gamma^t R_t \right].$$

Expectation is over ρ_0, T^{π_B}, π_R . Here T is natural completion (possibly infinite); H limits sampled rollouts. A truncated return is partial, so its successor value is retained. For $\gamma = 1$, this is respectively the native terminal Survival payoff or negative total asset damage. For $\gamma < 1$, it is the discounted task selected by the caller. Actors use local histories only. CTDE critics can use $g(s)$, joint actions and recorded task/memory features:

$$y_t = R_t + \gamma b_t Q^-(g(s'), a'_R).$$

Here Q^- is a target critic and successor actions follow the target actor policies. Use global completion, not individual death. A recurrent critic or augmented state may be needed because g omits history. This defines an objective; it does not prescribe an implemented optimizer.

Theme 2 / Particle / Acceleration / Damage

Physical state, navigation action and discrete transition

A. Scenario and complete state

Red defenders $\mathcal{R}$ protect stationary assets $\mathcal{P}$ against Blue $\mathcal{B}$. Set $\mathcal{U} = \mathcal{R} \cup \mathcal{B}$, $\mathcal{E} = \mathcal{U} \cup \mathcal{P}$, $N = |\mathcal{U}|$, $K = |\mathcal{P}|$, $M = N + K$. Episode-fixed roles are Attack, Scout or Disturb.

$$s_t = (\{x_i, h_i, r_i, c_i, \text{id}_i\}_{i \in \mathcal{U}}, \\ \{p_k, h_k, D_k\}_{k \in \mathcal{P}}, t, \zeta_t).$$

$x_i = (p_i, v_i) \in \mathcal{W} \times \mathbb{R}^3$ is a six-value kinematic state; no quaternion, inertia, wing or rotor is attached to a particle. Participation is $\ell_i = \mathbf{1}[h_i > 0]$. Reset UAV health is 1, target health $h_k^0 = 2$ by default; $v_k = 0$, $D_k = 0$ initially. D_k accumulates raw asset damage.

Configuration C fixes roster, bounds, controls and task. ζ_t contains persistent assignments and opponent memory, if used. Reactive controllers have no hidden integrators. Last command, functional bits, pre-step positions and step damage are overwritten transition records; retain them for exact reward/event replay. RNG state is required for exact replay, not actor input.

B. Implemented kinematic map

For a living, noncolliding particle, $\Delta = 1$ s,

$$\begin{aligned} p_i^* &= p_i + \Delta v_i, & p_i' &= \Pi_{\mathcal{W}}(p_i^*), \\ w_i &= v_i + \Delta \alpha_i, & \tilde{v}_i &= \mathcal{S}_{35,120}(w_i; v_i). \end{aligned}$$

For $\|w\| \geq 10^{-3}$,

$$\mathcal{S}_{m,V}(w; v) = \frac{w}{\|w\|} \text{clip}(\|w\|, m, V).$$

For $\|w\| < 10^{-3}$, use $mv/\|v\|$ when $\|v\| \geq 10^{-3}$ and me_1 otherwise. The cancellation fallback retains heading. After speed normalization, an outward component is set to zero on each clipped axis: upper wall $v_a > 0$ or lower wall $v_a < 0$.

The final 3D turn map limits the angle between old and candidate velocities to $\theta_{\max} = \pi$ per step, so the default map is identity. For a smaller cap, rotate the old heading toward the candidate about $v_i \times \tilde{v}_i$, preserving candidate speed. A deterministic orthogonal axis resolves the antiparallel case; near-zero products $\|v\|_i \|\tilde{v}_i\| \leq 10^{-3}$ have angle zero.

Position uses old velocity, with no $\frac{1}{2}\alpha\Delta^2$ term. Boundary projection can reduce the final speed below 35 m/s. Agents killed by collision do not move; agents killed by combat complete this step's motion, and their velocity is zeroed on the next inactive update.

Interpretation

This is a kinematic benchmark. Its action controls future velocity and thus subsequent interception geometry, rather than instantaneous translation. Flight decisions and automatic functional decisions are distinct.

C. Acceleration intent

Each living agent chooses either $a_i \in [-1, 1]^3$ or an index into

$$\mathcal{V} = \{0\} \cup \{z/\|z\| : z \in \{-1, 0, 1\}^3 \setminus \{0\}\},$$

giving 27 discrete primitives, zero first. The physical intent is $\alpha_i = 40a_i$ m/s². Continuous inputs have component bounds, so diagonal intents can have norm $\sqrt{3}40$; discrete nonzero intents have norm 40. 40 is a control scale, not a clamp on actual rigid-body acceleration. The chosen vector directly enters the particle map as $\alpha_i = 40a_i$. No inner rigid-body controller or actuator state is introduced. Zero input preserves unconstrained velocity; it does not command a stop.

Episode geometry and initial law

The default scale is $\lambda = 1$, with

$$\mathcal{W} = \lambda([-2500, 2500]^2 \times [0, 2500]) \text{ m}.$$

Assets are sampled independently from $\lambda([-2300, -1900] \times [-1200, 1200] \times [500, 1500])$, or supplied as a fixed layout. Red chooses a target uniformly and samples an area-uniform XY annulus: $\rho^2 \sim U((800\lambda)^2, (2200\lambda)^2)$, $\vartheta \sim U(0, 2\pi)$; XY is clipped to $\mathcal{W}$. Blue samples $x \sim U(0, 2500\lambda)$ and $y \sim U(-2500\lambda, 2500\lambda)$. UAV altitude is sampled in $[200, 1500]\lambda$; target altitude is separate.

Random velocity has a Gaussian unit direction and speed $\|(u_1, u_2, u_3)\|/\sqrt{3}$, where $u_a \sim U(35, 120)$. Initial positions and velocities are generated by the environment-local RNG. The default collision distance is $d_c = 20\lambda = 20$ m. The full/zero-damage radii are $(r_0, r_1) = (300, 600)\lambda = (300, 600)$ m. The default firing threshold is $r_f = r_0$; configuration may select $0 < r_f \leq r_0$.

2D compatibility

Only particle themes support $d = 2$. The three coordinate slots remain, $p_z = 100\lambda$, $v_z = \alpha_z = 0$. Acceleration uses a 2-vector or nine planar primitives; position remains a 3-vector with prescribed altitude. Radii become $(200, 400)\lambda$. Planar resets reject too-close openings, and the heading cap is applied on the circle. These are compatibility settings of this theme, not additional themes.

Task dependence

Navigation and observations are identical between Survival and Damage. Asset health evolution, completion, and defender reward differ as specified on the next page. Neither task changes the particle into a rigid-body vehicle.

Theme 2 / Particle / Acceleration / Damage

Task interaction, exact information structure and defender learning objective

E. Collision and automatic interaction

Predict living UAV paths: particles use $p_i + \tau \Delta v_i$, rigid models use synchronized RK4 endpoints clipped to $\mathcal{W}$. On each matched segment, let $r = p_i - p_j$, $d = (q_i - p_i) - (q_j - p_j)$:

$$\tau_* = \begin{cases} 0 & \|d\|^2 \leq 10^{-12}, \\ \text{clip}(-r^T d / \|d\|^2, 0, 1) & \text{otherwise,} \end{cases} \quad d_* = \|r + \tau_* d\|.$$

Collision sets both UAV HP to zero when $\min d_* \leq d_c$; targets are excluded. Pairs use i -major, $j < i$ order. Dead outer agents are skipped at entry and dead inner agents are skipped. An outer agent killed during its inner sweep continues that sweep. After collision, surviving Attack agents automatically fire at step-start positions:

$$I_i = \begin{cases} \mathbf{1}[\min_{k \in \mathcal{P}: h_k > 0} \|p_i - p_k\| < r_f], & i \in \mathcal{B}, \\ \mathbf{1}[\min_{j \in \mathcal{B}_A: h_j > 0} \|p_i - p_j\| < r_f], & i \in \mathcal{R}. \end{cases}$$

An empty minimum is $+\infty$; dead sources have $I_i = 0$. A denotes Attack. Fire is a separate rule, not a navigation action. With unit attack intensity,

$$\eta(d) = \begin{cases} 1 & d < r_0, \\ (r_1 - d)/(r_1 - r_0) & r_0 \leq d < r_1, \\ 0 & d \geq r_1, \end{cases}$$

$$L_k = \sum_{b \in \mathcal{B}_A: h_b > 0} I_b \eta(\|p_b - p_k\|),$$

$$L_i^{\text{atk}} = \sum_{j \in \mathcal{U}_A: c_j \neq c_i, h_j > 0} I_j \eta(\|p_j - p_i\|).$$

One source can affect multiple enemies; friendly explosions are excluded. All losses use a common post-collision, pre-motion snapshot and add before HP clipping.

Disturb and Scout

For source j , $d_{ji} = \|p_i - p_j\|$ and $\theta_{ji} = \angle(v_j, p_i - p_j)$; angle is zero if either norm is below 10^{-3} . Let

$$G_{ji} = \mathbf{1}[10^{-3} < d_{ji} < 600\lambda, \theta_{ji} < \pi/6] \\ \times e^{-(d_{ji}/\lambda)^2/2} / \sqrt{2\pi},$$

$$S_c(j) = \sum_{i \in \mathcal{U}: c_i = c, h_i > 0} G_{ji}.$$

Living Red Disturb uses $J_j = \mathbf{1}[S_R(j) < S_B(j)]$, Blue uses $J_j = \mathbf{1}[S_R(j) \geq S_B(j)]$. Dead sources contribute zero.

$$L_i^{\text{dis}} = .1 \sum_{j \in \mathcal{U}_D: h_j > 0} J_j G_{ji},$$

$$\tilde{h}_i = \max(0, h_i - L_i^{\text{atk}} - L_i^{\text{dis}}).$$

Set $h'_i = 0$ if $r_i = A, I_i = 1$, otherwise $h'_i = \tilde{h}_i$. Disturb damages either side, not assets. Scouts use $d < 2000\lambda$, $\theta < \pi/3$; this sector does not restrict native actor rows. Collision victims do not move. Combat victims complete their motion; inactive velocity is zeroed next step.

G. Damage payoff and completion

Assets keep finite reset health $h'_k = h_k^0$. Count raw contributions before HP clipping, including simultaneous overkill:

$$D'_k = D_k + L_k, \quad d_t = \sum_k L_k, \quad R_t = -d_t.$$

Blue reward is d_t ; there is no team-size, asset-count or HP normalization and no terminal Survival bonus.

$$d = \mathbf{1}[\forall b \in \mathcal{B}_A : h'_b \leq 0].$$

All agent slots remain until global completion/truncation. Dead slots have zero observations, ignored actions and still receive full team R_t , including later damage after Red extinction. Do not sum duplicate teammate rewards. Undiscounted returns are $(-\sum_k D_k, +\sum_k D_k)$. H only truncates; global bootstrap $b_t = 1 - d$ remains one at truncation. Actor action-loss masks use pre-step participation.

F. Exact observations and CTDE information

For each living entity $j \neq i$, the actor row is

$$e_{ij} = [(p_j - p_i)/L, (v_j - v_i)/(2V_{\max}), \\ h_j, 1, \mathbf{1}_{c_j=R}, \mathbf{1}_{c_j=B}, \mathbf{1}_{j \in \mathcal{P}}],$$

$$L = \sqrt{\sum_a (b_a^+ - b_a^-)^2}.$$

Here $V_{\max}$ is the preset maximum speed, not the reference speed. Rows have 11 values, in fixed roster order: Red Scout/Disturb/Attack, Blue Scout/Disturb/Attack, then targets, excluding self. Dead rows are zero. Every living entity is included; Attack, Scout and Disturb share this packing. Side flags do not encode role. Own health is absent from the self block.

The actor input is only an $(M - 1) \times 11$ entity table ($V_{\max} = 120$); it has no own-state block. In Damage, an already-dead observer has a zero complete table. With deterministic encoding, $Z_i(o | s) = \delta_{\text{encode}_i(s)}(o)$. Other UAV attitudes/rates, opponent memory and time are omitted; hence local information is partial despite noiseless sensing.

Privileged $g(s) = \text{state}()$ packs 11 values per entity,

$$[(p - b^-)/(b^+ - b^-), v/(2V_{\max}), h, \ell, R, B, P],$$

with dead blocks zero. Rigid models append 13 raw state values per UAV (dead zero), then $[t, \sum_k D_k, \mathbf{1}_{\text{Damage}}]$: size $11M + 13N + 3$. Particle size is $11M$, omitting time and damage. g is a CTDE feature, not the complete Markov state.

H. Defender Dec-POMDP and MARL objective

For a specified frozen opposing policy π_B , define

$$\mathcal{M}_R = (\mathcal{R}, \mathcal{S}, \{\mathcal{A}_i\}, T^{\pi_B}, \\ \{\mathcal{O}_i\}, Z, R, \gamma, \rho_0, H),$$

$$T^{\pi_B}(s' | s, a_R) = \int \delta_{F_C(s, a_R, a_B)}(s') \\ \times \pi_B(da_B | o_B, \zeta_t).$$

F_C is the stated collision, simultaneous interaction and controlled-motion map, including any memory evolution; ρ_0 is the reset law. Native execution accepts both teams' actions. Freezing a rule or learned opponent is an experiment choice; with both teams learning, the model is a POSG.

$$\tau_i = (o_{i,0}, a_{i,0}, \dots, o_{i,t}),$$

$$\pi_R(a_R | \tau_R) = \prod_{i \in \mathcal{R}} \pi_i(a_i | \tau_i),$$

$$\pi_R^* \in \arg \max_{\pi_R} \mathbb{E} \left[\sum_{t=0}^{T-1} \gamma^t R_t \right].$$

Expectation is over ρ_0, T^{π_B}, π_R . Here T is natural completion (possibly infinite); H limits sampled rollouts. A truncated return is partial, so its successor value is retained. For $\gamma = 1$, this is respectively the native terminal Survival payoff or negative total asset damage. For $\gamma < 1$, it is the discounted task selected by the caller. Actors use local histories only. CTDE critics can use $g(s)$, joint actions and recorded task/memory features:

$$y_t = R_t + \gamma b_t Q^-(g(s'), a'_R).$$

Here Q^- is a target critic and successor actions follow the target actor policies. Use global completion, not individual death. A recurrent critic or augmented state may be needed because g omits history. This defines an objective; it does not prescribe an implemented optimizer.

Theme 3 / Particle / Position / Survival

Physical state, navigation action and discrete transition

A. Scenario and complete state

Red defenders $\mathcal{R}$ protect stationary assets $\mathcal{P}$ against Blue $\mathcal{B}$. Set $\mathcal{U} = \mathcal{R} \cup \mathcal{B}$, $\mathcal{E} = \mathcal{U} \cup \mathcal{P}$, $N = |\mathcal{U}|$, $K = |\mathcal{P}|$, $M = N + K$. Episode-fixed roles are Attack, Scout or Disturb.

$$s_t = (\{x_i, h_i, r_i, c_i, \text{id}_i\}_{i \in \mathcal{U}}, \{p_k, h_k, D_k\}_{k \in \mathcal{P}}, t, \zeta_t).$$

$x_i = (p_i, v_i) \in \mathcal{W} \times \mathbb{R}^3$ is a six-value kinematic state; no quaternion, inertia, wing or rotor is attached to a particle. Participation is $\ell_i = \mathbf{1}[h_i > 0]$. Reset UAV health is 1, target health $h_k^0 = 2$ by default; $v_k = 0$, $D_k = 0$ initially. D_k accumulates raw asset damage.

Configuration C fixes roster, bounds, controls and task. ζ_t contains persistent assignments and opponent memory, if used. Reactive controllers have no hidden integrators. Last command, functional bits, pre-step positions and step damage are overwritten transition records; retain them for exact reward/event replay. RNG state is required for exact replay, not actor input.

B. Implemented kinematic map

For a living, noncolliding particle, $\Delta = 1$ s,

$$\begin{aligned} p_i^* &= p_i + \Delta v_i, & p_i' &= \Pi_{\mathcal{W}}(p_i^*), \\ w_i &= v_i + \Delta \alpha_i, & \tilde{v}_i &= \mathcal{S}_{35,120}(w_i; v_i). \end{aligned}$$

For $\|w\| \geq 10^{-3}$,

$$\mathcal{S}_{m,V}(w; v) = \frac{w}{\|w\|} \text{clip}(\|w\|, m, V).$$

For $\|w\| < 10^{-3}$, use $mv/\|v\|$ when $\|v\| \geq 10^{-3}$ and me_1 otherwise. The cancellation fallback retains heading. After speed normalization, an outward component is set to zero on each clipped axis: upper wall $v_a > 0$ or lower wall $v_a < 0$.

The final 3D turn map limits the angle between old and candidate velocities to $\theta_{\max} = \pi$ per step, so the default map is identity. For a smaller cap, rotate the old heading toward the candidate about $v_i \times \tilde{v}_i$, preserving candidate speed. A deterministic orthogonal axis resolves the antiparallel case; near-zero products $\|v\|_i \|\tilde{v}_i\| \leq 10^{-3}$ have angle zero.

Position uses old velocity, with no $\frac{1}{2}\alpha\Delta^2$ term. Boundary projection can reduce the final speed below 35 m/s. Agents killed by collision do not move; agents killed by combat complete this step's motion, and their velocity is zeroed on the next inactive update.

Interpretation

This is a kinematic benchmark. Its action controls future velocity and thus subsequent interception geometry, rather than instantaneous translation. Flight decisions and automatic functional decisions are distinct.

C. Absolute position intent

Each agent selects $a_i = p_i^{\text{des}} \in \mathcal{W}$ in metres. The stateless outer loop computes

$$\begin{aligned} d_i &= p_i^{\text{des}} - p_i, \\ v_i^{\text{des}} &= d_i \min\{.5, 120 / \max(\|d_i\|, 10^{-12})\}, \\ \alpha_i &= \text{sat}_{40}(2(v_i^{\text{des}} - v_i)), \\ \text{sat}_L(z) &= z \min\{1, L/\|z\|\}, \quad \text{sat}_L(0) = 0. \end{aligned}$$

The command is an absolute point, not a displacement or teleport. The velocity reference can approach zero, but the particle speed map enforces its nonzero free-flight minimum; this interface does not promise arrival with rest. There are three continuous action coordinates and no position-action quantization.

Episode geometry and initial law

The default scale is $\lambda = 1$, with

$$\mathcal{W} = \lambda([-2500, 2500]^2 \times [0, 2500]) \text{ m.}$$

Assets are sampled independently from $\lambda([-2300, -1900] \times [-1200, 1200] \times [500, 1500])$, or supplied as a fixed layout. Red chooses a target uniformly and samples an area-uniform XY annulus: $\rho^2 \sim U((800\lambda)^2, (2200\lambda)^2)$, $\vartheta \sim U(0, 2\pi)$; XY is clipped to $\mathcal{W}$. Blue samples $x \sim U(0, 2500\lambda)$ and $y \sim U(-2500\lambda, 2500\lambda)$. UAV altitude is sampled in $[200, 1500]\lambda$; target altitude is separate.

Random velocity has a Gaussian unit direction and speed $\|(u_1, u_2, u_3)\|/\sqrt{3}$, where $u_a \sim U(35, 120)$. Initial positions and velocities are generated by the environment-local RNG. The default collision distance is $d_c = 20\lambda = 20$ m. The full/zero-damage radii are $(r_0, r_1) = (300, 600)\lambda = (300, 600)$ m. The default firing threshold is $r_f = r_0$; configuration may select $0 < r_f \leq r_0$.

2D compatibility

Only particle themes support $d = 2$. The three coordinate slots remain, $p_z = 100\lambda$, $v_z = \alpha_z = 0$. Acceleration uses a 2-vector or nine planar primitives; position remains a 3-vector with prescribed altitude. Radii become $(200, 400)\lambda$. Planar resets reject too-close openings, and the heading cap is applied on the circle. These are compatibility settings of this theme, not additional themes.

Task dependence

Navigation and observations are identical between Survival and Damage. Asset health evolution, completion, and defender reward differ as specified on the next page. Neither task changes the particle into a rigid-body vehicle.

Theme 3 / Particle / Position / Survival

Task interaction, exact information structure and defender learning objective

E. Collision and automatic interaction

Predict living UAV paths: particles use $p_i + \tau \Delta v_i$, rigid models use synchronized RK4 endpoints clipped to $\mathcal{W}$. On each matched segment, let $r = p_i - p_j$, $d = (q_i - p_i) - (q_j - p_j)$:

$$\tau_* = \begin{cases} 0 & \|d\|^2 \leq 10^{-12}, \\ \text{clip}(-r^T d / \|d\|^2, 0, 1) & \text{otherwise,} \end{cases} \quad d_* = \|r + \tau_* d\|.$$

Collision sets both UAV HP to zero when $\min d_* \leq d_c$; targets are excluded. Pairs use i -major, $j < i$ order. Dead outer agents are skipped at entry and dead inner agents are skipped. An outer agent killed during its inner sweep continues that sweep. After collision, surviving Attack agents automatically fire at step-start positions:

$$I_i = \begin{cases} \mathbf{1}[\min_{k \in \mathcal{P}: h_k > 0} \|p_i - p_k\| < r_f], & i \in \mathcal{B}, \\ \mathbf{1}[\min_{j \in \mathcal{B}_A: h_j > 0} \|p_i - p_j\| < r_f], & i \in \mathcal{R}. \end{cases}$$

An empty minimum is $+\infty$; dead sources have $I_i = 0$. A denotes Attack. Fire is a separate rule, not a navigation action. With unit attack intensity,

$$\eta(d) = \begin{cases} 1 & d < r_0, \\ (r_1 - d)/(r_1 - r_0) & r_0 \leq d < r_1, \\ 0 & d \geq r_1, \end{cases}$$

$$L_k = \sum_{b \in \mathcal{B}_A: h_b > 0} I_b \eta(\|p_b - p_k\|),$$

$$L_i^{\text{atk}} = \sum_{j \in \mathcal{U}_A: c_j \neq c_i, h_j > 0} I_j \eta(\|p_j - p_i\|).$$

One source can affect multiple enemies; friendly explosions are excluded. All losses use a common post-collision, pre-motion snapshot and add before HP clipping.

Disturb and Scout

For source j , $d_{ji} = \|p_i - p_j\|$ and $\theta_{ji} = \angle(v_j, p_i - p_j)$; angle is zero if either norm is below 10^{-3} . Let

$$G_{ji} = \mathbf{1}[10^{-3} < d_{ji} < 600\lambda, \theta_{ji} < \pi/6] \\ \times e^{-(d_{ji}/\lambda)^2/2} / \sqrt{2\pi},$$

$$S_c(j) = \sum_{i \in \mathcal{U}: c_i = c, h_i > 0} G_{ji}.$$

Living Red Disturb uses $J_j = \mathbf{1}[S_R(j) < S_B(j)]$, Blue uses $J_j = \mathbf{1}[S_R(j) \geq S_B(j)]$. Dead sources contribute zero.

$$L_i^{\text{dis}} = .1 \sum_{j \in \mathcal{U}_D: h_j > 0} J_j G_{ji},$$

$$\tilde{h}_i = \max(0, h_i - L_i^{\text{atk}} - L_i^{\text{dis}}).$$

Set $h'_i = 0$ if $r_i = A, I_i = 1$, otherwise $h'_i = \tilde{h}_i$. Disturb damages either side, not assets. Scouts use $d < 2000\lambda$, $\theta < \pi/3$; this sector does not restrict native actor rows. Collision victims do not move. Combat victims complete their motion; inactive velocity is zeroed next step.

G. Survival payoff and completion

Assets obey $h'_k = \max(0, h_k - L_k)$, $D'_k = D_k + L_k$.

$$z(s') = \begin{cases} -1 & \exists k : h'_k < 10^{-3}, \\ +1 & \forall b \in \mathcal{B}_A : h'_b < 10^{-3}, \\ 0 & \text{otherwise.} \end{cases}$$

Target loss has priority. Natural completion is $d = \mathbf{1}[z \neq 0]$; default Red reward $R_t = 10z$, Blue $-R_t$. Red elimination alone does not end the world. Individual $h'_i \leq 0$ retires a Survival actor after its final transition. Horizon H truncates without a win bonus; use global bootstrap $b_t = 1 - d$, including individual death and truncation. Actor action-loss masks use pre-step participation. Explicit reward weights replace the default payoff as defined on page 1.

F. Exact observations and CTDE information

For each living entity $j \neq i$, the actor row is

$$e_{ij} = [(p_j - p_i)/L, (v_j - v_i)/(2V_{\max}), \\ h_j, 1, \mathbf{1}_{c_j=R}, \mathbf{1}_{c_j=B}, \mathbf{1}_{j \in \mathcal{P}}], \\ L = \sqrt{\sum_a (b_a^+ - b_a^-)^2}.$$

Here $V_{\max}$ is the preset maximum speed, not the reference speed. Rows have 11 values, in fixed roster order: Red Scout/Disturb/Attack, Blue Scout/Disturb/Attack, then targets, excluding self. Dead rows are zero. Every living entity is included; Attack, Scout and Disturb share this packing. Side flags do not encode role. Own health is absent from the self block.

The actor receives the $(M-1) \times 11$ entity table ($V_{\max} = 120$) and the unnormalized six-value self block $[p_i, v_i]$. With deterministic encoding, $Z_i(o | s) = \delta_{\text{encode}_i(s)}(o)$. Other UAV attitudes/rates, opponent memory and time are omitted; hence local information is partial despite noiseless sensing.

Privileged $g(s) = \text{state}()$ packs 11 values per entity,

$$[(p - b^-)/(b^+ - b^-), v/(2V_{\max}), h, \ell, R, B, P],$$

with dead blocks zero. Rigid models append 13 raw state values per UAV (dead zero), then $[t, \sum_k D_k, \mathbf{1}_{\text{Damage}}]$: size $11M + 13N + 3$. Particle size is $11M$, omitting time and damage. g is a CTDE feature, not the complete Markov state.

H. Defender Dec-POMDP and MARL objective

For a specified frozen opposing policy π_B , define

$$\mathcal{M}_R = (\mathcal{R}, \mathcal{S}, \{\mathcal{A}_i\}, T^{\pi_B}, \\ \{\mathcal{O}_i\}, Z, R, \gamma, \rho_0, H),$$

$$T^{\pi_B}(s' | s, a_R) = \int \delta_{F_C(s, a_R, a_B)}(s') \\ \times \pi_B(da_B | o_B, \zeta_t).$$

F_C is the stated collision, simultaneous interaction and controlled-motion map, including any memory evolution; ρ_0 is the reset law. Native execution accepts both teams' actions. Freezing a rule or learned opponent is an experiment choice; with both teams learning, the model is a POSG.

$$\tau_i = (o_{i,0}, a_{i,0}, \dots, o_{i,t}),$$

$$\pi_R(a_R | \tau_R) = \prod_{i \in \mathcal{R}} \pi_i(a_i | \tau_i),$$

$$\pi_R^* \in \arg \max_{\pi_R} \mathbb{E} \left[\sum_{t=0}^{T-1} \gamma^t R_t \right].$$

Expectation is over ρ_0, T^{π_B}, π_R . Here T is natural completion (possibly infinite); H limits sampled rollouts. A truncated return is partial, so its successor value is retained. For $\gamma = 1$, this is respectively the native terminal Survival payoff or negative total asset damage. For $\gamma < 1$, it is the discounted task selected by the caller. Actors use local histories only. CTDE critics can use $g(s)$, joint actions and recorded task/memory features:

$$y_t = R_t + \gamma b_t Q^-(g(s'), a'_R).$$

Here Q^- is a target critic and successor actions follow the target actor policies. Use global completion, not individual death. A recurrent critic or augmented state may be needed because g omits history. This defines an objective; it does not prescribe an implemented optimizer.

Theme 4 / Particle / Position / Damage

Physical state, navigation action and discrete transition

A. Scenario and complete state

Red defenders $\mathcal{R}$ protect stationary assets $\mathcal{P}$ against Blue $\mathcal{B}$. Set $\mathcal{U} = \mathcal{R} \cup \mathcal{B}$, $\mathcal{E} = \mathcal{U} \cup \mathcal{P}$, $N = |\mathcal{U}|$, $K = |\mathcal{P}|$, $M = N + K$. Episode-fixed roles are Attack, Scout or Disturb.

$$s_t = (\{x_i, h_i, r_i, c_i, \text{id}_i\}_{i \in \mathcal{U}}, \{p_k, h_k, D_k\}_{k \in \mathcal{P}}, t, \zeta_t).$$

$x_i = (p_i, v_i) \in \mathcal{W} \times \mathbb{R}^3$ is a six-value kinematic state; no quaternion, inertia, wing or rotor is attached to a particle. Participation is $\ell_i = \mathbf{1}[h_i > 0]$. Reset UAV health is 1, target health $h_k^0 = 2$ by default; $v_k = 0$, $D_k = 0$ initially. D_k accumulates raw asset damage.

Configuration C fixes roster, bounds, controls and task. ζ_t contains persistent assignments and opponent memory, if used. Reactive controllers have no hidden integrators. Last command, functional bits, pre-step positions and step damage are overwritten transition records; retain them for exact reward/event replay. RNG state is required for exact replay, not actor input.

B. Implemented kinematic map

For a living, noncolliding particle, $\Delta = 1$ s,

$$\begin{aligned} p_i^* &= p_i + \Delta v_i, & p_i' &= \Pi_{\mathcal{W}}(p_i^*), \\ w_i &= v_i + \Delta \alpha_i, & \tilde{v}_i &= \mathcal{S}_{35,120}(w_i; v_i). \end{aligned}$$

For $\|w\| \geq 10^{-3}$,

$$\mathcal{S}_{m,V}(w; v) = \frac{w}{\|w\|} \text{clip}(\|w\|, m, V).$$

For $\|w\| < 10^{-3}$, use $mv/\|v\|$ when $\|v\| \geq 10^{-3}$ and me_1 otherwise. The cancellation fallback retains heading. After speed normalization, an outward component is set to zero on each clipped axis: upper wall $v_a > 0$ or lower wall $v_a < 0$.

The final 3D turn map limits the angle between old and candidate velocities to $\theta_{\max} = \pi$ per step, so the default map is identity. For a smaller cap, rotate the old heading toward the candidate about $v_i \times \tilde{v}_i$, preserving candidate speed. A deterministic orthogonal axis resolves the antiparallel case; near-zero products $\|v\|_i \|\tilde{v}_i\| \leq 10^{-3}$ have angle zero.

Position uses old velocity, with no $\frac{1}{2}\alpha\Delta^2$ term. Boundary projection can reduce the final speed below 35 m/s. Agents killed by collision do not move; agents killed by combat complete this step's motion, and their velocity is zeroed on the next inactive update.

Interpretation

This is a kinematic benchmark. Its action controls future velocity and thus subsequent interception geometry, rather than instantaneous translation. Flight decisions and automatic functional decisions are distinct.

C. Absolute position intent

Each agent selects $a_i = p_i^{\text{des}} \in \mathcal{W}$ in metres. The stateless outer loop computes

$$\begin{aligned} d_i &= p_i^{\text{des}} - p_i, \\ v_i^{\text{des}} &= d_i \min\{.5, 120 / \max(\|d_i\|, 10^{-12})\}, \\ \alpha_i &= \text{sat}_{40}(2(v_i^{\text{des}} - v_i)), \\ \text{sat}_L(z) &= z \min\{1, L/\|z\|\}, \quad \text{sat}_L(0) = 0. \end{aligned}$$

The command is an absolute point, not a displacement or teleport. The velocity reference can approach zero, but the particle speed map enforces its nonzero free-flight minimum; this interface does not promise arrival with rest. There are three continuous action coordinates and no position-action quantization.

Episode geometry and initial law

The default scale is $\lambda = 1$, with

$$\mathcal{W} = \lambda([-2500, 2500]^2 \times [0, 2500]) \text{ m.}$$

Assets are sampled independently from $\lambda([-2300, -1900] \times [-1200, 1200] \times [500, 1500])$, or supplied as a fixed layout. Red chooses a target uniformly and samples an area-uniform XY annulus: $\rho^2 \sim U((800\lambda)^2, (2200\lambda)^2)$, $\vartheta \sim U(0, 2\pi)$; XY is clipped to $\mathcal{W}$. Blue samples $x \sim U(0, 2500\lambda)$ and $y \sim U(-2500\lambda, 2500\lambda)$. UAV altitude is sampled in $[200, 1500]\lambda$; target altitude is separate.

Random velocity has a Gaussian unit direction and speed $\|(u_1, u_2, u_3)\|/\sqrt{3}$, where $u_a \sim U(35, 120)$. Initial positions and velocities are generated by the environment-local RNG. The default collision distance is $d_c = 20\lambda = 20$ m. The full/zero-damage radii are $(r_0, r_1) = (300, 600)\lambda = (300, 600)$ m. The default firing threshold is $r_f = r_0$; configuration may select $0 < r_f \leq r_0$.

2D compatibility

Only particle themes support $d = 2$. The three coordinate slots remain, $p_z = 100\lambda$, $v_z = \alpha_z = 0$. Acceleration uses a 2-vector or nine planar primitives; position remains a 3-vector with prescribed altitude. Radii become $(200, 400)\lambda$. Planar resets reject too-close openings, and the heading cap is applied on the circle. These are compatibility settings of this theme, not additional themes.

Task dependence

Navigation and observations are identical between Survival and Damage. Asset health evolution, completion, and defender reward differ as specified on the next page. Neither task changes the particle into a rigid-body vehicle.

Theme 4 / Particle / Position / Damage

Task interaction, exact information structure and defender learning objective

E. Collision and automatic interaction

Predict living UAV paths: particles use $p_i + \tau \Delta v_i$, rigid models use synchronized RK4 endpoints clipped to $\mathcal{W}$. On each matched segment, let $r = p_i - p_j$, $d = (q_i - p_i) - (q_j - p_j)$:

$$\tau_* = \begin{cases} 0 & \|d\|^2 \leq 10^{-12}, \\ \text{clip}(-r^T d / \|d\|^2, 0, 1) & \text{otherwise,} \end{cases} \quad d_* = \|r + \tau_* d\|.$$

Collision sets both UAV HP to zero when $\min d_* \leq d_c$; targets are excluded. Pairs use i -major, $j < i$ order. Dead outer agents are skipped at entry and dead inner agents are skipped. An outer agent killed during its inner sweep continues that sweep. After collision, surviving Attack agents automatically fire at step-start positions:

$$I_i = \begin{cases} \mathbf{1}[\min_{k \in \mathcal{P}: h_k > 0} \|p_i - p_k\| < r_f], & i \in \mathcal{B}, \\ \mathbf{1}[\min_{j \in \mathcal{B}_A: h_j > 0} \|p_i - p_j\| < r_f], & i \in \mathcal{R}. \end{cases}$$

An empty minimum is $+\infty$; dead sources have $I_i = 0$. A denotes Attack. Fire is a separate rule, not a navigation action. With unit attack intensity,

$$\eta(d) = \begin{cases} 1 & d < r_0, \\ (r_1 - d)/(r_1 - r_0) & r_0 \leq d < r_1, \\ 0 & d \geq r_1, \end{cases}$$

$$L_k = \sum_{b \in \mathcal{B}_A: h_b > 0} I_b \eta(\|p_b - p_k\|),$$

$$L_i^{\text{atk}} = \sum_{j \in \mathcal{U}_A: c_j \neq c_i, h_j > 0} I_j \eta(\|p_j - p_i\|).$$

One source can affect multiple enemies; friendly explosions are excluded. All losses use a common post-collision, pre-motion snapshot and add before HP clipping.

Disturb and Scout

For source j , $d_{ji} = \|p_i - p_j\|$ and $\theta_{ji} = \angle(v_j, p_i - p_j)$; angle is zero if either norm is below 10^{-3} . Let

$$G_{ji} = \mathbf{1}[10^{-3} < d_{ji} < 600\lambda, \theta_{ji} < \pi/6] \\ \times e^{-(d_{ji}/\lambda)^2/2} / \sqrt{2\pi},$$

$$S_c(j) = \sum_{i \in \mathcal{U}: c_i = c, h_i > 0} G_{ji}.$$

Living Red Disturb uses $J_j = \mathbf{1}[S_R(j) < S_B(j)]$, Blue uses $J_j = \mathbf{1}[S_R(j) \geq S_B(j)]$. Dead sources contribute zero.

$$L_i^{\text{dis}} = .1 \sum_{j \in \mathcal{U}_D: h_j > 0} J_j G_{ji},$$

$$\tilde{h}_i = \max(0, h_i - L_i^{\text{atk}} - L_i^{\text{dis}}).$$

Set $h'_i = 0$ if $r_i = A, I_i = 1$, otherwise $h'_i = \tilde{h}_i$. Disturb damages either side, not assets. Scouts use $d < 2000\lambda$, $\theta < \pi/3$; this sector does not restrict native actor rows. Collision victims do not move. Combat victims complete their motion; inactive velocity is zeroed next step.

G. Damage payoff and completion

Assets keep finite reset health $h'_k = h_k^0$. Count raw contributions before HP clipping, including simultaneous overkill:

$$D'_k = D_k + L_k, \quad dt = \sum_k L_k, \quad R_t = -dt.$$

Blue reward is d_t ; there is no team-size, asset-count or HP normalization and no terminal Survival bonus.

$$d = \mathbf{1}[\forall b \in \mathcal{B}_A : h'_b \leq 0].$$

All agent slots remain until global completion/truncation. Dead slots have zero observations, ignored actions and still receive full team R_t , including later damage after Red extinction. Do not sum duplicate teammate rewards. Undiscounted returns are $(-\sum_k D_k, +\sum_k D_k)$. H only truncates; global bootstrap $b_t = 1 - d$ remains one at truncation. Actor action-loss masks use pre-step participation.

F. Exact observations and CTDE information

For each living entity $j \neq i$, the actor row is

$$e_{ij} = [(p_j - p_i)/L, (v_j - v_i)/(2V_{\max}), \\ h_j, 1, \mathbf{1}_{c_j=R}, \mathbf{1}_{c_j=B}, \mathbf{1}_{j \in \mathcal{P}}],$$

$$L = \sqrt{\sum_a (b_a^+ - b_a^-)^2}.$$

Here $V_{\max}$ is the preset maximum speed, not the reference speed. Rows have 11 values, in fixed roster order: Red Scout/Disturb/Attack, Blue Scout/Disturb/Attack, then targets, excluding self. Dead rows are zero. Every living entity is included; Attack, Scout and Disturb share this packing. Side flags do not encode role. Own health is absent from the self block.

The actor receives the $(M - 1) \times 11$ entity table ($V_{\max} = 120$) and unnormalized $[p_i, v_i]$. Both blocks are zero for an already-dead Damage observer. With deterministic encoding, $Z_i(o | s) = \delta_{\text{encode}_i(s)}(o)$. Other UAV attitudes/rates, opponent memory and time are omitted; hence local information is partial despite noiseless sensing.

Privileged $g(s) = \text{state}()$ packs 11 values per entity,

$$[(p - b^-)/(b^+ - b^-), v/(2V_{\max}), h, \ell, R, B, P],$$

with dead blocks zero. Rigid models append 13 raw state values per UAV (dead zero), then $[t, \sum_k D_k, \mathbf{1}_{\text{Damage}}]$: size $11M + 13N + 3$. Particle size is $11M$, omitting time and damage. g is a CTDE feature, not the complete Markov state.

H. Defender Dec-POMDP and MARL objective

For a specified frozen opposing policy π_B , define

$$\mathcal{M}_R = (\mathcal{R}, \mathcal{S}, \{\mathcal{A}_i\}, T^{\pi_B}, \\ \{\mathcal{O}_i\}, Z, R, \gamma, \rho_0, H),$$

$$T^{\pi_B}(s' | s, a_R) = \int \delta_{F_C(s, a_R, a_B)}(s') \\ \times \pi_B(da_B | o_B, \zeta_t).$$

F_C is the stated collision, simultaneous interaction and controlled-motion map, including any memory evolution; ρ_0 is the reset law. Native execution accepts both teams' actions. Freezing a rule or learned opponent is an experiment choice; with both teams learning, the model is a POSG.

$$\tau_i = (o_{i,0}, a_{i,0}, \dots, o_{i,t}),$$

$$\pi_R(a_R | \tau_R) = \prod_{i \in \mathcal{R}} \pi_i(a_i | \tau_i),$$

$$\pi_R^* \in \arg \max_{\pi_R} \mathbb{E} \left[\sum_{t=0}^{T-1} \gamma^t R_t \right].$$

Expectation is over ρ_0, T^{π_B}, π_R . Here T is natural completion (possibly infinite); H limits sampled rollouts. A truncated return is partial, so its successor value is retained. For $\gamma = 1$, this is respectively the native terminal Survival payoff or negative total asset damage. For $\gamma < 1$, it is the discounted task selected by the caller. Actors use local histories only. CTDE critics can use $g(s)$, joint actions and recorded task/memory features:

$$y_t = R_t + \gamma b_t Q^-(g(s'), a'_R).$$

Here Q^- is a target critic and successor actions follow the target actor policies. Use global completion, not individual death. A recurrent critic or augmented state may be needed because g omits history. This defines an objective; it does not prescribe an implemented optimizer.

Theme 5 / Fixed wing / Acceleration / Survival

Rigid-body aerodynamic transition and navigation/actuator semantics

A. Scenario and complete state

Red defenders $\mathcal{R}$ protect stationary assets $\mathcal{P}$ against Blue $\mathcal{B}$. Set $\mathcal{U} = \mathcal{R} \cup \mathcal{B}$, $\mathcal{E} = \mathcal{U} \cup \mathcal{P}$, $N = |\mathcal{U}|$, $K = |\mathcal{P}|$, $M = N + K$. Episode-fixed roles are Attack, Scout or Disturb.

$$s_t = (\{x_i, h_i, r_i, c_i, \text{id}_i\}_{i \in \mathcal{U}}, \\ \{p_k, h_k, D_k\}_{k \in \mathcal{P}}, t, \zeta_t).$$

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. Participation is $\ell_i = \mathbf{1}[h_i > 0]$. Reset UAV health is 1, target health $h_k^0 = 2$ by default; $v_k = 0$, $D_k = 0$ initially. D_k accumulates raw asset damage.

Configuration C fixes roster, bounds, controls and task. ζ_t contains persistent assignments and opponent memory, if used. Reactive controllers have no hidden integrators. Last command, functional bits, pre-step positions and step damage are overwritten transition records; retain them for exact reward/event replay. RNG state is required for exact replay, not actor input.

B. Six-degree-of-freedom dynamics

World coordinates are east-north-up (ENU); body coordinates are forward-left-up (FLU). The scalar-first quaternion $q = (q_w, q_x, q_y, q_z)$ maps body to world through $R(q) \in SO(3)$, with $q \in S^3$, $\|q\| = 1$ and $q \sim -q$. Thus 13 stored values represent 12 physical degrees of freedom.

$$\dot{p} = v, \quad \dot{v} = R(q)F_B/m - g e_3, \\ \dot{q} = \frac{1}{2}q \otimes (0, \omega), \quad \dot{\omega} = J^{-1}(M_B - \omega \times J\omega).$$

Here $g = 9.81 \text{ m s}^{-2}$, $\omega \in \mathbb{R}^3$ is the body angular velocity in rad/s; F_B excludes gravity. No wind, sensor noise or actuator lag is present.

$$R(q) = I + 2q_w[q_v]_{\times} + 2[q_v]_{\times}^2, \quad q_v = (q_x, q_y, q_z).$$

Here $[z]_{\times} y = z \times y$.

C. Aerosonde aerodynamic wrench

Let $C_f = \text{diag}(1, -1, -1)$ and $(u, v_b, w) = C_f R^T v$, $(p_b, q_b, r_b) = C_f \omega$. Define $V = \|(u, v_b, w)\|$, $\bar{V} = \max(V, 10^{-6})$, $\alpha = \text{atan2}(w, u)$, $\beta = \arcsin \text{clip}(v_b/\bar{V}, -1, 1)$, $\bar{Q} = \rho V^2/2$, $\hat{p} = bp_b/(2\bar{V})$, $\hat{q} = cq_b/(2\bar{V})$, $\hat{r} = br_b/(2\bar{V})$.

$$E_- = e^{\text{clip}(-50(\alpha-.47), -500, 500)}, \\ E_+ = e^{\text{clip}(50(\alpha+.47), -500, 500)}, \\ \sigma = (1 + E_- + E_+)/[(1 + E_-)(1 + E_+)], \\ C_L = (1 - \sigma)(.23 + 5.61\alpha) \\ + 2\sigma \text{sign}(\alpha) \sin^2 \alpha \cos \alpha + 7.95\hat{q} + .13\delta_e, \\ C_D = .043 + \frac{(.23 + 5.61\alpha)^2}{\pi(.9)(b^2/S)} + .0135\delta_e, \\ C_Y = -.98\beta + .075\delta_a + .19\delta_r, \\ C_\ell = -.13\beta - .51\hat{p} + .25\hat{r} + .17\delta_a + .0024\delta_r, \\ C_m = .0135 - 2.74\alpha - 38.21\hat{q} - .99\delta_e, \\ C_n = .073\beta + .069\hat{p} - .095\hat{r} - .011\delta_a - .069\delta_r.$$

$$F_B = C_f \left\{ \bar{Q} S \begin{bmatrix} -C_D \cos \alpha + C_L \sin \alpha \\ C_Y \\ -C_D \sin \alpha - C_L \cos \alpha \end{bmatrix} + \begin{bmatrix} T_p \\ 0 \\ 0 \end{bmatrix} \right\}, \\ M_B = C_f \left\{ \bar{Q} S \begin{bmatrix} bC_\ell \\ cC_m \\ bC_n \end{bmatrix} - \begin{bmatrix} Q_p \\ 0 \\ 0 \end{bmatrix} \right\}.$$

Motor, propeller and equilibrium

The physical constants are $m = 11 \text{ kg}$, $S = .55 \text{ m}^2$, $b = 2.8956 \text{ m}$, $c = .18994 \text{ m}$, $\rho = 1.2682 \text{ kg/m}^3$,

$$J = C_f \begin{bmatrix} .8244 & 0 & -.1204 \\ 0 & 1.135 & 0 \\ -.1204 & 0 & 1.759 \end{bmatrix} C_f^T.$$

With $D_p = .508$, $R_m = .042$, $I_0 = 1.5$, $V_m = 44.4$, $k_m = 60/(145\pi)$, the steady motor root is

$$A = \rho D_p^5 C_{Q0}/(4\pi^2), \\ B = \rho D_p^4 C_{Q1} V/(2\pi) + k_m^2/R_m, \\ C = \rho D_p^3 C_{Q2} V^2 - k_m V_m \tau/R_m + k_m I_0, \\ \Omega_p = \max\{0, (-B + \sqrt{\max(0, B^2 - 4AC)})/(2A)\}.$$

Set $z = (D_p^4 \Omega_p^2/(4\pi^2), D_p^3 V \Omega_p/(2\pi), D_p^2 V^2)$,

$$T_p = \rho(.09357, -.06044, -.1079)z, \\ Q_p = \rho D_p(.005230, .004970, -.01664)z.$$

Reference trim $(x^*(V_d), u^*(V_d))$ solves all six equations $\dot{v} = 0$, $\dot{\omega} = 0$, including propeller reaction torque. Unknowns are pitch, heading, elevator, aileron, rudder and throttle, at velocity $(V_d, 0, 0)$. The reset uses the 30 m/s level trim rotated to the initial heading. Mass/inertia are not scaled with the scene [1].

Action and reference conversion

Choose $a_i \in [-1, 1]^3$ or one of the 27 unit primitives. Scale once to $\alpha_i = 5a_i$, then hold $v_d = v_i(t) + \Delta\alpha_i$. At each RK4 stage,

$$\chi_d = \text{atan2}(v_{dy}, v_{dx}), \\ \gamma_d = \text{atan2}(v_{dz}, \max(\|v_{d,xy}\|, 10^{-6})), \\ V_d = \|v_d\|.$$

For tiny horizontal reference speed, χ_d is current heading. Component-limited continuous intent need not have norm ≤ 5 .

D. Implemented reference controller

Define bank $\phi = \text{atan2}(R_{32}, R_{33})$, nose-up pitch $\theta = \arcsin R_{31}$, heading $\psi = \text{atan2}(R_{21}, R_{11})$. Actual course is $\chi = \text{atan2}(v_y, v_x)$, or ψ when horizontal speed is tiny; path $\gamma_f = \text{atan2}(v_z, \|v_{xy}\|)$ and $V_a = \|v\|$.

$$V_d = \text{clip}(V_d, 18, 30), \\ \gamma_d = \text{clip}(\gamma_d, -15^\circ, 15^\circ), \\ e_\chi = \text{wrap}(\chi_d - \chi), \\ \bar{\phi} = -\text{atan2}(\max(V_a, 18)e_\chi, g), \\ \phi_d = \text{clip}(\bar{\phi}, -45^\circ, 45^\circ), \\ \bar{\theta} = \theta^* + \gamma_d + .5(\gamma_d - \gamma_f), \\ \theta_d = \text{clip}(\bar{\theta}, -20^\circ, 20^\circ), \\ \delta_e = \delta_e^* - 2(\theta_d - \theta) + .25q_b, \\ \delta_a = \delta_a^* + .5(\phi_d - \phi) - .08p_b, \\ r_d = g \tan \phi_d / \max(V_a, 18), \\ \delta_r = \delta_r^* + .8(\beta - \psi^*) + .1(r_b - r_d), \\ \tau = \tau^* + .08(V_d - V_a) + .7 \sin \gamma_d.$$

All angles use radians. The scheduled trim heading ψ^* is the lateral reference. Divide surfaces by 25° , map throttle to $2\tau - 1$ and clip the four outputs to $[-1, 1]$. These project gains/bounds do not clamp the physical flight envelope.

Integration, geometry and initialization

RK4 uses $h = .01 \text{ s}$, 100 synchronized substeps per 1 s step. Each stage feeds back the held reference: $k_j = f(x_j, U(x_j))$ and $x^+ = x + h(k_1 + 2k_2 + 2k_3 + k_4)/6$, then normalize q^+ . Default $\lambda = .4$, $d_c = 8 \text{ m}$, $(r_0, r_1) = (120, 240) \text{ m}$, $\mathcal{W} = [-1000, 1000]^2 \times [0, 1000] \text{ m}$. Clip path positions and the endpoint to scene bounds; remove endpoint outward velocity. Reset geometry follows the area-uniform Red annulus / Blue rectangular region, scaled by .4, with separated UAV openings and solved 30 m/s level trim. Only 3D is supported.

Theme 5 / Fixed wing / Acceleration / Survival

Task interaction, exact information structure and defender learning objective

E. Collision and automatic interaction

Predict living UAV paths: particles use $p_i + \tau \Delta v_i$, rigid models use synchronized RK4 endpoints clipped to $\mathcal{W}$. On each matched segment, let $r = p_i - p_j$, $d = (q_i - p_i) - (q_j - p_j)$:

$$\tau_* = \begin{cases} 0 & \|d\|^2 \leq 10^{-12}, \\ \text{clip}(-r^T d / \|d\|^2, 0, 1) & \text{otherwise,} \end{cases} \quad d_* = \|r + \tau_* d\|.$$

Collision sets both UAV HP to zero when $\min d_* \leq d_c$; targets are excluded. Pairs use i -major, $j < i$ order. Dead outer agents are skipped at entry and dead inner agents are skipped. An outer agent killed during its inner sweep continues that sweep. After collision, surviving Attack agents automatically fire at step-start positions:

$$I_i = \begin{cases} \mathbf{1}[\min_{k \in \mathcal{P}: h_k > 0} \|p_i - p_k\| < r_f], & i \in \mathcal{B}, \\ \mathbf{1}[\min_{j \in \mathcal{B}_A: h_j > 0} \|p_i - p_j\| < r_f], & i \in \mathcal{R}. \end{cases}$$

An empty minimum is $+\infty$; dead sources have $I_i = 0$. A denotes Attack. Fire is a separate rule, not a navigation action. With unit attack intensity,

$$\eta(d) = \begin{cases} 1 & d < r_0, \\ (r_1 - d)/(r_1 - r_0) & r_0 \leq d < r_1, \\ 0 & d \geq r_1, \end{cases}$$

$$L_k = \sum_{b \in \mathcal{B}_A: h_b > 0} I_b \eta(\|p_b - p_k\|),$$

$$L_i^{\text{atk}} = \sum_{j \in \mathcal{U}_A: c_j \neq c_i, h_j > 0} I_j \eta(\|p_j - p_i\|).$$

One source can affect multiple enemies; friendly explosions are excluded. All losses use a common post-collision, pre-motion snapshot and add before HP clipping.

Disturb and Scout

For source j , $d_{ji} = \|p_i - p_j\|$ and $\theta_{ji} = \angle(v_j, p_i - p_j)$; angle is zero if either norm is below 10^{-3} . Let

$$G_{ji} = \mathbf{1}[10^{-3} < d_{ji} < 600\lambda, \theta_{ji} < \pi/6] \times e^{-(d_{ji}/\lambda)^2/2} / \sqrt{2\pi},$$

$$S_c(j) = \sum_{i \in \mathcal{U}: c_i = c, h_i > 0} G_{ji}.$$

Living Red Disturb uses $J_j = \mathbf{1}[S_R(j) < S_B(j)]$, Blue uses $J_j = \mathbf{1}[S_R(j) \geq S_B(j)]$. Dead sources contribute zero.

$$L_i^{\text{dis}} = .1 \sum_{j \in \mathcal{U}_D: h_j > 0} J_j G_{ji},$$

$$\tilde{h}_i = \max(0, h_i - L_i^{\text{atk}} - L_i^{\text{dis}}).$$

Set $h'_i = 0$ if $r_i = A, I_i = 1$, otherwise $h'_i = \tilde{h}_i$. Disturb damages either side, not assets. Scouts use $d < 2000\lambda$, $\theta < \pi/3$; this sector does not restrict native actor rows. Collision victims do not move. Combat victims complete their motion; inactive velocity is zeroed next step.

G. Survival payoff and completion

Assets obey $h'_k = \max(0, h_k - L_k)$, $D'_k = D_k + L_k$.

$$z(s') = \begin{cases} -1 & \exists k : h'_k < 10^{-3}, \\ +1 & \forall b \in \mathcal{B}_A : h'_b < 10^{-3}, \\ 0 & \text{otherwise.} \end{cases}$$

Target loss has priority. Natural completion is $d = \mathbf{1}[z \neq 0]$; default Red reward $R_t = 10z$, Blue $-R_t$. Red elimination alone does not end the world. Individual $h'_i \leq 0$ retires a Survival actor after its final transition. Horizon H truncates without a win bonus; use global bootstrap $b_t = 1 - d$, including individual death and truncation. Actor action-loss masks use pre-step participation. Explicit reward weights replace the default payoff as defined on page 1.

F. Exact observations and CTDE information

For each living entity $j \neq i$, the actor row is

$$e_{ij} = [(p_j - p_i)/L, (v_j - v_i)/(2V_{\max}), h_j, 1, \mathbf{1}_{c_j=R}, \mathbf{1}_{c_j=B}, \mathbf{1}_{j \in \mathcal{P}}],$$

$$L = \sqrt{\sum_a (b_a^+ - b_a^-)^2}.$$

Here $V_{\max}$ is the preset maximum speed, not the reference speed. Rows have 11 values, in fixed roster order: Red Scout/Disturb/Attack, Blue Scout/Disturb/Attack, then targets, excluding self. Dead rows are zero. Every living entity is included; Attack, Scout and Disturb share this packing. Side flags do not encode role. Own health is absent from the self block.

The actor receives the $(M-1) \times 11$ entity table ($V_{\max} = 30$) plus unnormalized $[p_i, v_i, q_i, \omega_i] \in \mathbb{R}^{13}$. With deterministic encoding, $Z_i(o | s) = \delta_{\text{encode}_i(s)}(o)$. Other UAV attitudes/rates, opponent memory and time are omitted; hence local information is partial despite noiseless sensing.

Privileged $g(s) = \text{state}()$ packs 11 values per entity,

$$[(p - b^-)/(b^+ - b^-), v/(2V_{\max}), h, \ell, R, B, P],$$

with dead blocks zero. Rigid models append 13 raw state values per UAV (dead zero), then $[t, \sum_k D_k, \mathbf{1}_{\text{Damage}}]$: size $11M + 13N + 3$. Particle size is $11M$, omitting time and damage. g is a CTDE feature, not the complete Markov state.

H. Defender Dec-POMDP and MARL objective

For a specified frozen opposing policy π_B , define

$$\mathcal{M}_R = (\mathcal{R}, \mathcal{S}, \{\mathcal{A}_i\}, T^{\pi_B}, \{\mathcal{O}_i\}, Z, R, \gamma, \rho_0, H),$$

$$T^{\pi_B}(s' | s, a_R) = \int \delta_{F_C(s, a_R, a_B)}(s') \times \pi_B(da_B | o_B, \zeta_t).$$

F_C is the stated collision, simultaneous interaction and controlled-motion map, including any memory evolution; ρ_0 is the reset law. Native execution accepts both teams' actions. Freezing a rule or learned opponent is an experiment choice; with both teams learning, the model is a POSG.

$$\tau_i = (o_{i,0}, a_{i,0}, \dots, o_{i,t}),$$

$$\pi_R(a_R | \tau_R) = \prod_{i \in \mathcal{R}} \pi_i(a_i | \tau_i),$$

$$\pi_R^* \in \arg \max_{\pi_R} \mathbb{E} \left[\sum_{t=0}^{T-1} \gamma^t R_t \right].$$

Expectation is over ρ_0, T^{π_B}, π_R . Here T is natural completion (possibly infinite); H limits sampled rollouts. A truncated return is partial, so its successor value is retained. For $\gamma = 1$, this is respectively the native terminal Survival payoff or negative total asset damage. For $\gamma < 1$, it is the discounted task selected by the caller. Actors use local histories only. CTDE critics can use $g(s)$, joint actions and recorded task/memory features:

$$y_t = R_t + \gamma b_t Q^-(g(s'), a'_R).$$

Here Q^- is a target critic and successor actions follow the target actor policies. Use global completion, not individual death. A recurrent critic or augmented state may be needed because g omits history. This defines an objective; it does not prescribe an implemented optimizer.

Theme 6 / Fixed wing / Acceleration / Damage

Rigid-body aerodynamic transition and navigation/actuator semantics

A. Scenario and complete state

Red defenders $\mathcal{R}$ protect stationary assets $\mathcal{P}$ against Blue $\mathcal{B}$. Set $\mathcal{U} = \mathcal{R} \cup \mathcal{B}$, $\mathcal{E} = \mathcal{U} \cup \mathcal{P}$, $N = |\mathcal{U}|$, $K = |\mathcal{P}|$, $M = N + K$. Episode-fixed roles are Attack, Scout or Disturb.

$$s_t = (\{x_i, h_i, r_i, c_i, \text{id}_i\}_{i \in \mathcal{U}}, \\ \{p_k, h_k, D_k\}_{k \in \mathcal{P}}, t, \zeta_t).$$

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. Participation is $\ell_i = \mathbf{1}[h_i > 0]$. Reset UAV health is 1, target health $h_k^0 = 2$ by default; $v_k = 0$, $D_k = 0$ initially. D_k accumulates raw asset damage.

Configuration C fixes roster, bounds, controls and task. ζ_t contains persistent assignments and opponent memory, if used. Reactive controllers have no hidden integrators. Last command, functional bits, pre-step positions and step damage are overwritten transition records; retain them for exact reward/event replay. RNG state is required for exact replay, not actor input.

B. Six-degree-of-freedom dynamics

World coordinates are east-north-up (ENU); body coordinates are forward-left-up (FLU). The scalar-first quaternion $q = (q_w, q_x, q_y, q_z)$ maps body to world through $R(q) \in SO(3)$, with $q \in S^3$, $\|q\| = 1$ and $q \sim -q$. Thus 13 stored values represent 12 physical degrees of freedom.

$$\dot{p} = v, \quad \dot{v} = R(q)F_B/m - g e_3, \\ \dot{q} = \frac{1}{2}q \otimes (0, \omega), \quad \dot{\omega} = J^{-1}(M_B - \omega \times J\omega).$$

Here $g = 9.81 \text{ m s}^{-2}$, $\omega \in \mathbb{R}^3$ is the body angular velocity in rad/s; F_B excludes gravity. No wind, sensor noise or actuator lag is present.

$$R(q) = I + 2q_w[q_v]_{\times} + 2[q_v]_{\times}^2, \quad q_v = (q_x, q_y, q_z).$$

Here $[z]_{\times} y = z \times y$.

C. Aerosonde aerodynamic wrench

Let $C_f = \text{diag}(1, -1, -1)$ and $(u, v_b, w) = C_f R^T v$, $(p_b, q_b, r_b) = C_f \omega$. Define $V = \|(u, v_b, w)\|$, $\bar{V} = \max(V, 10^{-6})$, $\alpha = \text{atan2}(w, u)$, $\beta = \arcsin \text{clip}(v_b/\bar{V}, -1, 1)$, $\bar{Q} = \rho V^2/2$, $\hat{p} = bp_b/(2\bar{V})$, $\hat{q} = cq_b/(2\bar{V})$, $\hat{r} = br_b/(2\bar{V})$.

$$E_- = e^{\text{clip}(-50(\alpha-.47), -500, 500)}, \\ E_+ = e^{\text{clip}(50(\alpha+.47), -500, 500)}, \\ \sigma = (1 + E_- + E_+)/[(1 + E_-)(1 + E_+)], \\ C_L = (1 - \sigma)(.23 + 5.61\alpha) \\ + 2\sigma \text{sign}(\alpha) \sin^2 \alpha \cos \alpha + 7.95\hat{q} + .13\delta_e, \\ C_D = .043 + \frac{(.23 + 5.61\alpha)^2}{\pi(.9)(b^2/S)} + .0135\delta_e, \\ C_Y = -.98\beta + .075\delta_a + .19\delta_r, \\ C_\ell = -.13\beta - .51\hat{p} + .25\hat{r} + .17\delta_a + .0024\delta_r, \\ C_m = .0135 - 2.74\alpha - 38.21\hat{q} - .99\delta_e, \\ C_n = .073\beta + .069\hat{p} - .095\hat{r} - .011\delta_a - .069\delta_r.$$

$$F_B = C_f \left\{ \bar{Q} S \begin{bmatrix} -C_D \cos \alpha + C_L \sin \alpha \\ C_Y \\ -C_D \sin \alpha - C_L \cos \alpha \end{bmatrix} + \begin{bmatrix} T_p \\ 0 \\ 0 \end{bmatrix} \right\}, \\ M_B = C_f \left\{ \bar{Q} S \begin{bmatrix} bC_\ell \\ cC_m \\ bC_n \end{bmatrix} - \begin{bmatrix} Q_p \\ 0 \\ 0 \end{bmatrix} \right\}.$$

Motor, propeller and equilibrium

The physical constants are $m = 11 \text{ kg}$, $S = .55 \text{ m}^2$, $b = 2.8956 \text{ m}$, $c = .18994 \text{ m}$, $\rho = 1.2682 \text{ kg/m}^3$,

$$J = C_f \begin{bmatrix} .8244 & 0 & -.1204 \\ 0 & 1.135 & 0 \\ -.1204 & 0 & 1.759 \end{bmatrix} C_f^T.$$

With $D_p = .508$, $R_m = .042$, $I_0 = 1.5$, $V_m = 44.4$, $k_m = 60/(145 \cdot 2\pi)$, the steady motor root is

$$A = \rho D_p^5 C_{Q0}/(4\pi^2), \\ B = \rho D_p^4 C_{Q1} V/(2\pi) + k_m^2/R_m, \\ C = \rho D_p^3 C_{Q2} V^2 - k_m V_m \tau/R_m + k_m I_0, \\ \Omega_p = \max\{0, (-B + \sqrt{\max(0, B^2 - 4AC)})/(2A)\}.$$

Set $z = (D_p^4 \Omega_p^2/(4\pi^2), D_p^3 V \Omega_p/(2\pi), D_p^2 V^2)$,

$$T_p = \rho(.09357, -.06044, -.1079)z, \\ Q_p = \rho D_p(.005230, .004970, -.01664)z.$$

Reference trim $(x^*(V_d), u^*(V_d))$ solves all six equations $\dot{v} = 0$, $\dot{\omega} = 0$, including propeller reaction torque. Unknowns are pitch, heading, elevator, aileron, rudder and throttle, at velocity $(V_d, 0, 0)$. The reset uses the 30 m/s level trim rotated to the initial heading. Mass/inertia are not scaled with the scene [1].

Action and reference conversion

Choose $a_i \in [-1, 1]^3$ or one of the 27 unit primitives. Scale once to $\alpha_i = 5a_i$, then hold $v_d = v_i(t) + \Delta\alpha_i$. At each RK4 stage,

$$\chi_d = \text{atan2}(v_{dy}, v_{dx}), \\ \gamma_d = \text{atan2}(v_{dz}, \max(\|v_{d,xy}\|, 10^{-6})), \\ V_d = \|v_d\|.$$

For tiny horizontal reference speed, χ_d is current heading. Component-limited continuous intent need not have norm ≤ 5 .

D. Implemented reference controller

Define bank $\phi = \text{atan2}(R_{32}, R_{33})$, nose-up pitch $\theta = \arcsin R_{31}$, heading $\psi = \text{atan2}(R_{21}, R_{11})$. Actual course is $\chi = \text{atan2}(v_y, v_x)$, or ψ when horizontal speed is tiny; path $\gamma_f = \text{atan2}(v_z, \|v_{xy}\|)$ and $V_a = \|v\|$.

$$V_d = \text{clip}(V_d, 18, 30), \\ \gamma_d = \text{clip}(\gamma_d, -15^\circ, 15^\circ), \\ e_\chi = \text{wrap}(\chi_d - \chi), \\ \bar{\phi} = -\text{atan2}(\max(V_a, 18)e_\chi, g), \\ \phi_d = \text{clip}(\bar{\phi}, -45^\circ, 45^\circ), \\ \bar{\theta} = \theta^* + \gamma_d + .5(\gamma_d - \gamma_f), \\ \theta_d = \text{clip}(\bar{\theta}, -20^\circ, 20^\circ), \\ \delta_e = \delta_e^* - 2(\theta_d - \theta) + .25q_b, \\ \delta_a = \delta_a^* + .5(\phi_d - \phi) - .08p_b, \\ r_d = g \tan \phi_d / \max(V_a, 18), \\ \delta_r = \delta_r^* + .8(\beta - \psi^*) + .1(r_b - r_d), \\ \tau = \tau^* + .08(V_d - V_a) + .7 \sin \gamma_d.$$

All angles use radians. The scheduled trim heading ψ^* is the lateral reference. Divide surfaces by 25° , map throttle to $2\tau - 1$ and clip the four outputs to $[-1, 1]$. These project gains/bounds do not clamp the physical flight envelope.

Integration, geometry and initialization

RK4 uses $h = .01 \text{ s}$, 100 synchronized substeps per 1 s step. Each stage feeds back the held reference: $k_j = f(x_j, U(x_j))$ and $x^+ = x + h(k_1 + 2k_2 + 2k_3 + k_4)/6$, then normalize q^+ . Default $\lambda = .4$, $d_c = 8 \text{ m}$, $(r_0, r_1) = (120, 240) \text{ m}$, $\mathcal{W} = [-1000, 1000]^2 \times [0, 1000] \text{ m}$. Clip path positions and the endpoint to scene bounds; remove endpoint outward velocity. Reset geometry follows the area-uniform Red annulus / Blue rectangular region, scaled by .4, with separated UAV openings and solved 30 m/s level trim. Only 3D is supported.

Theme 6 / Fixed wing / Acceleration / Damage

Task interaction, exact information structure and defender learning objective

E. Collision and automatic interaction

Predict living UAV paths: particles use $p_i + \tau \Delta v_i$, rigid models use synchronized RK4 endpoints clipped to $\mathcal{W}$. On each matched segment, let $r = p_i - p_j$, $d = (q_i - p_i) - (q_j - p_j)$:

$$\tau_* = \begin{cases} 0 & \|d\|^2 \leq 10^{-12}, \\ \text{clip}(-r^T d / \|d\|^2, 0, 1) & \text{otherwise,} \end{cases} \quad d_* = \|r + \tau_* d\|.$$

Collision sets both UAV HP to zero when $\min d_* \leq d_c$; targets are excluded. Pairs use i -major, $j < i$ order. Dead outer agents are skipped at entry and dead inner agents are skipped. An outer agent killed during its inner sweep continues that sweep. After collision, surviving Attack agents automatically fire at step-start positions:

$$I_i = \begin{cases} \mathbf{1}[\min_{k \in \mathcal{P}: h_k > 0} \|p_i - p_k\| < r_f], & i \in \mathcal{B}, \\ \mathbf{1}[\min_{j \in \mathcal{B}_A: h_j > 0} \|p_i - p_j\| < r_f], & i \in \mathcal{R}. \end{cases}$$

An empty minimum is $+\infty$; dead sources have $I_i = 0$. A denotes Attack. Fire is a separate rule, not a navigation action. With unit attack intensity,

$$\eta(d) = \begin{cases} 1 & d < r_0, \\ (r_1 - d)/(r_1 - r_0) & r_0 \leq d < r_1, \\ 0 & d \geq r_1, \end{cases}$$

$$L_k = \sum_{b \in \mathcal{B}_A: h_b > 0} I_b \eta(\|p_b - p_k\|),$$

$$L_i^{\text{atk}} = \sum_{j \in \mathcal{U}_A: c_j \neq c_i, h_j > 0} I_j \eta(\|p_j - p_i\|).$$

One source can affect multiple enemies; friendly explosions are excluded. All losses use a common post-collision, pre-motion snapshot and add before HP clipping.

Disturb and Scout

For source j , $d_{ji} = \|p_i - p_j\|$ and $\theta_{ji} = \angle(v_j, p_i - p_j)$; angle is zero if either norm is below 10^{-3} . Let

$$G_{ji} = \mathbf{1}[10^{-3} < d_{ji} < 600\lambda, \theta_{ji} < \pi/6] \\ \times e^{-(d_{ji}/\lambda)^2/2} / \sqrt{2\pi},$$

$$S_c(j) = \sum_{i \in \mathcal{U}: c_i = c, h_i > 0} G_{ji}.$$

Living Red Disturb uses $J_j = \mathbf{1}[S_R(j) < S_B(j)]$, Blue uses $J_j = \mathbf{1}[S_R(j) \geq S_B(j)]$. Dead sources contribute zero.

$$L_i^{\text{dis}} = .1 \sum_{j \in \mathcal{U}_D: h_j > 0} J_j G_{ji},$$

$$\tilde{h}_i = \max(0, h_i - L_i^{\text{atk}} - L_i^{\text{dis}}).$$

Set $h'_i = 0$ if $r_i = A, I_i = 1$, otherwise $h'_i = \tilde{h}_i$. Disturb damages either side, not assets. Scouts use $d < 2000\lambda$, $\theta < \pi/3$; this sector does not restrict native actor rows. Collision victims do not move. Combat victims complete their motion; inactive velocity is zeroed next step.

G. Damage payoff and completion

Assets keep finite reset health $h'_k = h_k^0$. Count raw contributions before HP clipping, including simultaneous overkill:

$$D'_k = D_k + L_k, \quad d_t = \sum_k L_k, \quad R_t = -d_t.$$

Blue reward is d_t ; there is no team-size, asset-count or HP normalization and no terminal Survival bonus.

$$d = \mathbf{1}[\forall b \in \mathcal{B}_A : h'_b \leq 0].$$

All agent slots remain until global completion/truncation. Dead slots have zero observations, ignored actions and still receive full team R_t , including later damage after Red extinction. Do not sum duplicate teammate rewards. Undiscounted returns are $(-\sum_k D_k, +\sum_k D_k)$. H only truncates; global bootstrap $b_t = 1 - d$ remains one at truncation. Actor action-loss masks use pre-step participation.

F. Exact observations and CTDE information

For each living entity $j \neq i$, the actor row is

$$e_{ij} = [(p_j - p_i)/L, (v_j - v_i)/(2V_{\max}), \\ h_j, 1, \mathbf{1}_{c_j=R}, \mathbf{1}_{c_j=B}, \mathbf{1}_{j \in \mathcal{P}}],$$

$$L = \sqrt{\sum_a (b_a^+ - b_a^-)^2}.$$

Here $V_{\max}$ is the preset maximum speed, not the reference speed. Rows have 11 values, in fixed roster order: Red Scout/Disturb/Attack, Blue Scout/Disturb/Attack, then targets, excluding self. Dead rows are zero. Every living entity is included; Attack, Scout and Disturb share this packing. Side flags do not encode role. Own health is absent from the self block.

The actor receives the $(M - 1) \times 11$ entity table ($V_{\max} = 30$) plus unnormalized $[p_i, v_i, q_i, \omega_i]$. Both blocks are zero for an already-dead Damage observer. With deterministic encoding, $Z_i(o | s) = \delta_{\text{encode}_i(s)}(o)$. Other UAV attitudes/rates, opponent memory and time are omitted; hence local information is partial despite noiseless sensing.

Privileged $g(s) = \text{state}()$ packs 11 values per entity,

$$[(p - b^-)/(b^+ - b^-), v/(2V_{\max}), h, \ell, R, B, P],$$

with dead blocks zero. Rigid models append 13 raw state values per UAV (dead zero), then $[t, \sum_k D_k, \mathbf{1}_{\text{Damage}}]$: size $11M + 13N + 3$. Particle size is $11M$, omitting time and damage. g is a CTDE feature, not the complete Markov state.

H. Defender Dec-POMDP and MARL objective

For a specified frozen opposing policy π_B , define

$$\mathcal{M}_R = (\mathcal{R}, \mathcal{S}, \{\mathcal{A}_i\}, T^{\pi_B}, \\ \{\mathcal{O}_i\}, Z, R, \gamma, \rho_0, H),$$

$$T^{\pi_B}(s' | s, a_R) = \int \delta_{F_C(s, a_R, a_B)}(s') \\ \times \pi_B(da_B | o_B, \zeta_t).$$

F_C is the stated collision, simultaneous interaction and controlled-motion map, including any memory evolution; ρ_0 is the reset law. Native execution accepts both teams' actions. Freezing a rule or learned opponent is an experiment choice; with both teams learning, the model is a POSG.

$$\tau_i = (o_{i,0}, a_{i,0}, \dots, o_{i,t}),$$

$$\pi_R(a_R | \tau_R) = \prod_{i \in \mathcal{R}} \pi_i(a_i | \tau_i),$$

$$\pi_R^* \in \arg \max_{\pi_R} \mathbb{E} \left[\sum_{t=0}^{T-1} \gamma^t R_t \right].$$

Expectation is over ρ_0, T^{π_B}, π_R . Here T is natural completion (possibly infinite); H limits sampled rollouts. A truncated return is partial, so its successor value is retained. For $\gamma = 1$, this is respectively the native terminal Survival payoff or negative total asset damage. For $\gamma < 1$, it is the discounted task selected by the caller. Actors use local histories only. CTDE critics can use $g(s)$, joint actions and recorded task/memory features:

$$y_t = R_t + \gamma b_t Q^-(g(s'), a'_R).$$

Here Q^- is a target critic and successor actions follow the target actor policies. Use global completion, not individual death. A recurrent critic or augmented state may be needed because g omits history. This defines an objective; it does not prescribe an implemented optimizer.

Theme 7 / Fixed wing / Actuator / Survival

Rigid-body aerodynamic transition and navigation/actuator semantics

A. Scenario and complete state

Red defenders $\mathcal{R}$ protect stationary assets $\mathcal{P}$ against Blue $\mathcal{B}$. Set $\mathcal{U} = \mathcal{R} \cup \mathcal{B}$, $\mathcal{E} = \mathcal{U} \cup \mathcal{P}$, $N = |\mathcal{U}|$, $K = |\mathcal{P}|$, $M = N + K$. Episode-fixed roles are Attack, Scout or Disturb.

$$s_t = (\{x_i, h_i, r_i, c_i, \text{id}_i\}_{i \in \mathcal{U}}, \{p_k, h_k, D_k\}_{k \in \mathcal{P}}, t, \zeta_t).$$

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. Participation is $\ell_i = \mathbf{1}[h_i > 0]$. Reset UAV health is 1, target health $h_k^0 = 2$ by default; $v_k = 0$, $D_k = 0$ initially. D_k accumulates raw asset damage.

Configuration C fixes roster, bounds, controls and task. ζ_t contains persistent assignments and opponent memory, if used. Reactive controllers have no hidden integrators. Last command, functional bits, pre-step positions and step damage are overwritten transition records; retain them for exact reward/event replay. RNG state is required for exact replay, not actor input.

B. Six-degree-of-freedom dynamics

World coordinates are east-north-up (ENU); body coordinates are forward-left-up (FLU). The scalar-first quaternion $q = (q_w, q_x, q_y, q_z)$ maps body to world through $R(q) \in SO(3)$, with $q \in S^3$, $\|q\| = 1$ and $q \sim -q$. Thus 13 stored values represent 12 physical degrees of freedom.

$$\begin{aligned} \dot{p} &= v, & \dot{v} &= R(q)F_B/m - ge_3, \\ \dot{q} &= \frac{1}{2}q \otimes (0, \omega), & \dot{\omega} &= J^{-1}(M_B - \omega \times J\omega). \end{aligned}$$

Here $g = 9.81 \text{ m s}^{-2}$, $\omega \in \mathbb{R}^3$ is the body angular velocity in rad/s; F_B excludes gravity. No wind, sensor noise or actuator lag is present.

$$R(q) = I + 2q_w[q_v]_{\times} + 2[q_v]_{\times}^2, \quad q_v = (q_x, q_y, q_z).$$

Here $[z]_{\times}y = z \times y$.

C. Aerosonde aerodynamic wrench

Let $C_f = \text{diag}(1, -1, -1)$ and $(u, v_b, w) = C_f R^T v$, $(p_b, q_b, r_b) = C_f \omega$. Define $V = \|(u, v_b, w)\|$, $\bar{V} = \max(V, 10^{-6})$, $\alpha = \text{atan2}(w, u)$, $\beta = \arcsin \text{clip}(v_b/\bar{V}, -1, 1)$, $\bar{Q} = \rho V^2/2$, $\hat{p} = bp_b/(2\bar{V})$, $\hat{q} = cq_b/(2\bar{V})$, $\hat{r} = br_b/(2\bar{V})$.

$$\begin{aligned} E_- &= e^{\text{clip}(-50(\alpha-.47), -500, 500)}, \\ E_+ &= e^{\text{clip}(50(\alpha+.47), -500, 500)}, \\ \sigma &= (1 + E_- + E_+)/[(1 + E_-)(1 + E_+)], \\ C_L &= (1 - \sigma)(.23 + 5.61\alpha) \\ &\quad + 2\sigma \text{sign}(\alpha) \sin^2 \alpha \cos \alpha + 7.95\hat{q} + .13\delta_e, \\ C_D &= .043 + \frac{(.23 + 5.61\alpha)^2}{\pi(.9)(b^2/S)} + .0135\delta_e, \\ C_Y &= -.98\beta + .075\delta_a + .19\delta_r, \\ C_\ell &= -.13\beta - .51\hat{p} + .25\hat{r} + .17\delta_a + .0024\delta_r, \\ C_m &= .0135 - 2.74\alpha - 38.21\hat{q} - .99\delta_e, \\ C_n &= .073\beta + .069\hat{p} - .095\hat{r} - .011\delta_a - .069\delta_r. \end{aligned}$$

$$\begin{aligned} F_B &= C_f \left\{ \bar{Q} S \begin{bmatrix} -C_D \cos \alpha + C_L \sin \alpha \\ C_Y \\ -C_D \sin \alpha - C_L \cos \alpha \end{bmatrix} + \begin{bmatrix} T_p \\ 0 \\ 0 \end{bmatrix} \right\}, \\ M_B &= C_f \left\{ \bar{Q} S \begin{bmatrix} bC_\ell \\ cC_m \\ bC_n \end{bmatrix} - \begin{bmatrix} Q_p \\ 0 \\ 0 \end{bmatrix} \right\}. \end{aligned}$$

Motor, propeller and equilibrium

The physical constants are $m = 11 \text{ kg}$, $S = .55 \text{ m}^2$, $b = 2.8956 \text{ m}$, $c = .18994 \text{ m}$, $\rho = 1.2682 \text{ kg/m}^3$,

$$J = C_f \begin{bmatrix} .8244 & 0 & -.1204 \\ 0 & 1.135 & 0 \\ -.1204 & 0 & 1.759 \end{bmatrix} C_f^T.$$

With $D_p = .508$, $R_m = .042$, $I_0 = 1.5$, $V_m = 44.4$, $k_m = 60/(145 \cdot 2\pi)$, the steady motor root is

$$\begin{aligned} A &= \rho D_p^5 C_{Q0}/(4\pi^2), \\ B &= \rho D_p^4 C_{Q1} V/(2\pi) + k_m^2/R_m, \\ C &= \rho D_p^3 C_{Q2} V^2 - k_m V_m \tau/R_m + k_m I_0, \\ \Omega_p &= \max\{0, (-B + \sqrt{\max(0, B^2 - 4AC)})/(2A)\}. \end{aligned}$$

Set $z = (D_p^4 \Omega_p^2/(4\pi^2), D_p^3 V \Omega_p/(2\pi), D_p^2 V^2)$,

$$\begin{aligned} T_p &= \rho(.09357, -.06044, -.1079)z, \\ Q_p &= \rho D_p(.005230, .004970, -.01664)z. \end{aligned}$$

Reference trim $(x^*(V_d), u^*(V_d))$ solves all six equations $\dot{v} = 0$, $\dot{\omega} = 0$, including propeller reaction torque. Unknowns are pitch, heading, elevator, aileron, rudder and throttle, at velocity $(V_d, 0, 0)$. The reset uses the 30 m/s level trim rotated to the initial heading. Mass/inertia are not scaled with the scene [1].

D. Direct actuator action

Every living agent selects $a_i \in [-1, 1]^4$, ordered elevator, aileron, rudder, throttle:

$$(\delta_e, \delta_a, \delta_r) = 25^\circ(a_1, a_2, a_3), \quad \tau = (a_4 + 1)/2.$$

The actuator vector is constant for the external step and feeds the aerodynamic/motor equations directly. There is no automatic stabilization or navigation controller. In particular $a = 0$ is neutral surfaces with half throttle, not solved trim. Automatic fire still uses the same separate task rules.

Numerical transition and boundary

Write $f(x, u)$ for the above derivative, evaluated with normalized q . An external step is $\Delta = 1 \text{ s}$, $n = \lceil \Delta/.01 \rceil = 100$, $h = \Delta/n$. With stage feedback U ,

$$\begin{aligned} k_1 &= f(x, U(x)), \\ k_2 &= f(x + hk_1/2, U(x + hk_1/2)), \\ k_3 &= f(x + hk_2/2, U(x + hk_2/2)), \\ k_4 &= f(x + hk_3, U(x + hk_3)), \\ x^+ &= x + h(k_1 + 2k_2 + 2k_3 + k_4)/6, \\ q^+ &\leftarrow q^+/\|q^+\|. \end{aligned}$$

Repeat n times. A raw actuator is held constant; an intent is held but feedback is recomputed at every RK4 stage. Controllers have no hidden integrators.

For a candidate endpoint p^* , set $p'_a = \text{clip}(p_a^*, b_a^-, b_a^+)$. If $p_a^* > b_a^+$, $v'_a = \min(v_a^*, 0)$; if $p_a^* < b_a^-$, $v'_a = \max(v_a^*, 0)$; otherwise $v'_a = v_a^*$. Collision paths use clipped substep positions. Boundaries are task constraints; physical airspeed and attitude are not clamped to controller reference limits.

Geometry and initialization

Default $\lambda = .4$, $\mathcal{W} = [-1000, 1000]^2 \times [0, 1000] \text{ m}$, $d_c = 8 \text{ m}$, $(r_0, r_1) = (120, 240) \text{ m}$. Targets use the scaled region $[-920, -760] \times [-480, 480] \times [200, 600] \text{ m}$. Red uses an area-uniform XY annulus $[320, 880] \text{ m}$ around a sampled target; Blue uses $x \in [0, 1000]$ and full scene y ; agent altitude lies in $[80, 600] \text{ m}$. UAV openings reject distances $\leq \max(30\lambda, d_c + 10^{-6})$. Both teams reset at solved 30 m/s level flight; Blue heads west, Red east under random initialization. Only 3D is supported.

Theme 7 / Fixed wing / Actuator / Survival

Task interaction, exact information structure and defender learning objective

E. Collision and automatic interaction

Predict living UAV paths: particles use $p_i + \tau \Delta v_i$, rigid models use synchronized RK4 endpoints clipped to $\mathcal{W}$. On each matched segment, let $r = p_i - p_j$, $d = (q_i - p_i) - (q_j - p_j)$:

$$\tau_* = \begin{cases} 0 & \|d\|^2 \leq 10^{-12}, \\ \text{clip}(-r^T d / \|d\|^2, 0, 1) & \text{otherwise,} \end{cases} \quad d_* = \|r + \tau_* d\|.$$

Collision sets both UAV HP to zero when $\min d_* \leq d_c$; targets are excluded. Pairs use i -major, $j < i$ order. Dead outer agents are skipped at entry and dead inner agents are skipped. An outer agent killed during its inner sweep continues that sweep. After collision, surviving Attack agents automatically fire at step-start positions:

$$I_i = \begin{cases} \mathbf{1}[\min_{k \in \mathcal{P}: h_k > 0} \|p_i - p_k\| < r_f], & i \in \mathcal{B}, \\ \mathbf{1}[\min_{j \in \mathcal{B}_A: h_j > 0} \|p_i - p_j\| < r_f], & i \in \mathcal{R}. \end{cases}$$

An empty minimum is $+\infty$; dead sources have $I_i = 0$. A denotes Attack. Fire is a separate rule, not a navigation action. With unit attack intensity,

$$\eta(d) = \begin{cases} 1 & d < r_0, \\ (r_1 - d)/(r_1 - r_0) & r_0 \leq d < r_1, \\ 0 & d \geq r_1, \end{cases}$$

$$L_k = \sum_{b \in \mathcal{B}_A: h_b > 0} I_b \eta(\|p_b - p_k\|),$$

$$L_i^{\text{atk}} = \sum_{j \in \mathcal{U}_A: c_j \neq c_i, h_j > 0} I_j \eta(\|p_j - p_i\|).$$

One source can affect multiple enemies; friendly explosions are excluded. All losses use a common post-collision, pre-motion snapshot and add before HP clipping.

Disturb and Scout

For source j , $d_{ji} = \|p_i - p_j\|$ and $\theta_{ji} = \angle(v_j, p_i - p_j)$; angle is zero if either norm is below 10^{-3} . Let

$$G_{ji} = \mathbf{1}[10^{-3} < d_{ji} < 600\lambda, \theta_{ji} < \pi/6] \\ \times e^{-(d_{ji}/\lambda)^2/2} / \sqrt{2\pi},$$

$$S_c(j) = \sum_{i \in \mathcal{U}: c_i = c, h_i > 0} G_{ji}.$$

Living Red Disturb uses $J_j = \mathbf{1}[S_R(j) < S_B(j)]$, Blue uses $J_j = \mathbf{1}[S_R(j) \geq S_B(j)]$. Dead sources contribute zero.

$$L_i^{\text{dis}} = .1 \sum_{j \in \mathcal{U}_D: h_j > 0} J_j G_{ji},$$

$$\tilde{h}_i = \max(0, h_i - L_i^{\text{atk}} - L_i^{\text{dis}}).$$

Set $h'_i = 0$ if $r_i = A, I_i = 1$, otherwise $h'_i = \tilde{h}_i$. Disturb damages either side, not assets. Scouts use $d < 2000\lambda$, $\theta < \pi/3$; this sector does not restrict native actor rows. Collision victims do not move. Combat victims complete their motion; inactive velocity is zeroed next step.

G. Survival payoff and completion

Assets obey $h'_k = \max(0, h_k - L_k)$, $D'_k = D_k + L_k$.

$$z(s') = \begin{cases} -1 & \exists k : h'_k < 10^{-3}, \\ +1 & \forall b \in \mathcal{B}_A : h'_b < 10^{-3}, \\ 0 & \text{otherwise.} \end{cases}$$

Target loss has priority. Natural completion is $d = \mathbf{1}[z \neq 0]$; default Red reward $R_t = 10z$, Blue $-R_t$. Red elimination alone does not end the world. Individual $h'_i \leq 0$ retires a Survival actor after its final transition. Horizon H truncates without a win bonus; use global bootstrap $b_t = 1 - d$, including individual death and truncation. Actor action-loss masks use pre-step participation. Explicit reward weights replace the default payoff as defined on page 1.

F. Exact observations and CTDE information

For each living entity $j \neq i$, the actor row is

$$e_{ij} = [(p_j - p_i)/L, (v_j - v_i)/(2V_{\max}), \\ h_j, 1, \mathbf{1}_{c_j=R}, \mathbf{1}_{c_j=B}, \mathbf{1}_{j \in \mathcal{P}}],$$

$$L = \sqrt{\sum_a (b_a^+ - b_a^-)^2}.$$

Here $V_{\max}$ is the preset maximum speed, not the reference speed. Rows have 11 values, in fixed roster order: Red Scout/Disturb/Attack, Blue Scout/Disturb/Attack, then targets, excluding self. Dead rows are zero. Every living entity is included; Attack, Scout and Disturb share this packing. Side flags do not encode role. Own health is absent from the self block.

The actor receives the $(M-1) \times 11$ entity table ($V_{\max} = 30$) plus unnormalized $[p_i, v_i, q_i, \omega_i] \in \mathbb{R}^{13}$. Actuator controls do not add an observation block. With deterministic encoding, $Z_i(o | s) = \delta_{\text{encode}_i(s)}(o)$. Other UAV attitudes/rates, opponent memory and time are omitted; hence local information is partial despite noiseless sensing.

Privileged $g(s) = \text{state}()$ packs 11 values per entity,

$$[(p - b^-)/(b^+ - b^-), v/(2V_{\max}), h, \ell, R, B, P],$$

with dead blocks zero. Rigid models append 13 raw state values per UAV (dead zero), then $[t, \sum_k D_k, \mathbf{1}_{\text{Damage}}]$: size $11M + 13N + 3$. Particle size is $11M$, omitting time and damage. g is a CTDE feature, not the complete Markov state.

H. Defender Dec-POMDP and MARL objective

For a specified frozen opposing policy π_B , define

$$\mathcal{M}_R = (\mathcal{R}, \mathcal{S}, \{\mathcal{A}_i\}, T^{\pi_B}, \\ \{\mathcal{O}_i\}, Z, R, \gamma, \rho_0, H),$$

$$T^{\pi_B}(s' | s, a_R) = \int \delta_{F_C(s, a_R, a_B)}(s') \\ \times \pi_B(da_B | o_B, \zeta_t).$$

F_C is the stated collision, simultaneous interaction and controlled-motion map, including any memory evolution; ρ_0 is the reset law. Native execution accepts both teams' actions. Freezing a rule or learned opponent is an experiment choice; with both teams learning, the model is a POSG.

$$\tau_i = (o_{i,0}, a_{i,0}, \dots, o_{i,t}),$$

$$\pi_R(a_R | \tau_R) = \prod_{i \in \mathcal{R}} \pi_i(a_i | \tau_i),$$

$$\pi_R^* \in \arg \max_{\pi_R} \mathbb{E} \left[\sum_{t=0}^{T-1} \gamma^t R_t \right].$$

Expectation is over ρ_0, T^{π_B}, π_R . Here T is natural completion (possibly infinite); H limits sampled rollouts. A truncated return is partial, so its successor value is retained. For $\gamma = 1$, this is respectively the native terminal Survival payoff or negative total asset damage. For $\gamma < 1$, it is the discounted task selected by the caller. Actors use local histories only. CTDE critics can use $g(s)$, joint actions and recorded task/memory features:

$$y_t = R_t + \gamma b_t Q^-(g(s'), a'_R).$$

Here Q^- is a target critic and successor actions follow the target actor policies. Use global completion, not individual death. A recurrent critic or augmented state may be needed because g omits history. This defines an objective; it does not prescribe an implemented optimizer.

Theme 8 / Fixed wing / Actuator / Damage

Rigid-body aerodynamic transition and navigation/actuator semantics

A. Scenario and complete state

Red defenders $\mathcal{R}$ protect stationary assets $\mathcal{P}$ against Blue $\mathcal{B}$. Set $\mathcal{U} = \mathcal{R} \cup \mathcal{B}$, $\mathcal{E} = \mathcal{U} \cup \mathcal{P}$, $N = |\mathcal{U}|$, $K = |\mathcal{P}|$, $M = N + K$. Episode-fixed roles are Attack, Scout or Disturb.

$$s_t = (\{x_i, h_i, r_i, c_i, \text{id}_i\}_{i \in \mathcal{U}}, \{p_k, h_k, D_k\}_{k \in \mathcal{P}}, t, \zeta_t).$$

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. Participation is $\ell_i = \mathbf{1}[h_i > 0]$. Reset UAV health is 1, target health $h_k^0 = 2$ by default; $v_k = 0$, $D_k = 0$ initially. D_k accumulates raw asset damage.

Configuration C fixes roster, bounds, controls and task. ζ_t contains persistent assignments and opponent memory, if used. Reactive controllers have no hidden integrators. Last command, functional bits, pre-step positions and step damage are overwritten transition records; retain them for exact reward/event replay. RNG state is required for exact replay, not actor input.

B. Six-degree-of-freedom dynamics

World coordinates are east-north-up (ENU); body coordinates are forward-left-up (FLU). The scalar-first quaternion $q = (q_w, q_x, q_y, q_z)$ maps body to world through $R(q) \in SO(3)$, with $q \in S^3$, $\|q\| = 1$ and $q \sim -q$. Thus 13 stored values represent 12 physical degrees of freedom.

$$\begin{aligned} \dot{p} &= v, & \dot{v} &= R(q)F_B/m - ge_3, \\ \dot{q} &= \frac{1}{2}q \otimes (0, \omega), & \dot{\omega} &= J^{-1}(M_B - \omega \times J\omega). \end{aligned}$$

Here $g = 9.81 \text{ m s}^{-2}$, $\omega \in \mathbb{R}^3$ is the body angular velocity in rad/s; F_B excludes gravity. No wind, sensor noise or actuator lag is present.

$$R(q) = I + 2q_w[q_v]_{\times} + 2[q_v]_{\times}^2, \quad q_v = (q_x, q_y, q_z).$$

Here $[z]_{\times}y = z \times y$.

C. Aerosonde aerodynamic wrench

Let $C_f = \text{diag}(1, -1, -1)$ and $(u, v_b, w) = C_f R^T v$, $(p_b, q_b, r_b) = C_f \omega$. Define $V = \|(u, v_b, w)\|$, $\bar{V} = \max(V, 10^{-6})$, $\alpha = \text{atan2}(w, u)$, $\beta = \arcsin \text{clip}(v_b/\bar{V}, -1, 1)$, $\bar{Q} = \rho V^2/2$, $\hat{p} = bp_b/(2\bar{V})$, $\hat{q} = cq_b/(2\bar{V})$, $\hat{r} = br_b/(2\bar{V})$.

$$\begin{aligned} E_- &= e^{\text{clip}(-50(\alpha-.47), -500, 500)}, \\ E_+ &= e^{\text{clip}(50(\alpha+.47), -500, 500)}, \\ \sigma &= (1 + E_- + E_+)/[(1 + E_-)(1 + E_+)], \\ C_L &= (1 - \sigma)(.23 + 5.61\alpha) \\ &\quad + 2\sigma \text{sign}(\alpha) \sin^2 \alpha \cos \alpha + 7.95\hat{q} + .13\delta_e, \\ C_D &= .043 + \frac{(.23 + 5.61\alpha)^2}{\pi(.9)(b^2/S)} + .0135\delta_e, \\ C_Y &= -.98\beta + .075\delta_a + .19\delta_r, \\ C_\ell &= -.13\beta - .51\hat{p} + .25\hat{r} + .17\delta_a + .0024\delta_r, \\ C_m &= .0135 - 2.74\alpha - 38.21\hat{q} - .99\delta_e, \\ C_n &= .073\beta + .069\hat{p} - .095\hat{r} - .011\delta_a - .069\delta_r. \end{aligned}$$

$$\begin{aligned} F_B &= C_f \left\{ \bar{Q} S \begin{bmatrix} -C_D \cos \alpha + C_L \sin \alpha \\ C_Y \\ -C_D \sin \alpha - C_L \cos \alpha \end{bmatrix} + \begin{bmatrix} T_p \\ 0 \\ 0 \end{bmatrix} \right\}, \\ M_B &= C_f \left\{ \bar{Q} S \begin{bmatrix} bC_\ell \\ cC_m \\ bC_n \end{bmatrix} - \begin{bmatrix} Q_p \\ 0 \\ 0 \end{bmatrix} \right\}. \end{aligned}$$

Motor, propeller and equilibrium

The physical constants are $m = 11 \text{ kg}$, $S = .55 \text{ m}^2$, $b = 2.8956 \text{ m}$, $c = .18994 \text{ m}$, $\rho = 1.2682 \text{ kg/m}^3$,

$$J = C_f \begin{bmatrix} .8244 & 0 & -.1204 \\ 0 & 1.135 & 0 \\ -.1204 & 0 & 1.759 \end{bmatrix} C_f^T.$$

With $D_p = .508$, $R_m = .042$, $I_0 = 1.5$, $V_m = 44.4$, $k_m = 60/(145 \cdot 2\pi)$, the steady motor root is

$$\begin{aligned} A &= \rho D_p^5 C_{Q0}/(4\pi^2), \\ B &= \rho D_p^4 C_{Q1} V/(2\pi) + k_m^2/R_m, \\ C &= \rho D_p^3 C_{Q2} V^2 - k_m V_m \tau/R_m + k_m I_0, \\ \Omega_p &= \max\{0, (-B + \sqrt{\max(0, B^2 - 4AC)})/(2A)\}. \end{aligned}$$

Set $z = (D_p^4 \Omega_p^2/(4\pi^2), D_p^3 V \Omega_p/(2\pi), D_p^2 V^2)$,

$$\begin{aligned} T_p &= \rho(.09357, -.06044, -.1079)z, \\ Q_p &= \rho D_p(.005230, .004970, -.01664)z. \end{aligned}$$

Reference trim $(x^*(V_d), u^*(V_d))$ solves all six equations $\dot{v} = 0$, $\dot{\omega} = 0$, including propeller reaction torque. Unknowns are pitch, heading, elevator, aileron, rudder and throttle, at velocity $(V_d, 0, 0)$. The reset uses the 30 m/s level trim rotated to the initial heading. Mass/inertia are not scaled with the scene [1].

D. Direct actuator action

Every living agent selects $a_i \in [-1, 1]^4$, ordered elevator, aileron, rudder, throttle:

$$(\delta_e, \delta_a, \delta_r) = 25^\circ(a_1, a_2, a_3), \quad \tau = (a_4 + 1)/2.$$

The actuator vector is constant for the external step and feeds the aerodynamic/motor equations directly. There is no automatic stabilization or navigation controller. In particular $a = 0$ is neutral surfaces with half throttle, not solved trim. Automatic fire still uses the same separate task rules.

Numerical transition and boundary

Write $f(x, u)$ for the above derivative, evaluated with normalized q . An external step is $\Delta = 1 \text{ s}$, $n = \lceil \Delta/.01 \rceil = 100$, $h = \Delta/n$. With stage feedback U ,

$$\begin{aligned} k_1 &= f(x, U(x)), \\ k_2 &= f(x + hk_1/2, U(x + hk_1/2)), \\ k_3 &= f(x + hk_2/2, U(x + hk_2/2)), \\ k_4 &= f(x + hk_3, U(x + hk_3)), \\ x^+ &= x + h(k_1 + 2k_2 + 2k_3 + k_4)/6, \\ q^+ &\leftarrow q^+/\|q^+\|. \end{aligned}$$

Repeat n times. A raw actuator is held constant; an intent is held but feedback is recomputed at every RK4 stage. Controllers have no hidden integrators.

For a candidate endpoint p^* , set $p'_a = \text{clip}(p_a^*, b_a^-, b_a^+)$. If $p_a^* > b_a^+$, $v'_a = \min(v_a^*, 0)$; if $p_a^* < b_a^-$, $v'_a = \max(v_a^*, 0)$; otherwise $v'_a = v_a^*$. Collision paths use clipped substep positions. Boundaries are task constraints; physical airspeed and attitude are not clamped to controller reference limits.

Geometry and initialization

Default $\lambda = .4$, $\mathcal{W} = [-1000, 1000]^2 \times [0, 1000] \text{ m}$, $d_c = 8 \text{ m}$, $(r_0, r_1) = (120, 240) \text{ m}$. Targets use the scaled region $[-920, -760] \times [-480, 480] \times [200, 600] \text{ m}$. Red uses an area-uniform XY annulus $[320, 880] \text{ m}$ around a sampled target; Blue uses $x \in [0, 1000]$ and full scene y ; agent altitude lies in $[80, 600] \text{ m}$. UAV openings reject distances $\leq \max(30\lambda, d_c + 10^{-6})$. Both teams reset at solved 30 m/s level flight; Blue heads west, Red east under random initialization. Only 3D is supported.

Theme 8 / Fixed wing / Actuator / Damage

Task interaction, exact information structure and defender learning objective

E. Collision and automatic interaction

Predict living UAV paths: particles use $p_i + \tau \Delta v_i$, rigid models use synchronized RK4 endpoints clipped to $\mathcal{W}$. On each matched segment, let $r = p_i - p_j$, $d = (q_i - p_i) - (q_j - p_j)$:

$$\tau_* = \begin{cases} 0 & \|d\|^2 \leq 10^{-12}, \\ \text{clip}(-r^T d / \|d\|^2, 0, 1) & \text{otherwise,} \end{cases} \quad d_* = \|r + \tau_* d\|.$$

Collision sets both UAV HP to zero when $\min d_* \leq d_c$; targets are excluded. Pairs use i -major, $j < i$ order. Dead outer agents are skipped at entry and dead inner agents are skipped. An outer agent killed during its inner sweep continues that sweep. After collision, surviving Attack agents automatically fire at step-start positions:

$$I_i = \begin{cases} \mathbf{1}[\min_{k \in \mathcal{P}: h_k > 0} \|p_i - p_k\| < r_f], & i \in \mathcal{B}, \\ \mathbf{1}[\min_{j \in \mathcal{B}_A: h_j > 0} \|p_i - p_j\| < r_f], & i \in \mathcal{R}. \end{cases}$$

An empty minimum is $+\infty$; dead sources have $I_i = 0$. A denotes Attack. Fire is a separate rule, not a navigation action. With unit attack intensity,

$$\eta(d) = \begin{cases} 1 & d < r_0, \\ (r_1 - d)/(r_1 - r_0) & r_0 \leq d < r_1, \\ 0 & d \geq r_1, \end{cases}$$

$$L_k = \sum_{b \in \mathcal{B}_A: h_b > 0} I_b \eta(\|p_b - p_k\|),$$

$$L_i^{\text{atk}} = \sum_{j \in \mathcal{U}_A: c_j \neq c_i, h_j > 0} I_j \eta(\|p_j - p_i\|).$$

One source can affect multiple enemies; friendly explosions are excluded. All losses use a common post-collision, pre-motion snapshot and add before HP clipping.

Disturb and Scout

For source j , $d_{ji} = \|p_i - p_j\|$ and $\theta_{ji} = \angle(v_j, p_i - p_j)$; angle is zero if either norm is below 10^{-3} . Let

$$G_{ji} = \mathbf{1}[10^{-3} < d_{ji} < 600\lambda, \theta_{ji} < \pi/6] \\ \times e^{-(d_{ji}/\lambda)^2/2} / \sqrt{2\pi},$$

$$S_c(j) = \sum_{i \in \mathcal{U}: c_i = c, h_i > 0} G_{ji}.$$

Living Red Disturb uses $J_j = \mathbf{1}[S_R(j) < S_B(j)]$, Blue uses $J_j = \mathbf{1}[S_R(j) \geq S_B(j)]$. Dead sources contribute zero.

$$L_i^{\text{dis}} = .1 \sum_{j \in \mathcal{U}_D: h_j > 0} J_j G_{ji},$$

$$\tilde{h}_i = \max(0, h_i - L_i^{\text{atk}} - L_i^{\text{dis}}).$$

Set $h'_i = 0$ if $r_i = A, I_i = 1$, otherwise $h'_i = \tilde{h}_i$. Disturb damages either side, not assets. Scouts use $d < 2000\lambda$, $\theta < \pi/3$; this sector does not restrict native actor rows. Collision victims do not move. Combat victims complete their motion; inactive velocity is zeroed next step.

G. Damage payoff and completion

Assets keep finite reset health $h'_k = h_k^0$. Count raw contributions before HP clipping, including simultaneous overkill:

$$D'_k = D_k + L_k, \quad dt = \sum_k L_k, \quad R_t = -dt.$$

Blue reward is d_t ; there is no team-size, asset-count or HP normalization and no terminal Survival bonus.

$$d = \mathbf{1}[\forall b \in \mathcal{B}_A : h'_b \leq 0].$$

All agent slots remain until global completion/truncation. Dead slots have zero observations, ignored actions and still receive full team R_t , including later damage after Red extinction. Do not sum duplicate teammate rewards. Undiscounted returns are $(-\sum_k D_k, +\sum_k D_k)$. H only truncates; global bootstrap $b_t = 1 - d$ remains one at truncation. Actor action-loss masks use pre-step participation.

F. Exact observations and CTDE information

For each living entity $j \neq i$, the actor row is

$$e_{ij} = [(p_j - p_i)/L, (v_j - v_i)/(2V_{\max}), \\ h_j, 1, \mathbf{1}_{c_j=R}, \mathbf{1}_{c_j=B}, \mathbf{1}_{j \in \mathcal{P}}],$$

$$L = \sqrt{\sum_a (b_a^+ - b_a^-)^2}.$$

Here $V_{\max}$ is the preset maximum speed, not the reference speed. Rows have 11 values, in fixed roster order: Red Scout/Disturb/Attack, Blue Scout/Disturb/Attack, then targets, excluding self. Dead rows are zero. Every living entity is included; Attack, Scout and Disturb share this packing. Side flags do not encode role. Own health is absent from the self block.

The actor receives the $(M - 1) \times 11$ entity table ($V_{\max} = 30$) plus unnormalized $[p_i, v_i, q_i, \omega_i]$. Both blocks are zero for an already-dead Damage observer. With deterministic encoding, $Z_i(o | s) = \delta_{\text{encode}_i(s)}(o)$. Other UAV attitudes/rates, opponent memory and time are omitted; hence local information is partial despite noiseless sensing.

Privileged $g(s) = \text{state}()$ packs 11 values per entity,

$$[(p - b^-)/(b^+ - b^-), v/(2V_{\max}), h, \ell, R, B, P],$$

with dead blocks zero. Rigid models append 13 raw state values per UAV (dead zero), then $[t, \sum_k D_k, \mathbf{1}_{\text{Damage}}]$: size $11M + 13N + 3$. Particle size is $11M$, omitting time and damage. g is a CTDE feature, not the complete Markov state.

H. Defender Dec-POMDP and MARL objective

For a specified frozen opposing policy π_B , define

$$\mathcal{M}_R = (\mathcal{R}, \mathcal{S}, \{\mathcal{A}_i\}, T^{\pi_B}, \\ \{\mathcal{O}_i\}, Z, R, \gamma, \rho_0, H),$$

$$T^{\pi_B}(s' | s, a_R) = \int \delta_{F_C(s, a_R, a_B)}(s') \\ \times \pi_B(da_B | o_B, \zeta_t).$$

F_C is the stated collision, simultaneous interaction and controlled-motion map, including any memory evolution; ρ_0 is the reset law. Native execution accepts both teams' actions. Freezing a rule or learned opponent is an experiment choice; with both teams learning, the model is a POSG.

$$\tau_i = (o_{i,0}, a_{i,0}, \dots, o_{i,t}),$$

$$\pi_R(a_R | \tau_R) = \prod_{i \in \mathcal{R}} \pi_i(a_i | \tau_i),$$

$$\pi_R^* \in \arg \max_{\pi_R} \mathbb{E} \left[\sum_{t=0}^{T-1} \gamma^t R_t \right].$$

Expectation is over ρ_0, T^{π_B}, π_R . Here T is natural completion (possibly infinite); H limits sampled rollouts. A truncated return is partial, so its successor value is retained. For $\gamma = 1$, this is respectively the native terminal Survival payoff or negative total asset damage. For $\gamma < 1$, it is the discounted task selected by the caller. Actors use local histories only. CTDE critics can use $g(s)$, joint actions and recorded task/memory features:

$$y_t = R_t + \gamma b_t Q^-(g(s'), a'_R).$$

Here Q^- is a target critic and successor actions follow the target actor policies. Use global completion, not individual death. A recurrent critic or augmented state may be needed because g omits history. This defines an objective; it does not prescribe an implemented optimizer.

Theme 9 / Fixed wing / Position / Survival

Rigid-body aerodynamic transition and navigation/actuator semantics

A. Scenario and complete state

Red defenders $\mathcal{R}$ protect stationary assets $\mathcal{P}$ against Blue $\mathcal{B}$. Set $\mathcal{U} = \mathcal{R} \cup \mathcal{B}$, $\mathcal{E} = \mathcal{U} \cup \mathcal{P}$, $N = |\mathcal{U}|$, $K = |\mathcal{P}|$, $M = N + K$. Episode-fixed roles are Attack, Scout or Disturb.

$$s_t = (\{x_i, h_i, r_i, c_i, \text{id}_i\}_{i \in \mathcal{U}}, \{p_k, h_k, D_k\}_{k \in \mathcal{P}}, t, \zeta_t).$$

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. Participation is $\ell_i = \mathbf{1}[h_i > 0]$. Reset UAV health is 1, target health $h_k^0 = 2$ by default; $v_k = 0$, $D_k = 0$ initially. D_k accumulates raw asset damage.

Configuration C fixes roster, bounds, controls and task. ζ_t contains persistent assignments and opponent memory, if used. Reactive controllers have no hidden integrators. Last command, functional bits, pre-step positions and step damage are overwritten transition records; retain them for exact reward/event replay. RNG state is required for exact replay, not actor input.

B. Six-degree-of-freedom dynamics

World coordinates are east-north-up (ENU); body coordinates are forward-left-up (FLU). The scalar-first quaternion $q = (q_w, q_x, q_y, q_z)$ maps body to world through $R(q) \in SO(3)$, with $q \in S^3$, $\|q\| = 1$ and $q \sim -q$. Thus 13 stored values represent 12 physical degrees of freedom.

$$\begin{aligned} \dot{p} &= v, & \dot{v} &= R(q)F_B/m - g e_3, \\ \dot{q} &= \frac{1}{2}q \otimes (0, \omega), & \dot{\omega} &= J^{-1}(M_B - \omega \times J\omega). \end{aligned}$$

Here $g = 9.81 \text{ m s}^{-2}$, $\omega \in \mathbb{R}^3$ is the body angular velocity in rad/s; F_B excludes gravity. No wind, sensor noise or actuator lag is present.

$$R(q) = I + 2q_w[q_v]_{\times} + 2[q_v]_{\times}^2, \quad q_v = (q_x, q_y, q_z).$$

Here $[z]_{\times} y = z \times y$.

C. Aerosonde aerodynamic wrench

Let $C_f = \text{diag}(1, -1, -1)$ and $(u, v_b, w) = C_f R^T v$, $(p_b, q_b, r_b) = C_f \omega$. Define $V = \|(u, v_b, w)\|$, $\bar{V} = \max(V, 10^{-6})$, $\alpha = \text{atan2}(w, u)$, $\beta = \arcsin \text{clip}(v_b/\bar{V}, -1, 1)$, $\bar{Q} = \rho V^2/2$, $\hat{p} = bp_b/(2\bar{V})$, $\hat{q} = cq_b/(2\bar{V})$, $\hat{r} = br_b/(2\bar{V})$.

$$\begin{aligned} E_- &= e^{\text{clip}(-50(\alpha-.47), -500, 500)}, \\ E_+ &= e^{\text{clip}(50(\alpha+.47), -500, 500)}, \\ \sigma &= (1 + E_- + E_+)/[(1 + E_-)(1 + E_+)], \\ C_L &= (1 - \sigma)(.23 + 5.61\alpha) \\ &\quad + 2\sigma \text{sign}(\alpha) \sin^2 \alpha \cos \alpha + 7.95\hat{q} + .13\delta_e, \\ C_D &= .043 + \frac{(.23 + 5.61\alpha)^2}{\pi(.9)(b^2/S)} + .0135\delta_e, \\ C_Y &= -.98\beta + .075\delta_a + .19\delta_r, \\ C_\ell &= -.13\beta - .51\hat{p} + .25\hat{r} + .17\delta_a + .0024\delta_r, \\ C_m &= .0135 - 2.74\alpha - 38.21\hat{q} - .99\delta_e, \\ C_n &= .073\beta + .069\hat{p} - .095\hat{r} - .011\delta_a - .069\delta_r. \end{aligned}$$

$$\begin{aligned} F_B &= C_f \left\{ \bar{Q} S \begin{bmatrix} -C_D \cos \alpha + C_L \sin \alpha \\ C_Y \\ -C_D \sin \alpha - C_L \cos \alpha \end{bmatrix} + \begin{bmatrix} T_p \\ 0 \\ 0 \end{bmatrix} \right\}, \\ \bar{M}_B &= C_f \left\{ \bar{Q} S \begin{bmatrix} bC_\ell \\ cC_m \\ bC_n \end{bmatrix} - \begin{bmatrix} Q_p \\ 0 \\ 0 \end{bmatrix} \right\}. \end{aligned}$$

Motor, propeller and equilibrium

The physical constants are $m = 11 \text{ kg}$, $S = .55 \text{ m}^2$, $b = 2.8956 \text{ m}$, $c = .18994 \text{ m}$, $\rho = 1.2682 \text{ kg/m}^3$,

$$J = C_f \begin{bmatrix} .8244 & 0 & -.1204 \\ 0 & 1.135 & 0 \\ -.1204 & 0 & 1.759 \end{bmatrix} C_f^T.$$

With $D_p = .508$, $R_m = .042$, $I_0 = 1.5$, $V_m = 44.4$, $k_m = 60/(145 \cdot 2\pi)$, the steady motor root is

$$\begin{aligned} A &= \rho D_p^5 C_{Q0}/(4\pi^2), \\ B &= \rho D_p^4 C_{Q1} V/(2\pi) + k_m^2/R_m, \\ C &= \rho D_p^3 C_{Q2} V^2 - k_m V_m \tau/R_m + k_m I_0, \\ \Omega_p &= \max\{0, (-B + \sqrt{\max(0, B^2 - 4AC)})/(2A)\}. \end{aligned}$$

Set $z = (D_p^4 \Omega_p^2/(4\pi^2), D_p^3 V \Omega_p/(2\pi), D_p^2 V^2)$,

$$\begin{aligned} T_p &= \rho(.09357, -.06044, -.1079)z, \\ Q_p &= \rho D_p(.005230, .004970, -.01664)z. \end{aligned}$$

Reference trim $(x^*(V_d), u^*(V_d))$ solves all six equations $\dot{v} = 0$, $\dot{\omega} = 0$, including propeller reaction torque. Unknowns are pitch, heading, elevator, aileron, rudder and throttle, at velocity $(V_d, 0, 0)$. The reset uses the 30 m/s level trim rotated to the initial heading. Mass/inertia are not scaled with the scene [1].

Absolute position action

Choose $a_i = p_d \in \mathcal{W}$ in metres. At every stage, $d = p_d - p$ determines

$$\begin{aligned} \chi_d &= \text{atan2}(d_y, d_x), \\ \gamma_d &= \text{atan2}(d_z, \max(\|d_{xy}\|, 30)), \quad V_d = 30. \end{aligned}$$

Tiny horizontal displacement uses current heading. The aircraft flies toward and through the target; it cannot hover there.

D. Implemented reference controller

Define bank $\phi = \text{atan2}(R_{32}, R_{33})$, nose-up pitch $\theta = \arcsin R_{31}$, heading $\psi = \text{atan2}(R_{21}, R_{11})$. Actual course is $\chi = \text{atan2}(v_y, v_x)$, or ψ when horizontal speed is tiny; path $\gamma_f = \text{atan2}(v_z, \|v_{xy}\|)$ and $V_a = \|v\|$.

$$\begin{aligned} V_d &= \text{clip}(V_d, 18, 30), \\ \gamma_d &= \text{clip}(\gamma_d, -15^\circ, 15^\circ), \\ e_\chi &= \text{wrap}(\chi_d - \chi), \\ \bar{\phi} &= -\text{atan2}(\max(V_a, 18)e_\chi, g), \\ \phi_d &= \text{clip}(\bar{\phi}, -45^\circ, 45^\circ), \\ \bar{\theta} &= \theta^* + \gamma_d + .5(\gamma_d - \gamma_f), \\ \theta_d &= \text{clip}(\bar{\theta}, -20^\circ, 20^\circ), \\ \delta_e &= \delta_e^* - 2(\theta_d - \theta) + .25q_b, \\ \delta_a &= \delta_a^* + .5(\phi_d - \phi) - .08p_b, \\ r_d &= g \tan \phi_d / \max(V_a, 18), \\ \delta_r &= \delta_r^* + .8(\beta - \psi^*) + .1(r_b - r_d), \\ \tau &= \tau^* + .08(V_d - V_a) + .7 \sin \gamma_d. \end{aligned}$$

All angles use radians. The scheduled trim heading ψ^* is the lateral reference. Divide surfaces by 25° , map throttle to $2\tau - 1$ and clip the four outputs to $[-1, 1]$. These project gains/bounds do not clamp the physical flight envelope.

Integration, geometry and initialization

RK4 uses $h = .01 \text{ s}$, 100 synchronized substeps per 1 s step. Each stage feeds back the held reference: $k_j = f(x_j, U(x_j))$ and $x^+ = x + h(k_1 + 2k_2 + 2k_3 + k_4)/6$, then normalize q^+ . Default $\lambda = .4$, $d_c = 8 \text{ m}$, $(r_0, r_1) = (120, 240) \text{ m}$, $\mathcal{W} = [-1000, 1000]^2 \times [0, 1000] \text{ m}$. Clip path positions and the endpoint to scene bounds; remove endpoint outward velocity. Reset geometry follows the area-uniform Red annulus / Blue rectangular region, scaled by .4, with separated UAV openings and solved 30 m/s level trim. Only 3D is supported.

Theme 9 / Fixed wing / Position / Survival

Task interaction, exact information structure and defender learning objective

E. Collision and automatic interaction

Predict living UAV paths: particles use $p_i + \tau \Delta v_i$, rigid models use synchronized RK4 endpoints clipped to $\mathcal{W}$. On each matched segment, let $r = p_i - p_j$, $d = (q_i - p_i) - (q_j - p_j)$:

$$\tau_* = \begin{cases} 0 & \|d\|^2 \leq 10^{-12}, \\ \text{clip}(-r^T d / \|d\|^2, 0, 1) & \text{otherwise,} \end{cases} \quad d_* = \|r + \tau_* d\|.$$

Collision sets both UAV HP to zero when $\min d_* \leq d_c$; targets are excluded. Pairs use i -major, $j < i$ order. Dead outer agents are skipped at entry and dead inner agents are skipped. An outer agent killed during its inner sweep continues that sweep. After collision, surviving Attack agents automatically fire at step-start positions:

$$I_i = \begin{cases} \mathbf{1}[\min_{k \in \mathcal{P}: h_k > 0} \|p_i - p_k\| < r_f], & i \in \mathcal{B}, \\ \mathbf{1}[\min_{j \in \mathcal{B}_A: h_j > 0} \|p_i - p_j\| < r_f], & i \in \mathcal{R}. \end{cases}$$

An empty minimum is $+\infty$; dead sources have $I_i = 0$. A denotes Attack. Fire is a separate rule, not a navigation action. With unit attack intensity,

$$\eta(d) = \begin{cases} 1 & d < r_0, \\ (r_1 - d)/(r_1 - r_0) & r_0 \leq d < r_1, \\ 0 & d \geq r_1, \end{cases}$$

$$L_k = \sum_{b \in \mathcal{B}_A: h_b > 0} I_b \eta(\|p_b - p_k\|),$$

$$L_i^{\text{atk}} = \sum_{j \in \mathcal{U}_A: c_j \neq c_i, h_j > 0} I_j \eta(\|p_j - p_i\|).$$

One source can affect multiple enemies; friendly explosions are excluded. All losses use a common post-collision, pre-motion snapshot and add before HP clipping.

Disturb and Scout

For source j , $d_{ji} = \|p_i - p_j\|$ and $\theta_{ji} = \angle(v_j, p_i - p_j)$; angle is zero if either norm is below 10^{-3} . Let

$$G_{ji} = \mathbf{1}[10^{-3} < d_{ji} < 600\lambda, \theta_{ji} < \pi/6] \times e^{-(d_{ji}/\lambda)^2/2} / \sqrt{2\pi},$$

$$S_c(j) = \sum_{i \in \mathcal{U}: c_i = c, h_i > 0} G_{ji}.$$

Living Red Disturb uses $J_j = \mathbf{1}[S_R(j) < S_B(j)]$, Blue uses $J_j = \mathbf{1}[S_R(j) \geq S_B(j)]$. Dead sources contribute zero.

$$L_i^{\text{dis}} = .1 \sum_{j \in \mathcal{U}_D: h_j > 0} J_j G_{ji},$$

$$\tilde{h}_i = \max(0, h_i - L_i^{\text{atk}} - L_i^{\text{dis}}).$$

Set $h'_i = 0$ if $r_i = A, I_i = 1$, otherwise $h'_i = \tilde{h}_i$. Disturb damages either side, not assets. Scouts use $d < 2000\lambda$, $\theta < \pi/3$; this sector does not restrict native actor rows. Collision victims do not move. Combat victims complete their motion; inactive velocity is zeroed next step.

G. Survival payoff and completion

Assets obey $h'_k = \max(0, h_k - L_k)$, $D'_k = D_k + L_k$.

$$z(s') = \begin{cases} -1 & \exists k : h'_k < 10^{-3}, \\ +1 & \forall b \in \mathcal{B}_A : h'_b < 10^{-3}, \\ 0 & \text{otherwise.} \end{cases}$$

Target loss has priority. Natural completion is $d = \mathbf{1}[z \neq 0]$; default Red reward $R_t = 10z$, Blue $-R_t$. Red elimination alone does not end the world. Individual $h'_i \leq 0$ retires a Survival actor after its final transition. Horizon H truncates without a win bonus; use global bootstrap $b_t = 1 - d$, including individual death and truncation. Actor action-loss masks use pre-step participation. Explicit reward weights replace the default payoff as defined on page 1.

F. Exact observations and CTDE information

For each living entity $j \neq i$, the actor row is

$$e_{ij} = \begin{bmatrix} (p_j - p_i)/L, (v_j - v_i)/(2V_{\max}), \\ h_j, 1, \mathbf{1}_{c_j=R}, \mathbf{1}_{c_j=B}, \mathbf{1}_{j \in \mathcal{P}} \end{bmatrix},$$

$$L = \sqrt{\sum_a (b_a^+ - b_a^-)^2}.$$

Here $V_{\max}$ is the preset maximum speed, not the reference speed. Rows have 11 values, in fixed roster order: Red Scout/Disturb/Attack, Blue Scout/Disturb/Attack, then targets, excluding self. Dead rows are zero. Every living entity is included; Attack, Scout and Disturb share this packing. Side flags do not encode role. Own health is absent from the self block.

The actor receives the $(M-1) \times 11$ entity table ($V_{\max} = 30$) plus unnormalized $[p_i, v_i, q_i, \omega_i] \in \mathbb{R}^{13}$. With deterministic encoding, $Z_i(o | s) = \delta_{\text{encode}_i(s)}(o)$. Other UAV attitudes/rates, opponent memory and time are omitted; hence local information is partial despite noiseless sensing.

Privileged $g(s) = \text{state}()$ packs 11 values per entity,

$$[(p - b^-)/(b^+ - b^-), v/(2V_{\max}), h, \ell, R, B, P],$$

with dead blocks zero. Rigid models append 13 raw state values per UAV (dead zero), then $[t, \sum_k D_k, \mathbf{1}_{\text{Damage}}]$: size $11M + 13N + 3$. Particle size is $11M$, omitting time and damage. g is a CTDE feature, not the complete Markov state.

H. Defender Dec-POMDP and MARL objective

For a specified frozen opposing policy π_B , define

$$\mathcal{M}_R = (\mathcal{R}, \mathcal{S}, \{\mathcal{A}_i\}, T^{\pi_B}, \{\mathcal{O}_i\}, Z, R, \gamma, \rho_0, H),$$

$$T^{\pi_B}(s' | s, a_R) = \int \delta_{F_C(s, a_R, a_B)}(s') \times \pi_B(da_B | o_B, \zeta_t).$$

F_C is the stated collision, simultaneous interaction and controlled-motion map, including any memory evolution; ρ_0 is the reset law. Native execution accepts both teams' actions. Freezing a rule or learned opponent is an experiment choice; with both teams learning, the model is a POSG.

$$\tau_i = (o_{i,0}, a_{i,0}, \dots, o_{i,t}),$$

$$\pi_R(a_R | \tau_R) = \prod_{i \in \mathcal{R}} \pi_i(a_i | \tau_i),$$

$$\pi_R^* \in \arg \max_{\pi_R} \mathbb{E} \left[\sum_{t=0}^{T-1} \gamma^t R_t \right].$$

Expectation is over ρ_0, T^{π_B}, π_R . Here T is natural completion (possibly infinite); H limits sampled rollouts. A truncated return is partial, so its successor value is retained. For $\gamma = 1$, this is respectively the native terminal Survival payoff or negative total asset damage. For $\gamma < 1$, it is the discounted task selected by the caller. Actors use local histories only. CTDE critics can use $g(s)$, joint actions and recorded task/memory features:

$$y_t = R_t + \gamma b_t Q^-(g(s'), a'_R).$$

Here Q^- is a target critic and successor actions follow the target actor policies. Use global completion, not individual death. A recurrent critic or augmented state may be needed because g omits history. This defines an objective; it does not prescribe an implemented optimizer.

Theme 10 / Fixed wing / Position / Damage

Rigid-body aerodynamic transition and navigation/actuator semantics

A. Scenario and complete state

Red defenders $\mathcal{R}$ protect stationary assets $\mathcal{P}$ against Blue $\mathcal{B}$. Set $\mathcal{U} = \mathcal{R} \cup \mathcal{B}$, $\mathcal{E} = \mathcal{U} \cup \mathcal{P}$, $N = |\mathcal{U}|$, $K = |\mathcal{P}|$, $M = N + K$. Episode-fixed roles are Attack, Scout or Disturb.

$$s_t = (\{x_i, h_i, r_i, c_i, \text{id}_i\}_{i \in \mathcal{U}}, \{p_k, h_k, D_k\}_{k \in \mathcal{P}}, t, \zeta_t).$$

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. Participation is $\ell_i = \mathbf{1}[h_i > 0]$. Reset UAV health is 1, target health $h_k^0 = 2$ by default; $v_k = 0$, $D_k = 0$ initially. D_k accumulates raw asset damage.

Configuration C fixes roster, bounds, controls and task. ζ_t contains persistent assignments and opponent memory, if used. Reactive controllers have no hidden integrators. Last command, functional bits, pre-step positions and step damage are overwritten transition records; retain them for exact reward/event replay. RNG state is required for exact replay, not actor input.

B. Six-degree-of-freedom dynamics

World coordinates are east-north-up (ENU); body coordinates are forward-left-up (FLU). The scalar-first quaternion $q = (q_w, q_x, q_y, q_z)$ maps body to world through $R(q) \in SO(3)$, with $q \in S^3$, $\|q\| = 1$ and $q \sim -q$. Thus 13 stored values represent 12 physical degrees of freedom.

$$\begin{aligned} \dot{p} &= v, & \dot{v} &= R(q)F_B/m - g e_3, \\ \dot{q} &= \frac{1}{2}q \otimes (0, \omega), & \dot{\omega} &= J^{-1}(M_B - \omega \times J\omega). \end{aligned}$$

Here $g = 9.81 \text{ m s}^{-2}$, $\omega \in \mathbb{R}^3$ is the body angular velocity in rad/s; F_B excludes gravity. No wind, sensor noise or actuator lag is present.

$$R(q) = I + 2q_w[q_v]_{\times} + 2[q_v]_{\times}^2, \quad q_v = (q_x, q_y, q_z).$$

Here $[z]_{\times} y = z \times y$.

C. Aerosonde aerodynamic wrench

Let $C_f = \text{diag}(1, -1, -1)$ and $(u, v_b, w) = C_f R^T v$, $(p_b, q_b, r_b) = C_f \omega$. Define $V = \|(u, v_b, w)\|$, $\bar{V} = \max(V, 10^{-6})$, $\alpha = \text{atan2}(w, u)$, $\beta = \arcsin \text{clip}(v_b/\bar{V}, -1, 1)$, $\bar{Q} = \rho V^2/2$, $\hat{p} = bp_b/(2\bar{V})$, $\hat{q} = cq_b/(2\bar{V})$, $\hat{r} = br_b/(2\bar{V})$.

$$\begin{aligned} E_- &= e^{\text{clip}(-50(\alpha-.47), -500, 500)}, \\ E_+ &= e^{\text{clip}(50(\alpha+.47), -500, 500)}, \\ \sigma &= (1 + E_- + E_+)/[(1 + E_-)(1 + E_+)], \\ C_L &= (1 - \sigma)(.23 + 5.61\alpha) \\ &\quad + 2\sigma \text{sign}(\alpha) \sin^2 \alpha \cos \alpha + 7.95\hat{q} + .13\delta_e, \\ C_D &= .043 + \frac{(.23 + 5.61\alpha)^2}{\pi(.9)(b^2/S)} + .0135\delta_e, \\ C_Y &= -.98\beta + .075\delta_a + .19\delta_r, \\ C_\ell &= -.13\beta - .51\hat{p} + .25\hat{r} + .17\delta_a + .0024\delta_r, \\ C_m &= .0135 - 2.74\alpha - 38.21\hat{q} - .99\delta_e, \\ C_n &= .073\beta + .069\hat{p} - .095\hat{r} - .011\delta_a - .069\delta_r. \end{aligned}$$

$$\begin{aligned} F_B &= C_f \left\{ \bar{Q} S \begin{bmatrix} -C_D \cos \alpha + C_L \sin \alpha \\ C_Y \\ -C_D \sin \alpha - C_L \cos \alpha \end{bmatrix} + \begin{bmatrix} T_p \\ 0 \\ 0 \end{bmatrix} \right\}, \\ \bar{M}_B &= C_f \left\{ \bar{Q} S \begin{bmatrix} bC_\ell \\ cC_m \\ bC_n \end{bmatrix} - \begin{bmatrix} Q_p \\ 0 \\ 0 \end{bmatrix} \right\}. \end{aligned}$$

Motor, propeller and equilibrium

The physical constants are $m = 11 \text{ kg}$, $S = .55 \text{ m}^2$, $b = 2.8956 \text{ m}$, $c = .18994 \text{ m}$, $\rho = 1.2682 \text{ kg/m}^3$,

$$J = C_f \begin{bmatrix} .8244 & 0 & -.1204 \\ 0 & 1.135 & 0 \\ -.1204 & 0 & 1.759 \end{bmatrix} C_f^T.$$

With $D_p = .508$, $R_m = .042$, $I_0 = 1.5$, $V_m = 44.4$, $k_m = 60/(145 \cdot 2\pi)$, the steady motor root is

$$\begin{aligned} A &= \rho D_p^5 C_{Q0}/(4\pi^2), \\ B &= \rho D_p^4 C_{Q1} V/(2\pi) + k_m^2/R_m, \\ C &= \rho D_p^3 C_{Q2} V^2 - k_m V_m \tau/R_m + k_m I_0, \\ \Omega_p &= \max\{0, (-B + \sqrt{\max(0, B^2 - 4AC)})/(2A)\}. \end{aligned}$$

Set $z = (D_p^4 \Omega_p^2/(4\pi^2), D_p^3 V \Omega_p/(2\pi), D_p^2 V^2)$,

$$\begin{aligned} T_p &= \rho(.09357, -.06044, -.1079)z, \\ Q_p &= \rho D_p(.005230, .004970, -.01664)z. \end{aligned}$$

Reference trim $(x^*(V_d), u^*(V_d))$ solves all six equations $\dot{v} = 0$, $\dot{\omega} = 0$, including propeller reaction torque. Unknowns are pitch, heading, elevator, aileron, rudder and throttle, at velocity $(V_d, 0, 0)$. The reset uses the 30 m/s level trim rotated to the initial heading. Mass/inertia are not scaled with the scene [1].

Absolute position action

Choose $a_i = p_d \in \mathcal{W}$ in metres. At every stage, $d = p_d - p$ determines

$$\begin{aligned} \chi_d &= \text{atan2}(d_y, d_x), \\ \gamma_d &= \text{atan2}(d_z, \max(\|d_{xy}\|, 30)), \quad V_d = 30. \end{aligned}$$

Tiny horizontal displacement uses current heading. The aircraft flies toward and through the target; it cannot hover there.

D. Implemented reference controller

Define bank $\phi = \text{atan2}(R_{32}, R_{33})$, nose-up pitch $\theta = \arcsin R_{31}$, heading $\psi = \text{atan2}(R_{21}, R_{11})$. Actual course is $\chi = \text{atan2}(v_y, v_x)$, or ψ when horizontal speed is tiny; path $\gamma_f = \text{atan2}(v_z, \|v_{xy}\|)$ and $V_a = \|v\|$.

$$\begin{aligned} V_d &= \text{clip}(V_d, 18, 30), \\ \gamma_d &= \text{clip}(\gamma_d, -15^\circ, 15^\circ), \\ e_\chi &= \text{wrap}(\chi_d - \chi), \\ \bar{\phi} &= -\text{atan2}(\max(V_a, 18)e_\chi, g), \\ \phi_d &= \text{clip}(\bar{\phi}, -45^\circ, 45^\circ), \\ \bar{\theta} &= \theta^* + \gamma_d + .5(\gamma_d - \gamma_f), \\ \theta_d &= \text{clip}(\bar{\theta}, -20^\circ, 20^\circ), \\ \delta_e &= \delta_e^* - 2(\theta_d - \theta) + .25q_b, \\ \delta_a &= \delta_a^* + .5(\phi_d - \phi) - .08p_b, \\ r_d &= g \tan \phi_d / \max(V_a, 18), \\ \delta_r &= \delta_r^* + .8(\beta - \psi^*) + .1(r_b - r_d), \\ \tau &= \tau^* + .08(V_d - V_a) + .7 \sin \gamma_d. \end{aligned}$$

All angles use radians. The scheduled trim heading ψ^* is the lateral reference. Divide surfaces by 25° , map throttle to $2\tau - 1$ and clip the four outputs to $[-1, 1]$. These project gains/bounds do not clamp the physical flight envelope.

Integration, geometry and initialization

RK4 uses $h = .01 \text{ s}$, 100 synchronized substeps per 1 s step. Each stage feeds back the held reference: $k_j = f(x_j, U(x_j))$ and $x^+ = x + h(k_1 + 2k_2 + 2k_3 + k_4)/6$, then normalize q^+ . Default $\lambda = .4$, $d_c = 8 \text{ m}$, $(r_0, r_1) = (120, 240) \text{ m}$, $\mathcal{W} = [-1000, 1000]^2 \times [0, 1000] \text{ m}$. Clip path positions and the endpoint to scene bounds; remove endpoint outward velocity. Reset geometry follows the area-uniform Red annulus / Blue rectangular region, scaled by .4, with separated UAV openings and solved 30 m/s level trim. Only 3D is supported.

Theme 10 / Fixed wing / Position / Damage

Task interaction, exact information structure and defender learning objective

E. Collision and automatic interaction

Predict living UAV paths: particles use $p_i + \tau \Delta v_i$, rigid models use synchronized RK4 endpoints clipped to $\mathcal{W}$. On each matched segment, let $r = p_i - p_j$, $d = (q_i - p_i) - (q_j - p_j)$:

$$\tau_* = \begin{cases} 0 & \|d\|^2 \leq 10^{-12}, \\ \text{clip}(-r^T d / \|d\|^2, 0, 1) & \text{otherwise,} \end{cases} \quad d_* = \|r + \tau_* d\|.$$

Collision sets both UAV HP to zero when $\min d_* \leq d_c$; targets are excluded. Pairs use i -major, $j < i$ order. Dead outer agents are skipped at entry and dead inner agents are skipped. An outer agent killed during its inner sweep continues that sweep. After collision, surviving Attack agents automatically fire at step-start positions:

$$I_i = \begin{cases} \mathbf{1}[\min_{k \in \mathcal{P}: h_k > 0} \|p_i - p_k\| < r_f], & i \in \mathcal{B}, \\ \mathbf{1}[\min_{j \in \mathcal{B}_A: h_j > 0} \|p_i - p_j\| < r_f], & i \in \mathcal{R}. \end{cases}$$

An empty minimum is $+\infty$; dead sources have $I_i = 0$. A denotes Attack. Fire is a separate rule, not a navigation action. With unit attack intensity,

$$\eta(d) = \begin{cases} 1 & d < r_0, \\ (r_1 - d)/(r_1 - r_0) & r_0 \leq d < r_1, \\ 0 & d \geq r_1, \end{cases}$$

$$L_k = \sum_{b \in \mathcal{B}_A: h_b > 0} I_b \eta(\|p_b - p_k\|),$$

$$L_i^{\text{atk}} = \sum_{j \in \mathcal{U}_A: c_j \neq c_i, h_j > 0} I_j \eta(\|p_j - p_i\|).$$

One source can affect multiple enemies; friendly explosions are excluded. All losses use a common post-collision, pre-motion snapshot and add before HP clipping.

Disturb and Scout

For source j , $d_{ji} = \|p_i - p_j\|$ and $\theta_{ji} = \angle(v_j, p_i - p_j)$; angle is zero if either norm is below 10^{-3} . Let

$$G_{ji} = \mathbf{1}[10^{-3} < d_{ji} < 600\lambda, \theta_{ji} < \pi/6] \\ \times e^{-(d_{ji}/\lambda)^2/2} / \sqrt{2\pi},$$

$$S_c(j) = \sum_{i \in \mathcal{U}: c_i = c, h_i > 0} G_{ji}.$$

Living Red Disturb uses $J_j = \mathbf{1}[S_R(j) < S_B(j)]$, Blue uses $J_j = \mathbf{1}[S_R(j) \geq S_B(j)]$. Dead sources contribute zero.

$$L_i^{\text{dis}} = .1 \sum_{j \in \mathcal{U}_D: h_j > 0} J_j G_{ji},$$

$$\tilde{h}_i = \max(0, h_i - L_i^{\text{atk}} - L_i^{\text{dis}}).$$

Set $h'_i = 0$ if $r_i = A, I_i = 1$, otherwise $h'_i = \tilde{h}_i$. Disturb damages either side, not assets. Scouts use $d < 2000\lambda$, $\theta < \pi/3$; this sector does not restrict native actor rows. Collision victims do not move. Combat victims complete their motion; inactive velocity is zeroed next step.

G. Damage payoff and completion

Assets keep finite reset health $h'_k = h_k^0$. Count raw contributions before HP clipping, including simultaneous overkill:

$$D'_k = D_k + L_k, \quad d_t = \sum_k L_k, \quad R_t = -d_t.$$

Blue reward is d_t ; there is no team-size, asset-count or HP normalization and no terminal Survival bonus.

$$d = \mathbf{1}[\forall b \in \mathcal{B}_A : h'_b \leq 0].$$

All agent slots remain until global completion/truncation. Dead slots have zero observations, ignored actions and still receive full team R_t , including later damage after Red extinction. Do not sum duplicate teammate rewards. Undiscounted returns are $(-\sum_k D_k, +\sum_k D_k)$. H only truncates; global bootstrap $b_t = 1 - d$ remains one at truncation. Actor action-loss masks use pre-step participation.

F. Exact observations and CTDE information

For each living entity $j \neq i$, the actor row is

$$e_{ij} = [(p_j - p_i)/L, (v_j - v_i)/(2V_{\max}), \\ h_j, 1, \mathbf{1}_{c_j=R}, \mathbf{1}_{c_j=B}, \mathbf{1}_{j \in \mathcal{P}}], \\ L = \sqrt{\sum_a (b_a^+ - b_a^-)^2}.$$

Here $V_{\max}$ is the preset maximum speed, not the reference speed. Rows have 11 values, in fixed roster order: Red Scout/Disturb/Attack, Blue Scout/Disturb/Attack, then targets, excluding self. Dead rows are zero. Every living entity is included; Attack, Scout and Disturb share this packing. Side flags do not encode role. Own health is absent from the self block.

The actor receives the $(M - 1) \times 11$ entity table ($V_{\max} = 30$) plus unnormalized $[p_i, v_i, q_i, \omega_i]$. Both blocks are zero for an already-dead Damage observer. With deterministic encoding, $Z_i(o | s) = \delta_{\text{encode}_i(s)}(o)$. Other UAV attitudes/rates, opponent memory and time are omitted; hence local information is partial despite noiseless sensing.

Privileged $g(s) = \text{state}()$ packs 11 values per entity,

$$[(p - b^-)/(b^+ - b^-), v/(2V_{\max}), h, \ell, R, B, P],$$

with dead blocks zero. Rigid models append 13 raw state values per UAV (dead zero), then $[t, \sum_k D_k, \mathbf{1}_{\text{Damage}}]$: size $11M + 13N + 3$. Particle size is $11M$, omitting time and damage. g is a CTDE feature, not the complete Markov state.

H. Defender Dec-POMDP and MARL objective

For a specified frozen opposing policy π_B , define

$$\mathcal{M}_R = (\mathcal{R}, \mathcal{S}, \{\mathcal{A}_i\}, T^{\pi_B}, \\ \{\mathcal{O}_i\}, Z, R, \gamma, \rho_0, H),$$

$$T^{\pi_B}(s' | s, a_R) = \int \delta_{F_C(s, a_R, a_B)}(s') \\ \times \pi_B(da_B | o_B, \zeta_t).$$

F_C is the stated collision, simultaneous interaction and controlled-motion map, including any memory evolution; ρ_0 is the reset law. Native execution accepts both teams' actions. Freezing a rule or learned opponent is an experiment choice; with both teams learning, the model is a POSG.

$$\tau_i = (o_{i,0}, a_{i,0}, \dots, o_{i,t}),$$

$$\pi_R(a_R | \tau_R) = \prod_{i \in \mathcal{R}} \pi_i(a_i | \tau_i),$$

$$\pi_R^* \in \arg \max_{\pi_R} \mathbb{E} \left[\sum_{t=0}^{T-1} \gamma^t R_t \right].$$

Expectation is over ρ_0, T^{π_B}, π_R . Here T is natural completion (possibly infinite); H limits sampled rollouts. A truncated return is partial, so its successor value is retained. For $\gamma = 1$, this is respectively the native terminal Survival payoff or negative total asset damage. For $\gamma < 1$, it is the discounted task selected by the caller. Actors use local histories only. CTDE critics can use $g(s)$, joint actions and recorded task/memory features:

$$y_t = R_t + \gamma b_t Q^-(g(s'), a'_R).$$

Here Q^- is a target critic and successor actions follow the target actor policies. Use global completion, not individual death. A recurrent critic or augmented state may be needed because g omits history. This defines an objective; it does not prescribe an implemented optimizer.

Theme 11 / Quadrotor / Acceleration / Survival

Rigid-body thrust dynamics, reference feedback and feasible actuation

A. Scenario and complete state

Red defenders $\mathcal{R}$ protect stationary assets $\mathcal{P}$ against Blue $\mathcal{B}$. Set $\mathcal{U} = \mathcal{R} \cup \mathcal{B}$, $\mathcal{E} = \mathcal{U} \cup \mathcal{P}$, $N = |\mathcal{U}|$, $K = |\mathcal{P}|$, $M = N + K$. Episode-fixed roles are Attack, Scout or Disturb.

$$s_t = (\{x_i, h_i, r_i, c_i, \text{id}_i\}_{i \in \mathcal{U}}, \{p_k, h_k, D_k\}_{k \in \mathcal{P}}, t, \zeta_t).$$

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. Participation is $\ell_i = \mathbf{1}[h_i > 0]$. Reset UAV health is 1, target health $h_k^0 = 2$ by default; $v_k = 0$, $D_k = 0$ initially. D_k accumulates raw asset damage.

Configuration C fixes roster, bounds, controls and task. ζ_t contains persistent assignments and opponent memory, if used. Reactive controllers have no hidden integrators. Last command, functional bits, pre-step positions and step damage are overwritten transition records; retain them for exact reward/event replay. RNG state is required for exact replay, not actor input.

B. Six-degree-of-freedom dynamics

World coordinates are east-north-up (ENU); body coordinates are forward-left-up (FLU). The scalar-first quaternion $q = (q_w, q_x, q_y, q_z)$ maps body to world through $R(q) \in SO(3)$, with $q \in S^3$, $\|q\| = 1$ and $q \sim -q$. Thus 13 stored values represent 12 physical degrees of freedom.

$$\begin{aligned} \dot{p} &= v, & \dot{v} &= R(q)F_B/m - ge_3, \\ \dot{q} &= \frac{1}{2}q \otimes (0, \omega), & \dot{\omega} &= J^{-1}(M_B - \omega \times J\omega). \end{aligned}$$

Here $g = 9.81 \text{ m s}^{-2}$, $\omega \in \mathbb{R}^3$ is the body angular velocity in rad/s; F_B excludes gravity. No wind, sensor noise or actuator lag is present.

$$R(q) = I + 2q_w[q_v]_{\times} + 2[q_v]_{\times}^2, \quad q_v = (q_x, q_y, q_z).$$

Here $[z]_{\times}y = z \times y$.

C. Four-rotor wrench and parameters

The rotor order is front, right, rear, left, and normalized thrust fractions are $u_j \in [0, 1]$. With $m = 4.34 \text{ kg}$, $J = \text{diag}(.0820, .0845, .1377) \text{ kg m}^2$, arm $l = .315 \text{ m}$ and yaw ratio $c_\tau = .008004 \text{ m}$ [2],

$$f_{\max} = 2.5mg/4, \quad f_j = f_{\max}u_j,$$

$$\begin{aligned} \begin{bmatrix} f \\ M_x \\ M_y \\ M_z \end{bmatrix} &= A \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}, \\ A &= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -l & 0 & l \\ -l & 0 & l & 0 \\ c_\tau & -c_\tau & c_\tau & -c_\tau \end{bmatrix}. \end{aligned}$$

The rigid-body force is $F_B = (0, 0, f)$ and moment is (M_x, M_y, M_z) . Ideal level hover has $u_j = .4$. The maximum thrust-to-weight ratio 2.5 is a simulation assumption. No aerodynamic/rotor drag, gyroscopic rotor torque or motor time constant is included [3].

Episode geometry and reset law

Default $\lambda = .16$, $\mathcal{W} = [-400, 400]^2 \times [0, 400] \text{ m}$, $d_c = 3.2 \text{ m}$, $(r_0, r_1) = (48, 96) \text{ m}$ and $r_f = r_0$ (configurable $0 < r_f \leq r_0$). Assets are independently uniform in $[-368, -304] \times [-192, 192] \times [80, 240] \text{ m}$ or use a fixed layout. Red samples an area-uniform XY annulus $[128, 352] \text{ m}$ around a uniformly chosen asset, clipped to scene bounds; Blue is uniform over $[0, 400] \times [-400, 400] \text{ m}$. UAV altitude is uniform in $[32, 240] \text{ m}$.

Reset UAVs at $v = 0$, $q = (1, 0, 0, 0)$, $\omega = 0$. Reject UAV openings within $\max(4.8, d_c + 10^{-6}) \text{ m}$ of any previously placed entity; later collisions are allowed. Only 3D is supported.

C. Acceleration intent

Each living agent chooses either $a_i \in [-1, 1]^3$ or an index into

$$\mathcal{V} = \{0\} \cup \{z/\|z\| : z \in \{-1, 0, 1\}^3 \setminus \{0\}\},$$

giving 27 discrete primitives, zero first. The physical intent is $\alpha_i = 6a_i \text{ m/s}^2$. Continuous inputs have component bounds, so diagonal intents can have norm $\sqrt{3}6$; discrete nonzero intents have norm 6. 6 is a control scale, not a clamp on actual rigid-body acceleration. The external intent establishes $v_d = v(t) + 6\Delta a_i$ once. This reference is held for the full external step. It is not recomputed as $v_{\text{stage}} + 6\Delta a_i$ at each stage. The controller below maps the reference to feasible rotor commands; actual acceleration depends on the evolving attitude and rotor limits.

D. Geometric feedback and feasible allocation

At a stage state, clip a held velocity reference to 20 m/s, then compute

$$\begin{aligned} a_d &= \text{sat}_6(2(v_d - v)), & F_d &= m(a_d + ge_3), \\ b_{3d} &= F_d/\|F_d\|, \\ b_{2d} &= (b_{3d} \times e_1)/\|b_{3d} \times e_1\|, \\ R_d &= [b_{2d} \times b_{3d}, b_{2d}, b_{3d}], \\ e_R &= \frac{1}{2}(R_d^T R - R^T R_d)^\vee, \\ M_d &= -8.81e_R - 2.54\omega + \omega \times J\omega, \\ f_d &= F_d^T R e_3. \end{aligned}$$

Fixed world-east heading avoids hidden yaw memory. The 6 m/s² bound is below g , so F_d and its heading construction are nonsingular. This uses attitude PD and gyro compensation, without the source paper's trajectory derivative or desired angular-rate feedforward.

To preserve feasible collective thrust, set $\bar{f} = \text{clip}(f_d, 0, 4f_{\max})/4$ and $d = A^{-1}(0, M_d)^T$. Define

$$\begin{aligned} \kappa_+ &= \min_{d_j > 0} (f_{\max} - \bar{f})/d_j, \\ \kappa_- &= \min_{d_j < 0} (-\bar{f})/d_j, \quad \kappa = \min(1, \kappa_+, \kappa_-). \end{aligned}$$

$$u_j = \text{clip}\left(\frac{\bar{f} + \kappa d_j}{f_{\max}}, 0, 1\right).$$

Empty minima are ignored. Uniform differential scaling sacrifices moment while retaining the clipped collective; independent rotor clipping is not assumed to preserve it.

Numerical transition and boundary

Write $f(x, u)$ for the above derivative, evaluated with normalized q . An external step is $\Delta = 1 \text{ s}$, $n = \lceil \Delta/.01 \rceil = 100$, $h = \Delta/n$. With stage feedback U ,

$$\begin{aligned} k_1 &= f(x, U(x)), \\ k_2 &= f(x + hk_1/2, U(x + hk_1/2)), \\ k_3 &= f(x + hk_2/2, U(x + hk_2/2)), \\ k_4 &= f(x + hk_3, U(x + hk_3)), \\ x^+ &= x + h(k_1 + 2k_2 + 2k_3 + k_4)/6, \\ q^+ &\leftarrow q^+/\|q^+\|. \end{aligned}$$

Repeat n times. A raw actuator is held constant; an intent is held but feedback is recomputed at every RK4 stage. Controllers have no hidden integrators.

For a candidate endpoint p^* , set $p'_a = \text{clip}(p_a^*, b_a^-, b_a^+)$. If $p_a^* > b_a^+$, $v'_a = \min(v_a^*, 0)$; if $p_a^* < b_a^-$, $v'_a = \max(v_a^*, 0)$; otherwise $v'_a = v_a^*$. Collision paths use clipped substep positions. Boundaries are task constraints; physical airspeed and attitude are not clamped to controller reference limits.

Theme 11 / Quadrotor / Acceleration / Survival

Task interaction, exact information structure and defender learning objective

E. Collision and automatic interaction

Predict living UAV paths: particles use $p_i + \tau \Delta v_i$, rigid models use synchronized RK4 endpoints clipped to $\mathcal{W}$. On each matched segment, let $r = p_i - p_j$, $d = (q_i - p_i) - (q_j - p_j)$:

$$\tau_* = \begin{cases} 0 & \|d\|^2 \leq 10^{-12}, \\ \text{clip}(-r^T d / \|d\|^2, 0, 1) & \text{otherwise,} \end{cases} \quad d_* = \|r + \tau_* d\|.$$

Collision sets both UAV HP to zero when $\min d_* \leq d_c$; targets are excluded. Pairs use i -major, $j < i$ order. Dead outer agents are skipped at entry and dead inner agents are skipped. An outer agent killed during its inner sweep continues that sweep. After collision, surviving Attack agents automatically fire at step-start positions:

$$I_i = \begin{cases} \mathbf{1}[\min_{k \in \mathcal{P}: h_k > 0} \|p_i - p_k\| < r_f], & i \in \mathcal{B}, \\ \mathbf{1}[\min_{j \in \mathcal{B}_A: h_j > 0} \|p_i - p_j\| < r_f], & i \in \mathcal{R}. \end{cases}$$

An empty minimum is $+\infty$; dead sources have $I_i = 0$. A denotes Attack. Fire is a separate rule, not a navigation action. With unit attack intensity,

$$\eta(d) = \begin{cases} 1 & d < r_0, \\ (r_1 - d)/(r_1 - r_0) & r_0 \leq d < r_1, \\ 0 & d \geq r_1, \end{cases}$$

$$L_k = \sum_{b \in \mathcal{B}_A: h_b > 0} I_b \eta(\|p_b - p_k\|),$$

$$L_i^{\text{atk}} = \sum_{j \in \mathcal{U}_A: c_j \neq c_i, h_j > 0} I_j \eta(\|p_j - p_i\|).$$

One source can affect multiple enemies; friendly explosions are excluded. All losses use a common post-collision, pre-motion snapshot and add before HP clipping.

Disturb and Scout

For source j , $d_{ji} = \|p_i - p_j\|$ and $\theta_{ji} = \angle(v_j, p_i - p_j)$; angle is zero if either norm is below 10^{-3} . Let

$$G_{ji} = \mathbf{1}[10^{-3} < d_{ji} < 600\lambda, \theta_{ji} < \pi/6] \times e^{-(d_{ji}/\lambda)^2/2} / \sqrt{2\pi},$$

$$S_c(j) = \sum_{i \in \mathcal{U}: c_i = c, h_i > 0} G_{ji}.$$

Living Red Disturb uses $J_j = \mathbf{1}[S_R(j) < S_B(j)]$, Blue uses $J_j = \mathbf{1}[S_R(j) \geq S_B(j)]$. Dead sources contribute zero.

$$L_i^{\text{dis}} = .1 \sum_{j \in \mathcal{U}_D: h_j > 0} J_j G_{ji},$$

$$\tilde{h}_i = \max(0, h_i - L_i^{\text{atk}} - L_i^{\text{dis}}).$$

Set $h'_i = 0$ if $r_i = A, I_i = 1$, otherwise $h'_i = \tilde{h}_i$. Disturb damages either side, not assets. Scouts use $d < 2000\lambda$, $\theta < \pi/3$; this sector does not restrict native actor rows. Collision victims do not move. Combat victims complete their motion; inactive velocity is zeroed next step.

G. Survival payoff and completion

Assets obey $h'_k = \max(0, h_k - L_k)$, $D'_k = D_k + L_k$.

$$z(s') = \begin{cases} -1 & \exists k : h'_k < 10^{-3}, \\ +1 & \forall b \in \mathcal{B}_A : h'_b < 10^{-3}, \\ 0 & \text{otherwise.} \end{cases}$$

Target loss has priority. Natural completion is $d = \mathbf{1}[z \neq 0]$; default Red reward $R_t = 10z$, Blue $-R_t$. Red elimination alone does not end the world. Individual $h'_i \leq 0$ retires a Survival actor after its final transition. Horizon H truncates without a win bonus; use global bootstrap $b_t = 1 - d$, including individual death and truncation. Actor action-loss masks use pre-step participation. Explicit reward weights replace the default payoff as defined on page 1.

F. Exact observations and CTDE information

For each living entity $j \neq i$, the actor row is

$$e_{ij} = [(p_j - p_i)/L, (v_j - v_i)/(2V_{\max}), h_j, 1, \mathbf{1}_{c_j=R}, \mathbf{1}_{c_j=B}, \mathbf{1}_{j \in \mathcal{P}}],$$

$$L = \sqrt{\sum_a (b_a^+ - b_a^-)^2}.$$

Here $V_{\max}$ is the preset maximum speed, not the reference speed. Rows have 11 values, in fixed roster order: Red Scout/Disturb/Attack, Blue Scout/Disturb/Attack, then targets, excluding self. Dead rows are zero. Every living entity is included; Attack, Scout and Disturb share this packing. Side flags do not encode role. Own health is absent from the self block.

The actor receives the $(M-1) \times 11$ entity table ($V_{\max} = 20$) plus unnormalized $[p_i, v_i, q_i, \omega_i] \in \mathbb{R}^{13}$. With deterministic encoding, $Z_i(o | s) = \delta_{\text{encode}_i(s)}(o)$. Other UAV attitudes/rates, opponent memory and time are omitted; hence local information is partial despite noiseless sensing.

Privileged $g(s) = \text{state}()$ packs 11 values per entity,

$$[(p - b^-)/(b^+ - b^-), v/(2V_{\max}), h, \ell, R, B, P],$$

with dead blocks zero. Rigid models append 13 raw state values per UAV (dead zero), then $[t, \sum_k D_k, \mathbf{1}_{\text{Damage}}]$: size $11M + 13N + 3$. Particle size is $11M$, omitting time and damage. g is a CTDE feature, not the complete Markov state.

H. Defender Dec-POMDP and MARL objective

For a specified frozen opposing policy π_B , define

$$\mathcal{M}_R = (\mathcal{R}, \mathcal{S}, \{\mathcal{A}_i\}, T^{\pi_B}, \{\mathcal{O}_i\}, Z, R, \gamma, \rho_0, H),$$

$$T^{\pi_B}(s' | s, a_R) = \int \delta_{F_C(s, a_R, a_B)}(s') \times \pi_B(da_B | o_B, \zeta_t).$$

F_C is the stated collision, simultaneous interaction and controlled-motion map, including any memory evolution; ρ_0 is the reset law. Native execution accepts both teams' actions. Freezing a rule or learned opponent is an experiment choice; with both teams learning, the model is a POSG.

$$\tau_i = (o_{i,0}, a_{i,0}, \dots, o_{i,t}),$$

$$\pi_R(a_R | \tau_R) = \prod_{i \in \mathcal{R}} \pi_i(a_i | \tau_i),$$

$$\pi_R^* \in \arg \max_{\pi_R} \mathbb{E} \left[\sum_{t=0}^{T-1} \gamma^t R_t \right].$$

Expectation is over ρ_0, T^{π_B}, π_R . Here T is natural completion (possibly infinite); H limits sampled rollouts. A truncated return is partial, so its successor value is retained. For $\gamma = 1$, this is respectively the native terminal Survival payoff or negative total asset damage. For $\gamma < 1$, it is the discounted task selected by the caller. Actors use local histories only. CTDE critics can use $g(s)$, joint actions and recorded task/memory features:

$$y_t = R_t + \gamma b_t Q^-(g(s'), a'_R).$$

Here Q^- is a target critic and successor actions follow the target actor policies. Use global completion, not individual death. A recurrent critic or augmented state may be needed because g omits history. This defines an objective; it does not prescribe an implemented optimizer.

Theme 12 / Quadrotor / Acceleration / Damage

Rigid-body thrust dynamics, reference feedback and feasible actuation

A. Scenario and complete state

Red defenders $\mathcal{R}$ protect stationary assets $\mathcal{P}$ against Blue $\mathcal{B}$. Set $\mathcal{U} = \mathcal{R} \cup \mathcal{B}$, $\mathcal{E} = \mathcal{U} \cup \mathcal{P}$, $N = |\mathcal{U}|$, $K = |\mathcal{P}|$, $M = N + K$. Episode-fixed roles are Attack, Scout or Disturb.

$$s_t = (\{x_i, h_i, r_i, c_i, \text{id}_i\}_{i \in \mathcal{U}}, \\ \{p_k, h_k, D_k\}_{k \in \mathcal{P}}, t, \zeta_t).$$

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. Participation is $\ell_i = \mathbf{1}[h_i > 0]$. Reset UAV health is 1, target health $h_k^0 = 2$ by default; $v_k = 0$, $D_k = 0$ initially. D_k accumulates raw asset damage.

Configuration C fixes roster, bounds, controls and task. ζ_t contains persistent assignments and opponent memory, if used. Reactive controllers have no hidden integrators. Last command, functional bits, pre-step positions and step damage are overwritten transition records; retain them for exact reward/event replay. RNG state is required for exact replay, not actor input.

B. Six-degree-of-freedom dynamics

World coordinates are east-north-up (ENU); body coordinates are forward-left-up (FLU). The scalar-first quaternion $q = (q_w, q_x, q_y, q_z)$ maps body to world through $R(q) \in SO(3)$, with $q \in S^3$, $\|q\| = 1$ and $q \sim -q$. Thus 13 stored values represent 12 physical degrees of freedom.

$$\dot{p} = v, \quad \dot{v} = R(q)F_B/m - ge_3, \\ \dot{q} = \frac{1}{2}q \otimes (0, \omega), \quad \dot{\omega} = J^{-1}(M_B - \omega \times J\omega).$$

Here $g = 9.81 \text{ m s}^{-2}$, $\omega \in \mathbb{R}^3$ is the body angular velocity in rad/s; F_B excludes gravity. No wind, sensor noise or actuator lag is present.

$$R(q) = I + 2q_w[q_v]_{\times} + 2[q_v]_{\times}^2, \quad q_v = (q_x, q_y, q_z).$$

Here $[z]_{\times}y = z \times y$.

C. Four-rotor wrench and parameters

The rotor order is front, right, rear, left, and normalized thrust fractions are $u_j \in [0, 1]$. With $m = 4.34 \text{ kg}$, $J = \text{diag}(.0820, .0845, .1377) \text{ kg m}^2$, arm $l = .315 \text{ m}$ and yaw ratio $c_\tau = .008004 \text{ m [2]}$,

$$f_{\max} = 2.5mg/4, \quad f_j = f_{\max}u_j,$$

$$\begin{bmatrix} f \\ M_x \\ M_y \\ M_z \end{bmatrix} = A \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}, \\ A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -l & 0 & l \\ -l & 0 & l & 0 \\ c_\tau & -c_\tau & c_\tau & -c_\tau \end{bmatrix}.$$

The rigid-body force is $F_B = (0, 0, f)$ and moment is (M_x, M_y, M_z) . Ideal level hover has $u_j = .4$. The maximum thrust-to-weight ratio 2.5 is a simulation assumption. No aerodynamic/rotor drag, gyroscopic rotor torque or motor time constant is included [3].

Episode geometry and reset law

Default $\lambda = .16$, $\mathcal{W} = [-400, 400]^2 \times [0, 400] \text{ m}$, $d_c = 3.2 \text{ m}$, $(r_0, r_1) = (48, 96) \text{ m}$ and $r_f = r_0$ (configurable $0 < r_f \leq r_0$). Assets are independently uniform in $[-368, -304] \times [-192, 192] \times [80, 240] \text{ m}$ or use a fixed layout. Red samples an area-uniform XY annulus $[128, 352] \text{ m}$ around a uniformly chosen asset, clipped to scene bounds; Blue is uniform over $[0, 400] \times [-400, 400] \text{ m}$. UAV altitude is uniform in $[32, 240] \text{ m}$.

Reset UAVs at $v = 0$, $q = (1, 0, 0, 0)$, $\omega = 0$. Reject UAV openings within $\max(4.8, d_c + 10^{-6}) \text{ m}$ of any previously placed entity; later collisions are allowed. Only 3D is supported.

C. Acceleration intent

Each living agent chooses either $a_i \in [-1, 1]^3$ or an index into

$$\mathcal{V} = \{0\} \cup \{z/\|z\| : z \in \{-1, 0, 1\}^3 \setminus \{0\}\},$$

giving 27 discrete primitives, zero first. The physical intent is $\alpha_i = 6a_i \text{ m/s}^2$. Continuous inputs have component bounds, so diagonal intents can have norm $\sqrt{3}6$; discrete nonzero intents have norm 6. 6 is a control scale, not a clamp on actual rigid-body acceleration. The external intent establishes $v_d = v(t) + 6\Delta a_i$ once. This reference is held for the full external step. It is not recomputed as $v_{\text{stage}} + 6\Delta a_i$ at each stage. The controller below maps the reference to feasible rotor commands; actual acceleration depends on the evolving attitude and rotor limits.

D. Geometric feedback and feasible allocation

At a stage state, clip a held velocity reference to 20 m/s, then compute

$$a_d = \text{sat}_6(2(v_d - v)), \quad F_d = m(a_d + ge_3), \\ b_{3d} = F_d/\|F_d\|, \\ b_{2d} = (b_{3d} \times e_1)/\|b_{3d} \times e_1\|, \\ R_d = [b_{2d} \times b_{3d}, b_{2d}, b_{3d}], \\ e_R = \frac{1}{2}(R_d^T R - R^T R_d)^\vee, \\ M_d = -8.81e_R - 2.54\omega + \omega \times J\omega, \\ f_d = F_d^T R e_3.$$

Fixed world-east heading avoids hidden yaw memory. The 6 m/s² bound is below g , so F_d and its heading construction are nonsingular. This uses attitude PD and gyro compensation, without the source paper's trajectory derivative or desired angular-rate feedforward.

To preserve feasible collective thrust, set $\bar{f} = \text{clip}(f_d, 0, 4f_{\max})/4$ and $d = A^{-1}(0, M_d)^T$. Define

$$\kappa_+ = \min_{d_j > 0} (f_{\max} - \bar{f})/d_j, \\ \kappa_- = \min_{d_j < 0} (-\bar{f})/d_j, \quad \kappa = \min(1, \kappa_+, \kappa_-).$$

$$u_j = \text{clip}\left(\frac{\bar{f} + \kappa d_j}{f_{\max}}, 0, 1\right).$$

Empty minima are ignored. Uniform differential scaling sacrifices moment while retaining the clipped collective; independent rotor clipping is not assumed to preserve it.

Numerical transition and boundary

Write $f(x, u)$ for the above derivative, evaluated with normalized q . An external step is $\Delta = 1 \text{ s}$, $n = \lceil \Delta/.01 \rceil = 100$, $h = \Delta/n$. With stage feedback U ,

$$k_1 = f(x, U(x)), \\ k_2 = f(x + hk_1/2, U(x + hk_1/2)), \\ k_3 = f(x + hk_2/2, U(x + hk_2/2)), \\ k_4 = f(x + hk_3, U(x + hk_3)), \\ x^+ = x + h(k_1 + 2k_2 + 2k_3 + k_4)/6, \\ q^+ \leftarrow q^+/\|q^+\|.$$

Repeat n times. A raw actuator is held constant; an intent is held but feedback is recomputed at every RK4 stage. Controllers have no hidden integrators.

For a candidate endpoint p^* , set $p'_a = \text{clip}(p_a^*, b_a^-, b_a^+)$. If $p_a^* > b_a^+$, $v'_a = \min(v_a^*, 0)$; if $p_a^* < b_a^-$, $v'_a = \max(v_a^*, 0)$; otherwise $v'_a = v_a^*$. Collision paths use clipped substep positions. Boundaries are task constraints; physical airspeed and attitude are not clamped to controller reference limits.

Theme 12 / Quadrotor / Acceleration / Damage

Task interaction, exact information structure and defender learning objective

E. Collision and automatic interaction

Predict living UAV paths: particles use $p_i + \tau \Delta v_i$, rigid models use synchronized RK4 endpoints clipped to $\mathcal{W}$. On each matched segment, let $r = p_i - p_j$, $d = (q_i - p_i) - (q_j - p_j)$:

$$\tau_* = \begin{cases} 0 & \|d\|^2 \leq 10^{-12}, \\ \text{clip}(-r^T d / \|d\|^2, 0, 1) & \text{otherwise,} \end{cases} \quad d_* = \|r + \tau_* d\|.$$

Collision sets both UAV HP to zero when $\min d_* \leq d_c$; targets are excluded. Pairs use i -major, $j < i$ order. Dead outer agents are skipped at entry and dead inner agents are skipped. An outer agent killed during its inner sweep continues that sweep. After collision, surviving Attack agents automatically fire at step-start positions:

$$I_i = \begin{cases} \mathbf{1}[\min_{k \in \mathcal{P}: h_k > 0} \|p_i - p_k\| < r_f], & i \in \mathcal{B}, \\ \mathbf{1}[\min_{j \in \mathcal{B}_A: h_j > 0} \|p_i - p_j\| < r_f], & i \in \mathcal{R}. \end{cases}$$

An empty minimum is $+\infty$; dead sources have $I_i = 0$. A denotes Attack. Fire is a separate rule, not a navigation action. With unit attack intensity,

$$\eta(d) = \begin{cases} 1 & d < r_0, \\ (r_1 - d)/(r_1 - r_0) & r_0 \leq d < r_1, \\ 0 & d \geq r_1, \end{cases}$$

$$L_k = \sum_{b \in \mathcal{B}_A: h_b > 0} I_b \eta(\|p_b - p_k\|),$$

$$L_i^{\text{atk}} = \sum_{j \in \mathcal{U}_A: c_j \neq c_i, h_j > 0} I_j \eta(\|p_j - p_i\|).$$

One source can affect multiple enemies; friendly explosions are excluded. All losses use a common post-collision, pre-motion snapshot and add before HP clipping.

Disturb and Scout

For source j , $d_{ji} = \|p_i - p_j\|$ and $\theta_{ji} = \angle(v_j, p_i - p_j)$; angle is zero if either norm is below 10^{-3} . Let

$$G_{ji} = \mathbf{1}[10^{-3} < d_{ji} < 600\lambda, \theta_{ji} < \pi/6] \\ \times e^{-(d_{ji}/\lambda)^2/2} / \sqrt{2\pi},$$

$$S_c(j) = \sum_{i \in \mathcal{U}: c_i = c, h_i > 0} G_{ji}.$$

Living Red Disturb uses $J_j = \mathbf{1}[S_R(j) < S_B(j)]$, Blue uses $J_j = \mathbf{1}[S_R(j) \geq S_B(j)]$. Dead sources contribute zero.

$$L_i^{\text{dis}} = .1 \sum_{j \in \mathcal{U}_D: h_j > 0} J_j G_{ji},$$

$$\tilde{h}_i = \max(0, h_i - L_i^{\text{atk}} - L_i^{\text{dis}}).$$

Set $h'_i = 0$ if $r_i = A, I_i = 1$, otherwise $h'_i = \tilde{h}_i$. Disturb damages either side, not assets. Scouts use $d < 2000\lambda$, $\theta < \pi/3$; this sector does not restrict native actor rows. Collision victims do not move. Combat victims complete their motion; inactive velocity is zeroed next step.

G. Damage payoff and completion

Assets keep finite reset health $h'_k = h_k^0$. Count raw contributions before HP clipping, including simultaneous overkill:

$$D'_k = D_k + L_k, \quad d_t = \sum_k L_k, \quad R_t = -d_t.$$

Blue reward is d_t ; there is no team-size, asset-count or HP normalization and no terminal Survival bonus.

$$d = \mathbf{1}[\forall b \in \mathcal{B}_A : h'_b \leq 0].$$

All agent slots remain until global completion/truncation. Dead slots have zero observations, ignored actions and still receive full team R_t , including later damage after Red extinction. Do not sum duplicate teammate rewards. Undiscounted returns are $(-\sum_k D_k, +\sum_k D_k)$. H only truncates; global bootstrap $b_t = 1 - d$ remains one at truncation. Actor action-loss masks use pre-step participation.

F. Exact observations and CTDE information

For each living entity $j \neq i$, the actor row is

$$e_{ij} = [(p_j - p_i)/L, (v_j - v_i)/(2V_{\max}), \\ h_j, 1, \mathbf{1}_{c_j=R}, \mathbf{1}_{c_j=B}, \mathbf{1}_{j \in \mathcal{P}}], \\ L = \sqrt{\sum_a (b_a^+ - b_a^-)^2}.$$

Here $V_{\max}$ is the preset maximum speed, not the reference speed. Rows have 11 values, in fixed roster order: Red Scout/Disturb/Attack, Blue Scout/Disturb/Attack, then targets, excluding self. Dead rows are zero. Every living entity is included; Attack, Scout and Disturb share this packing. Side flags do not encode role. Own health is absent from the self block.

The actor receives the $(M - 1) \times 11$ entity table ($V_{\max} = 20$) plus unnormalized $[p_i, v_i, q_i, \omega_i]$. Both blocks are zero for an already-dead Damage observer. With deterministic encoding, $Z_i(o | s) = \delta_{\text{encode}_i(s)}(o)$. Other UAV attitudes/rates, opponent memory and time are omitted; hence local information is partial despite noiseless sensing.

Privileged $g(s) = \text{state}()$ packs 11 values per entity,

$$[(p - b^-)/(b^+ - b^-), v/(2V_{\max}), h, \ell, R, B, P],$$

with dead blocks zero. Rigid models append 13 raw state values per UAV (dead zero), then $[t, \sum_k D_k, \mathbf{1}_{\text{Damage}}]$: size $11M + 13N + 3$. Particle size is $11M$, omitting time and damage. g is a CTDE feature, not the complete Markov state.

H. Defender Dec-POMDP and MARL objective

For a specified frozen opposing policy π_B , define

$$\mathcal{M}_R = (\mathcal{R}, \mathcal{S}, \{\mathcal{A}_i\}, T^{\pi_B}, \\ \{\mathcal{O}_i\}, Z, R, \gamma, \rho_0, H),$$

$$T^{\pi_B}(s' | s, a_R) = \int \delta_{F_C(s, a_R, a_B)}(s') \\ \times \pi_B(da_B | o_B, \zeta_t).$$

F_C is the stated collision, simultaneous interaction and controlled-motion map, including any memory evolution; ρ_0 is the reset law. Native execution accepts both teams' actions. Freezing a rule or learned opponent is an experiment choice; with both teams learning, the model is a POSG.

$$\tau_i = (o_{i,0}, a_{i,0}, \dots, o_{i,t}),$$

$$\pi_R(a_R | \tau_R) = \prod_{i \in \mathcal{R}} \pi_i(a_i | \tau_i),$$

$$\pi_R^* \in \arg \max_{\pi_R} \mathbb{E} \left[\sum_{t=0}^{T-1} \gamma^t R_t \right].$$

Expectation is over ρ_0, T^{π_B}, π_R . Here T is natural completion (possibly infinite); H limits sampled rollouts. A truncated return is partial, so its successor value is retained. For $\gamma = 1$, this is respectively the native terminal Survival payoff or negative total asset damage. For $\gamma < 1$, it is the discounted task selected by the caller. Actors use local histories only. CTDE critics can use $g(s)$, joint actions and recorded task/memory features:

$$y_t = R_t + \gamma b_t Q^-(g(s'), a'_R).$$

Here Q^- is a target critic and successor actions follow the target actor policies. Use global completion, not individual death. A recurrent critic or augmented state may be needed because g omits history. This defines an objective; it does not prescribe an implemented optimizer.

Theme 13 / Quadrotor / Actuator / Survival

Rigid-body thrust dynamics, reference feedback and feasible actuation

A. Scenario and complete state

Red defenders $\mathcal{R}$ protect stationary assets $\mathcal{P}$ against Blue $\mathcal{B}$. Set $\mathcal{U} = \mathcal{R} \cup \mathcal{B}$, $\mathcal{E} = \mathcal{U} \cup \mathcal{P}$, $N = |\mathcal{U}|$, $K = |\mathcal{P}|$, $M = N + K$. Episode-fixed roles are Attack, Scout or Disturb.

$$s_t = (\{x_i, h_i, r_i, c_i, \text{id}_i\}_{i \in \mathcal{U}}, \{p_k, h_k, D_k\}_{k \in \mathcal{P}}, t, \zeta_t).$$

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. Participation is $\ell_i = \mathbf{1}[h_i > 0]$. Reset UAV health is 1, target health $h_k^0 = 2$ by default; $v_k = 0$, $D_k = 0$ initially. D_k accumulates raw asset damage.

Configuration C fixes roster, bounds, controls and task. ζ_t contains persistent assignments and opponent memory, if used. Reactive controllers have no hidden integrators. Last command, functional bits, pre-step positions and step damage are overwritten transition records; retain them for exact reward/event replay. RNG state is required for exact replay, not actor input.

B. Six-degree-of-freedom dynamics

World coordinates are east-north-up (ENU); body coordinates are forward-left-up (FLU). The scalar-first quaternion $q = (q_w, q_x, q_y, q_z)$ maps body to world through $R(q) \in SO(3)$, with $q \in S^3$, $\|q\| = 1$ and $q \sim -q$. Thus 13 stored values represent 12 physical degrees of freedom.

$$\begin{aligned} \dot{p} &= v, & \dot{v} &= R(q)F_B/m - ge_3, \\ \dot{q} &= \frac{1}{2}q \otimes (0, \omega), & \dot{\omega} &= J^{-1}(M_B - \omega \times J\omega). \end{aligned}$$

Here $g = 9.81 \text{ m s}^{-2}$, $\omega \in \mathbb{R}^3$ is the body angular velocity in rad/s; F_B excludes gravity. No wind, sensor noise or actuator lag is present.

$$R(q) = I + 2q_w[q_v]_{\times} + 2[q_v]_{\times}^2, \quad q_v = (q_x, q_y, q_z).$$

Here $[z]_{\times}y = z \times y$.

C. Four-rotor wrench and parameters

The rotor order is front, right, rear, left, and normalized thrust fractions are $u_j \in [0, 1]$. With $m = 4.34 \text{ kg}$, $J = \text{diag}(.0820, .0845, .1377) \text{ kg m}^2$, arm $l = .315 \text{ m}$ and yaw ratio $c_\tau = .008004 \text{ m}$ [2],

$$f_{\max} = 2.5mg/4, \quad f_j = f_{\max}u_j,$$

$$\begin{aligned} \begin{bmatrix} f \\ M_x \\ M_y \\ M_z \end{bmatrix} &= A \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}, \\ A &= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -l & 0 & l \\ -l & 0 & l & 0 \\ c_\tau & -c_\tau & c_\tau & -c_\tau \end{bmatrix}. \end{aligned}$$

The rigid-body force is $F_B = (0, 0, f)$ and moment is (M_x, M_y, M_z) . Ideal level hover has $u_j = .4$. The maximum thrust-to-weight ratio 2.5 is a simulation assumption. No aerodynamic/rotor drag, gyroscopic rotor torque or motor time constant is included [3].

Episode geometry and reset law

Default $\lambda = .16$, $\mathcal{W} = [-400, 400]^2 \times [0, 400] \text{ m}$, $d_c = 3.2 \text{ m}$, $(r_0, r_1) = (48, 96) \text{ m}$ and $r_f = r_0$ (configurable $0 < r_f \leq r_0$). Assets are independently uniform in $[-368, -304] \times [-192, 192] \times [80, 240] \text{ m}$ or use a fixed layout. Red samples an area-uniform XY annulus $[128, 352] \text{ m}$ around a uniformly chosen asset, clipped to scene bounds; Blue is uniform over $[0, 400] \times [-400, 400] \text{ m}$. UAV altitude is uniform in $[32, 240] \text{ m}$.

Reset UAVs at $v = 0$, $q = (1, 0, 0, 0)$, $\omega = 0$. Reject UAV openings within $\max(4.8, d_c + 10^{-6}) \text{ m}$ of any previously placed entity; later collisions are allowed. Only 3D is supported.

Direct rotor-thrust action

Each living agent selects $a_i \in [0, 1]^4$, ordered front, right, rear, left, and $u = a_i$ is held throughout the external step. The wrench map enters Newton-Euler dynamics directly, without automatic attitude or position stabilization. Zero thrust is free fall, not hover; .4 on every rotor is ideal level hover.

Numerical transition and boundary

Write $f(x, u)$ for the above derivative, evaluated with normalized q . An external step is $\Delta = 1 \text{ s}$, $n = \lceil \Delta/.01 \rceil = 100$, $h = \Delta/n$. With stage feedback U ,

$$\begin{aligned} k_1 &= f(x, U(x)), \\ k_2 &= f(x + hk_1/2, U(x + hk_1/2)), \\ k_3 &= f(x + hk_2/2, U(x + hk_2/2)), \\ k_4 &= f(x + hk_3, U(x + hk_3)), \\ x^+ &= x + h(k_1 + 2k_2 + 2k_3 + k_4)/6, \\ q^+ &\leftarrow q^+/\|q^+\|. \end{aligned}$$

Repeat n times. A raw actuator is held constant; an intent is held but feedback is recomputed at every RK4 stage. Controllers have no hidden integrators.

For a candidate endpoint p^* , set $p'_a = \text{clip}(p_a^*, b_a^-, b_a^+)$. If $p_a^* > b_a^+$, $v'_a = \min(v_a^*, 0)$; if $p_a^* < b_a^-$, $v'_a = \max(v_a^*, 0)$; otherwise $v'_a = v_a^*$. Collision paths use clipped substep positions. Boundaries are task constraints; physical airspeed and attitude are not clamped to controller reference limits.

Decision and task interpretation

The agent must learn a rotor-level feedback policy from its local information. Identical rotor actions can generate different world accelerations because thrust follows the current body + z axis. Attitude and angular velocity are therefore indispensable components of the physical state and are explicitly supplied in the actor's self block.

The default world is $[-400, 400]^2 \times [0, 400] \text{ m}$. Protected assets have positive finite health in both tasks; only the Survival task decrements asset health. A saturated or destabilizing action is propagated by the rigid-body equations, rather than repaired by a reference controller. Low altitude does not itself cause a crash: scene boundaries project position and remove outward velocity.

Theme 13 / Quadrotor / Actuator / Survival

Task interaction, exact information structure and defender learning objective

E. Collision and automatic interaction

Predict living UAV paths: particles use $p_i + \tau \Delta v_i$, rigid models use synchronized RK4 endpoints clipped to $\mathcal{W}$. On each matched segment, let $r = p_i - p_j$, $d = (q_i - p_i) - (q_j - p_j)$:

$$\tau_* = \begin{cases} 0 & \|d\|^2 \leq 10^{-12}, \\ \text{clip}(-r^T d / \|d\|^2, 0, 1) & \text{otherwise,} \end{cases} \quad d_* = \|r + \tau_* d\|.$$

Collision sets both UAV HP to zero when $\min d_* \leq d_c$; targets are excluded. Pairs use i -major, $j < i$ order. Dead outer agents are skipped at entry and dead inner agents are skipped. An outer agent killed during its inner sweep continues that sweep. After collision, surviving Attack agents automatically fire at step-start positions:

$$I_i = \begin{cases} \mathbf{1}[\min_{k \in \mathcal{P}: h_k > 0} \|p_i - p_k\| < r_f], & i \in \mathcal{B}, \\ \mathbf{1}[\min_{j \in \mathcal{B}_A: h_j > 0} \|p_i - p_j\| < r_f], & i \in \mathcal{R}. \end{cases}$$

An empty minimum is $+\infty$; dead sources have $I_i = 0$. A denotes Attack. Fire is a separate rule, not a navigation action. With unit attack intensity,

$$\eta(d) = \begin{cases} 1 & d < r_0, \\ (r_1 - d)/(r_1 - r_0) & r_0 \leq d < r_1, \\ 0 & d \geq r_1, \end{cases}$$

$$L_k = \sum_{b \in \mathcal{B}_A: h_b > 0} I_b \eta(\|p_b - p_k\|),$$

$$L_i^{\text{atk}} = \sum_{j \in \mathcal{U}_A: c_j \neq c_i, h_j > 0} I_j \eta(\|p_j - p_i\|).$$

One source can affect multiple enemies; friendly explosions are excluded. All losses use a common post-collision, pre-motion snapshot and add before HP clipping.

Disturb and Scout

For source j , $d_{ji} = \|p_i - p_j\|$ and $\theta_{ji} = \angle(v_j, p_i - p_j)$; angle is zero if either norm is below 10^{-3} . Let

$$G_{ji} = \mathbf{1}[10^{-3} < d_{ji} < 600\lambda, \theta_{ji} < \pi/6] \\ \times e^{-(d_{ji}/\lambda)^2/2} / \sqrt{2\pi},$$

$$S_c(j) = \sum_{i \in \mathcal{U}: c_i = c, h_i > 0} G_{ji}.$$

Living Red Disturb uses $J_j = \mathbf{1}[S_R(j) < S_B(j)]$, Blue uses $J_j = \mathbf{1}[S_R(j) \geq S_B(j)]$. Dead sources contribute zero.

$$L_i^{\text{dis}} = .1 \sum_{j \in \mathcal{U}_D: h_j > 0} J_j G_{ji},$$

$$\tilde{h}_i = \max(0, h_i - L_i^{\text{atk}} - L_i^{\text{dis}}).$$

Set $h'_i = 0$ if $r_i = A, I_i = 1$, otherwise $h'_i = \tilde{h}_i$. Disturb damages either side, not assets. Scouts use $d < 2000\lambda$, $\theta < \pi/3$; this sector does not restrict native actor rows. Collision victims do not move. Combat victims complete their motion; inactive velocity is zeroed next step.

G. Survival payoff and completion

Assets obey $h'_k = \max(0, h_k - L_k)$, $D'_k = D_k + L_k$.

$$z(s') = \begin{cases} -1 & \exists k : h'_k < 10^{-3}, \\ +1 & \forall b \in \mathcal{B}_A : h'_b < 10^{-3}, \\ 0 & \text{otherwise.} \end{cases}$$

Target loss has priority. Natural completion is $d = \mathbf{1}[z \neq 0]$; default Red reward $R_t = 10z$, Blue $-R_t$. Red elimination alone does not end the world. Individual $h'_i \leq 0$ retires a Survival actor after its final transition. Horizon H truncates without a win bonus; use global bootstrap $b_t = 1 - d$, including individual death and truncation. Actor action-loss masks use pre-step participation. Explicit reward weights replace the default payoff as defined on page 1.

F. Exact observations and CTDE information

For each living entity $j \neq i$, the actor row is

$$e_{ij} = [(p_j - p_i)/L, (v_j - v_i)/(2V_{\max}), \\ h_j, 1, \mathbf{1}_{c_j=R}, \mathbf{1}_{c_j=B}, \mathbf{1}_{j \in \mathcal{P}}],$$

$$L = \sqrt{\sum_a (b_a^+ - b_a^-)^2}.$$

Here $V_{\max}$ is the preset maximum speed, not the reference speed. Rows have 11 values, in fixed roster order: Red Scout/Disturb/Attack, Blue Scout/Disturb/Attack, then targets, excluding self. Dead rows are zero. Every living entity is included; Attack, Scout and Disturb share this packing. Side flags do not encode role. Own health is absent from the self block.

The actor receives the $(M-1) \times 11$ entity table ($V_{\max} = 20$) plus unnormalized $[p_i, v_i, q_i, \omega_i] \in \mathbb{R}^{13}$. Rotor controls do not add an observation block. With deterministic encoding, $Z_i(o | s) = \delta_{\text{encode}_i(s)}(o)$. Other UAV attitudes/rates, opponent memory and time are omitted; hence local information is partial despite noiseless sensing.

Privileged $g(s) = \text{state}()$ packs 11 values per entity,

$$[(p - b^-)/(b^+ - b^-), v/(2V_{\max}), h, \ell, R, B, P],$$

with dead blocks zero. Rigid models append 13 raw state values per UAV (dead zero), then $[t, \sum_k D_k, \mathbf{1}_{\text{Damage}}]$: size $11M + 13N + 3$. Particle size is $11M$, omitting time and damage. g is a CTDE feature, not the complete Markov state.

H. Defender Dec-POMDP and MARL objective

For a specified frozen opposing policy π_B , define

$$\mathcal{M}_R = (\mathcal{R}, \mathcal{S}, \{\mathcal{A}_i\}, T^{\pi_B}, \\ \{\mathcal{O}_i\}, Z, R, \gamma, \rho_0, H),$$

$$T^{\pi_B}(s' | s, a_R) = \int \delta_{F_C(s, a_R, a_B)}(s') \\ \times \pi_B(da_B | o_B, \zeta_t).$$

F_C is the stated collision, simultaneous interaction and controlled-motion map, including any memory evolution; ρ_0 is the reset law. Native execution accepts both teams' actions. Freezing a rule or learned opponent is an experiment choice; with both teams learning, the model is a POSG.

$$\tau_i = (o_{i,0}, a_{i,0}, \dots, o_{i,t}),$$

$$\pi_R(a_R | \tau_R) = \prod_{i \in \mathcal{R}} \pi_i(a_i | \tau_i),$$

$$\pi_R^* \in \arg \max_{\pi_R} \mathbb{E} \left[\sum_{t=0}^{T-1} \gamma^t R_t \right].$$

Expectation is over ρ_0, T^{π_B}, π_R . Here T is natural completion (possibly infinite); H limits sampled rollouts. A truncated return is partial, so its successor value is retained. For $\gamma = 1$, this is respectively the native terminal Survival payoff or negative total asset damage. For $\gamma < 1$, it is the discounted task selected by the caller. Actors use local histories only. CTDE critics can use $g(s)$, joint actions and recorded task/memory features:

$$y_t = R_t + \gamma b_t Q^-(g(s'), a'_R).$$

Here Q^- is a target critic and successor actions follow the target actor policies. Use global completion, not individual death. A recurrent critic or augmented state may be needed because g omits history. This defines an objective; it does not prescribe an implemented optimizer.

Theme 14 / Quadrotor / Actuator / Damage

Rigid-body thrust dynamics, reference feedback and feasible actuation

A. Scenario and complete state

Red defenders $\mathcal{R}$ protect stationary assets $\mathcal{P}$ against Blue $\mathcal{B}$. Set $\mathcal{U} = \mathcal{R} \cup \mathcal{B}$, $\mathcal{E} = \mathcal{U} \cup \mathcal{P}$, $N = |\mathcal{U}|$, $K = |\mathcal{P}|$, $M = N + K$. Episode-fixed roles are Attack, Scout or Disturb.

$$s_t = (\{x_i, h_i, r_i, c_i, \text{id}_i\}_{i \in \mathcal{U}}, \{p_k, h_k, D_k\}_{k \in \mathcal{P}}, t, \zeta_t).$$

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. Participation is $\ell_i = \mathbf{1}[h_i > 0]$. Reset UAV health is 1, target health $h_k^0 = 2$ by default; $v_k = 0$, $D_k = 0$ initially. D_k accumulates raw asset damage.

Configuration C fixes roster, bounds, controls and task. ζ_t contains persistent assignments and opponent memory, if used. Reactive controllers have no hidden integrators. Last command, functional bits, pre-step positions and step damage are overwritten transition records; retain them for exact reward/event replay. RNG state is required for exact replay, not actor input.

B. Six-degree-of-freedom dynamics

World coordinates are east-north-up (ENU); body coordinates are forward-left-up (FLU). The scalar-first quaternion $q = (q_w, q_x, q_y, q_z)$ maps body to world through $R(q) \in SO(3)$, with $q \in S^3$, $\|q\| = 1$ and $q \sim -q$. Thus 13 stored values represent 12 physical degrees of freedom.

$$\begin{aligned} \dot{p} &= v, & \dot{v} &= R(q)F_B/m - ge_3, \\ \dot{q} &= \frac{1}{2}q \otimes (0, \omega), & \dot{\omega} &= J^{-1}(M_B - \omega \times J\omega). \end{aligned}$$

Here $g = 9.81 \text{ m s}^{-2}$, $\omega \in \mathbb{R}^3$ is the body angular velocity in rad/s; F_B excludes gravity. No wind, sensor noise or actuator lag is present.

$$R(q) = I + 2q_w[q_v]_{\times} + 2[q_v]_{\times}^2, \quad q_v = (q_x, q_y, q_z).$$

Here $[z]_{\times} y = z \times y$.

C. Four-rotor wrench and parameters

The rotor order is front, right, rear, left, and normalized thrust fractions are $u_j \in [0, 1]$. With $m = 4.34 \text{ kg}$, $J = \text{diag}(.0820, .0845, .1377) \text{ kg m}^2$, arm $l = .315 \text{ m}$ and yaw ratio $c_\tau = .008004 \text{ m}$ [2],

$$f_{\max} = 2.5mg/4, \quad f_j = f_{\max}u_j,$$

$$\begin{aligned} \begin{bmatrix} f \\ M_x \\ M_y \\ M_z \end{bmatrix} &= A \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}, \\ A &= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -l & 0 & l \\ -l & 0 & l & 0 \\ c_\tau & -c_\tau & c_\tau & -c_\tau \end{bmatrix}. \end{aligned}$$

The rigid-body force is $F_B = (0, 0, f)$ and moment is (M_x, M_y, M_z) . Ideal level hover has $u_j = .4$. The maximum thrust-to-weight ratio 2.5 is a simulation assumption. No aerodynamic/rotor drag, gyroscopic rotor torque or motor time constant is included [3].

Episode geometry and reset law

Default $\lambda = .16$, $\mathcal{W} = [-400, 400]^2 \times [0, 400] \text{ m}$, $d_c = 3.2 \text{ m}$, $(r_0, r_1) = (48, 96) \text{ m}$ and $r_f = r_0$ (configurable $0 < r_f \leq r_0$). Assets are independently uniform in $[-368, -304] \times [-192, 192] \times [80, 240] \text{ m}$ or use a fixed layout. Red samples an area-uniform XY annulus $[128, 352] \text{ m}$ around a uniformly chosen asset, clipped to scene bounds; Blue is uniform over $[0, 400] \times [-400, 400] \text{ m}$. UAV altitude is uniform in $[32, 240] \text{ m}$.

Reset UAVs at $v = 0$, $q = (1, 0, 0, 0)$, $\omega = 0$. Reject UAV openings within $\max(4.8, d_c + 10^{-6}) \text{ m}$ of any previously placed entity; later collisions are allowed. Only 3D is supported.

Direct rotor-thrust action

Each living agent selects $a_i \in [0, 1]^4$, ordered front, right, rear, left, and $u = a_i$ is held throughout the external step. The wrench map enters Newton-Euler dynamics directly, without automatic attitude or position stabilization. Zero thrust is free fall, not hover; .4 on every rotor is ideal level hover.

Numerical transition and boundary

Write $f(x, u)$ for the above derivative, evaluated with normalized q . An external step is $\Delta = 1 \text{ s}$, $n = \lceil \Delta/.01 \rceil = 100$, $h = \Delta/n$. With stage feedback U ,

$$\begin{aligned} k_1 &= f(x, U(x)), \\ k_2 &= f(x + hk_1/2, U(x + hk_1/2)), \\ k_3 &= f(x + hk_2/2, U(x + hk_2/2)), \\ k_4 &= f(x + hk_3, U(x + hk_3)), \\ x^+ &= x + h(k_1 + 2k_2 + 2k_3 + k_4)/6, \\ q^+ &\leftarrow q^+/\|q^+\|. \end{aligned}$$

Repeat n times. A raw actuator is held constant; an intent is held but feedback is recomputed at every RK4 stage. Controllers have no hidden integrators.

For a candidate endpoint p^* , set $p'_a = \text{clip}(p_a^*, b_a^-, b_a^+)$. If $p_a^* > b_a^+$, $v'_a = \min(v_a^*, 0)$; if $p_a^* < b_a^-$, $v'_a = \max(v_a^*, 0)$; otherwise $v'_a = v_a^*$. Collision paths use clipped substep positions. Boundaries are task constraints; physical airspeed and attitude are not clamped to controller reference limits.

Decision and task interpretation

The agent must learn a rotor-level feedback policy from its local information. Identical rotor actions can generate different world accelerations because thrust follows the current body + z axis. Attitude and angular velocity are therefore indispensable components of the physical state and are explicitly supplied in the actor's self block.

The default world is $[-400, 400]^2 \times [0, 400] \text{ m}$. Protected assets have positive finite health in both tasks; only the Survival task decrements asset health. A saturated or destabilizing action is propagated by the rigid-body equations, rather than repaired by a reference controller. Low altitude does not itself cause a crash: scene boundaries project position and remove outward velocity.

Theme 14 / Quadrotor / Actuator / Damage

Task interaction, exact information structure and defender learning objective

E. Collision and automatic interaction

Predict living UAV paths: particles use $p_i + \tau \Delta v_i$, rigid models use synchronized RK4 endpoints clipped to $\mathcal{W}$. On each matched segment, let $r = p_i - p_j$, $d = (q_i - p_i) - (q_j - p_j)$:

$$\tau_* = \begin{cases} 0 & \|d\|^2 \leq 10^{-12}, \\ \text{clip}(-r^T d / \|d\|^2, 0, 1) & \text{otherwise,} \end{cases} \quad d_* = \|r + \tau_* d\|.$$

Collision sets both UAV HP to zero when $\min d_* \leq d_c$; targets are excluded. Pairs use i -major, $j < i$ order. Dead outer agents are skipped at entry and dead inner agents are skipped. An outer agent killed during its inner sweep continues that sweep. After collision, surviving Attack agents automatically fire at step-start positions:

$$I_i = \begin{cases} \mathbf{1}[\min_{k \in \mathcal{P}: h_k > 0} \|p_i - p_k\| < r_f], & i \in \mathcal{B}, \\ \mathbf{1}[\min_{j \in \mathcal{B}_A: h_j > 0} \|p_i - p_j\| < r_f], & i \in \mathcal{R}. \end{cases}$$

An empty minimum is $+\infty$; dead sources have $I_i = 0$. A denotes Attack. Fire is a separate rule, not a navigation action. With unit attack intensity,

$$\eta(d) = \begin{cases} 1 & d < r_0, \\ (r_1 - d)/(r_1 - r_0) & r_0 \leq d < r_1, \\ 0 & d \geq r_1, \end{cases}$$

$$L_k = \sum_{b \in \mathcal{B}_A: h_b > 0} I_b \eta(\|p_b - p_k\|),$$

$$L_i^{\text{atk}} = \sum_{j \in \mathcal{U}_A: c_j \neq c_i, h_j > 0} I_j \eta(\|p_j - p_i\|).$$

One source can affect multiple enemies; friendly explosions are excluded. All losses use a common post-collision, pre-motion snapshot and add before HP clipping.

Disturb and Scout

For source j , $d_{ji} = \|p_i - p_j\|$ and $\theta_{ji} = \angle(v_j, p_i - p_j)$; angle is zero if either norm is below 10^{-3} . Let

$$G_{ji} = \mathbf{1}[10^{-3} < d_{ji} < 600\lambda, \theta_{ji} < \pi/6] \\ \times e^{-(d_{ji}/\lambda)^2/2} / \sqrt{2\pi},$$

$$S_c(j) = \sum_{i \in \mathcal{U}: c_i = c, h_i > 0} G_{ji}.$$

Living Red Disturb uses $J_j = \mathbf{1}[S_R(j) < S_B(j)]$, Blue uses $J_j = \mathbf{1}[S_R(j) \geq S_B(j)]$. Dead sources contribute zero.

$$L_i^{\text{dis}} = .1 \sum_{j \in \mathcal{U}_D: h_j > 0} J_j G_{ji},$$

$$\tilde{h}_i = \max(0, h_i - L_i^{\text{atk}} - L_i^{\text{dis}}).$$

Set $h'_i = 0$ if $r_i = A, I_i = 1$, otherwise $h'_i = \tilde{h}_i$. Disturb damages either side, not assets. Scouts use $d < 2000\lambda$, $\theta < \pi/3$; this sector does not restrict native actor rows. Collision victims do not move. Combat victims complete their motion; inactive velocity is zeroed next step.

G. Damage payoff and completion

Assets keep finite reset health $h'_k = h_k^0$. Count raw contributions before HP clipping, including simultaneous overkill:

$$D'_k = D_k + L_k, \quad d_t = \sum_k L_k, \quad R_t = -d_t.$$

Blue reward is d_t ; there is no team-size, asset-count or HP normalization and no terminal Survival bonus.

$$d = \mathbf{1}[\forall b \in \mathcal{B}_A : h'_b \leq 0].$$

All agent slots remain until global completion/truncation. Dead slots have zero observations, ignored actions and still receive full team R_t , including later damage after Red extinction. Do not sum duplicate teammate rewards. Undiscounted returns are $(-\sum_k D_k, +\sum_k D_k)$. H only truncates; global bootstrap $b_t = 1 - d$ remains one at truncation. Actor action-loss masks use pre-step participation.

F. Exact observations and CTDE information

For each living entity $j \neq i$, the actor row is

$$e_{ij} = [(p_j - p_i)/L, (v_j - v_i)/(2V_{\max}), \\ h_j, 1, \mathbf{1}_{c_j=R}, \mathbf{1}_{c_j=B}, \mathbf{1}_{j \in \mathcal{P}}], \\ L = \sqrt{\sum_a (b_a^+ - b_a^-)^2}.$$

Here $V_{\max}$ is the preset maximum speed, not the reference speed. Rows have 11 values, in fixed roster order: Red Scout/Disturb/Attack, Blue Scout/Disturb/Attack, then targets, excluding self. Dead rows are zero. Every living entity is included; Attack, Scout and Disturb share this packing. Side flags do not encode role. Own health is absent from the self block.

The actor receives the $(M - 1) \times 11$ entity table ($V_{\max} = 20$) plus unnormalized $[p_i, v_i, q_i, \omega_i]$. Both blocks are zero for an already-dead Damage observer. With deterministic encoding, $Z_i(o | s) = \delta_{\text{encode}_i(s)}(o)$. Other UAV attitudes/rates, opponent memory and time are omitted; hence local information is partial despite noiseless sensing.

Privileged $g(s) = \text{state}()$ packs 11 values per entity,

$$[(p - b^-)/(b^+ - b^-), v/(2V_{\max}), h, \ell, R, B, P],$$

with dead blocks zero. Rigid models append 13 raw state values per UAV (dead zero), then $[t, \sum_k D_k, \mathbf{1}_{\text{Damage}}]$: size $11M + 13N + 3$. Particle size is $11M$, omitting time and damage. g is a CTDE feature, not the complete Markov state.

H. Defender Dec-POMDP and MARL objective

For a specified frozen opposing policy π_B , define

$$\mathcal{M}_R = (\mathcal{R}, \mathcal{S}, \{\mathcal{A}_i\}, T^{\pi_B}, \\ \{\mathcal{O}_i\}, Z, R, \gamma, \rho_0, H),$$

$$T^{\pi_B}(s' | s, a_R) = \int \delta_{F_C(s, a_R, a_B)}(s') \\ \times \pi_B(da_B | o_B, \zeta_t).$$

F_C is the stated collision, simultaneous interaction and controlled-motion map, including any memory evolution; ρ_0 is the reset law. Native execution accepts both teams' actions. Freezing a rule or learned opponent is an experiment choice; with both teams learning, the model is a POSG.

$$\tau_i = (o_{i,0}, a_{i,0}, \dots, o_{i,t}),$$

$$\pi_R(a_R | \tau_R) = \prod_{i \in \mathcal{R}} \pi_i(a_i | \tau_i),$$

$$\pi_R^* \in \arg \max_{\pi_R} \mathbb{E} \left[\sum_{t=0}^{T-1} \gamma^t R_t \right].$$

Expectation is over ρ_0, T^{π_B}, π_R . Here T is natural completion (possibly infinite); H limits sampled rollouts. A truncated return is partial, so its successor value is retained. For $\gamma = 1$, this is respectively the native terminal Survival payoff or negative total asset damage. For $\gamma < 1$, it is the discounted task selected by the caller. Actors use local histories only. CTDE critics can use $g(s)$, joint actions and recorded task/memory features:

$$y_t = R_t + \gamma b_t Q^-(g(s'), a'_R).$$

Here Q^- is a target critic and successor actions follow the target actor policies. Use global completion, not individual death. A recurrent critic or augmented state may be needed because g omits history. This defines an objective; it does not prescribe an implemented optimizer.

Theme 15 / Quadrotor / Position / Survival

Rigid-body thrust dynamics, reference feedback and feasible actuation

A. Scenario and complete state

Red defenders $\mathcal{R}$ protect stationary assets $\mathcal{P}$ against Blue $\mathcal{B}$. Set $\mathcal{U} = \mathcal{R} \cup \mathcal{B}$, $\mathcal{E} = \mathcal{U} \cup \mathcal{P}$, $N = |\mathcal{U}|$, $K = |\mathcal{P}|$, $M = N + K$. Episode-fixed roles are Attack, Scout or Disturb.

$$s_t = (\{x_i, h_i, r_i, c_i, \text{id}_i\}_{i \in \mathcal{U}}, \{p_k, h_k, D_k\}_{k \in \mathcal{P}}, t, \zeta_t).$$

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. Participation is $\ell_i = \mathbf{1}[h_i > 0]$. Reset UAV health is 1, target health $h_k^0 = 2$ by default; $v_k = 0$, $D_k = 0$ initially. D_k accumulates raw asset damage.

Configuration C fixes roster, bounds, controls and task. ζ_t contains persistent assignments and opponent memory, if used. Reactive controllers have no hidden integrators. Last command, functional bits, pre-step positions and step damage are overwritten transition records; retain them for exact reward/event replay. RNG state is required for exact replay, not actor input.

B. Six-degree-of-freedom dynamics

World coordinates are east-north-up (ENU); body coordinates are forward-left-up (FLU). The scalar-first quaternion $q = (q_w, q_x, q_y, q_z)$ maps body to world through $R(q) \in SO(3)$, with $q \in S^3$, $\|q\| = 1$ and $q \sim -q$. Thus 13 stored values represent 12 physical degrees of freedom.

$$\begin{aligned} \dot{p} &= v, & \dot{v} &= R(q)F_B/m - ge_3, \\ \dot{q} &= \frac{1}{2}q \otimes (0, \omega), & \dot{\omega} &= J^{-1}(M_B - \omega \times J\omega). \end{aligned}$$

Here $g = 9.81 \text{ m s}^{-2}$, $\omega \in \mathbb{R}^3$ is the body angular velocity in rad/s; F_B excludes gravity. No wind, sensor noise or actuator lag is present.

$$R(q) = I + 2q_w[q_v]_{\times} + 2[q_v]_{\times}^2, \quad q_v = (q_x, q_y, q_z).$$

Here $[z]_{\times}y = z \times y$.

C. Four-rotor wrench and parameters

The rotor order is front, right, rear, left, and normalized thrust fractions are $u_j \in [0, 1]$. With $m = 4.34 \text{ kg}$, $J = \text{diag}(.0820, .0845, .1377) \text{ kg m}^2$, arm $l = .315 \text{ m}$ and yaw ratio $c_\tau = .008004 \text{ m}$ [2],

$$f_{\max} = 2.5mg/4, \quad f_j = f_{\max}u_j,$$

$$\begin{aligned} \begin{bmatrix} f \\ M_x \\ M_y \\ M_z \end{bmatrix} &= A \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}, \\ A &= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -l & 0 & l \\ -l & 0 & l & 0 \\ c_\tau & -c_\tau & c_\tau & -c_\tau \end{bmatrix}. \end{aligned}$$

The rigid-body force is $F_B = (0, 0, f)$ and moment is (M_x, M_y, M_z) . Ideal level hover has $u_j = .4$. The maximum thrust-to-weight ratio 2.5 is a simulation assumption. No aerodynamic/rotor drag, gyroscopic rotor torque or motor time constant is included [3].

Episode geometry and reset law

Default $\lambda = .16$, $\mathcal{W} = [-400, 400]^2 \times [0, 400] \text{ m}$, $d_c = 3.2 \text{ m}$, $(r_0, r_1) = (48, 96) \text{ m}$ and $r_f = r_0$ (configurable $0 < r_f \leq r_0$). Assets are independently uniform in $[-368, -304] \times [-192, 192] \times [80, 240] \text{ m}$ or use a fixed layout. Red samples an area-uniform XY annulus $[128, 352] \text{ m}$ around a uniformly chosen asset, clipped to scene bounds; Blue is uniform over $[0, 400] \times [-400, 400] \text{ m}$. UAV altitude is uniform in $[32, 240] \text{ m}$.

Reset UAVs at $v = 0$, $q = (1, 0, 0, 0)$, $\omega = 0$. Reject UAV openings within $\max(4.8, d_c + 10^{-6}) \text{ m}$ of any previously placed entity; later collisions are allowed. Only 3D is supported.

Absolute position intent

Each living agent selects $p_d \in \mathcal{W}$ in metres. At every RK4 stage compute

$$v_d = \text{sat}_{12}(.5(p_d - p)), \quad \text{sat}_L(z) = z \min(1, L/\|z\|).$$

For $z = 0$, define $\text{sat}_L(0) = 0$. The position reference is held over the external step; feedback updates using stage position and velocity. Unlike the fixed wing and minimum-speed particle, this ideal quadrotor can command zero velocity near its target. Reaching the point is a feedback outcome, not an instantaneous assignment.

D. Geometric feedback and feasible allocation

At a stage state, clip a held velocity reference to 20 m/s, then compute

$$\begin{aligned} a_d &= \text{sat}_6(2(v_d - v)), & F_d &= m(a_d + ge_3), \\ b_{3d} &= F_d/\|F_d\|, \\ b_{2d} &= (b_{3d} \times e_1)/\|b_{3d} \times e_1\|, \\ R_d &= [b_{2d} \times b_{3d}, b_{2d}, b_{3d}], \\ e_R &= \frac{1}{2}(R_d^T R - R^T R_d)^\vee, \\ M_d &= -8.81e_R - 2.54\omega + \omega \times J\omega, \\ f_d &= F_d^T R e_3. \end{aligned}$$

Fixed world-east heading avoids hidden yaw memory. The 6 m/s² bound is below g , so F_d and its heading construction are nonsingular. This uses attitude PD and gyro compensation, without the source paper's trajectory derivative or desired angular-rate feedforward.

To preserve feasible collective thrust, set $\bar{f} = \text{clip}(f_d, 0, 4f_{\max})/4$ and $d = A^{-1}(0, M_d)^T$. Define

$$\begin{aligned} \kappa_+ &= \min_{d_j > 0} (f_{\max} - \bar{f})/d_j, \\ \kappa_- &= \min_{d_j < 0} (-\bar{f})/d_j, \quad \kappa = \min(1, \kappa_+, \kappa_-). \end{aligned}$$

$$u_j = \text{clip}\left(\frac{\bar{f} + \kappa d_j}{f_{\max}}, 0, 1\right).$$

Empty minima are ignored. Uniform differential scaling sacrifices moment while retaining the clipped collective; independent rotor clipping is not assumed to preserve it.

Numerical transition and boundary

Write $f(x, u)$ for the above derivative, evaluated with normalized q . An external step is $\Delta = 1 \text{ s}$, $n = \lceil \Delta/.01 \rceil = 100$, $h = \Delta/n$. With stage feedback U ,

$$\begin{aligned} k_1 &= f(x, U(x)), \\ k_2 &= f(x + hk_1/2, U(x + hk_1/2)), \\ k_3 &= f(x + hk_2/2, U(x + hk_2/2)), \\ k_4 &= f(x + hk_3, U(x + hk_3)), \\ x^+ &= x + h(k_1 + 2k_2 + 2k_3 + k_4)/6, \\ q^+ &\leftarrow q^+/\|q^+\|. \end{aligned}$$

Repeat n times. A raw actuator is held constant; an intent is held but feedback is recomputed at every RK4 stage. Controllers have no hidden integrators.

For a candidate endpoint p^* , set $p'_a = \text{clip}(p_a^*, b_a^-, b_a^+)$. If $p_a^* > b_a^+$, $v'_a = \min(v_a^*, 0)$; if $p_a^* < b_a^-$, $v'_a = \max(v_a^*, 0)$; otherwise $v'_a = v_a^*$. Collision paths use clipped substep positions. Boundaries are task constraints; physical airspeed and attitude are not clamped to controller reference limits.

Theme 15 / Quadrotor / Position / Survival

Task interaction, exact information structure and defender learning objective

E. Collision and automatic interaction

Predict living UAV paths: particles use $p_i + \tau \Delta v_i$, rigid models use synchronized RK4 endpoints clipped to $\mathcal{W}$. On each matched segment, let $r = p_i - p_j$, $d = (q_i - p_i) - (q_j - p_j)$:

$$\tau_* = \begin{cases} 0 & \|d\|^2 \leq 10^{-12}, \\ \text{clip}(-r^T d / \|d\|^2, 0, 1) & \text{otherwise,} \end{cases} \quad d_* = \|r + \tau_* d\|.$$

Collision sets both UAV HP to zero when $\min d_* \leq d_c$; targets are excluded. Pairs use i -major, $j < i$ order. Dead outer agents are skipped at entry and dead inner agents are skipped. An outer agent killed during its inner sweep continues that sweep. After collision, surviving Attack agents automatically fire at step-start positions:

$$I_i = \begin{cases} \mathbf{1}[\min_{k \in \mathcal{P}: h_k > 0} \|p_i - p_k\| < r_f], & i \in \mathcal{B}, \\ \mathbf{1}[\min_{j \in \mathcal{B}_A: h_j > 0} \|p_i - p_j\| < r_f], & i \in \mathcal{R}. \end{cases}$$

An empty minimum is $+\infty$; dead sources have $I_i = 0$. A denotes Attack. Fire is a separate rule, not a navigation action. With unit attack intensity,

$$\eta(d) = \begin{cases} 1 & d < r_0, \\ (r_1 - d)/(r_1 - r_0) & r_0 \leq d < r_1, \\ 0 & d \geq r_1, \end{cases}$$

$$L_k = \sum_{b \in \mathcal{B}_A: h_b > 0} I_b \eta(\|p_b - p_k\|),$$

$$L_i^{\text{atk}} = \sum_{j \in \mathcal{U}_A: c_j \neq c_i, h_j > 0} I_j \eta(\|p_j - p_i\|).$$

One source can affect multiple enemies; friendly explosions are excluded. All losses use a common post-collision, pre-motion snapshot and add before HP clipping.

Disturb and Scout

For source j , $d_{ji} = \|p_i - p_j\|$ and $\theta_{ji} = \angle(v_j, p_i - p_j)$; angle is zero if either norm is below 10^{-3} . Let

$$G_{ji} = \mathbf{1}[10^{-3} < d_{ji} < 600\lambda, \theta_{ji} < \pi/6] \times e^{-(d_{ji}/\lambda)^2/2} / \sqrt{2\pi},$$

$$S_c(j) = \sum_{i \in \mathcal{U}: c_i = c, h_i > 0} G_{ji}.$$

Living Red Disturb uses $J_j = \mathbf{1}[S_R(j) < S_B(j)]$, Blue uses $J_j = \mathbf{1}[S_R(j) \geq S_B(j)]$. Dead sources contribute zero.

$$L_i^{\text{dis}} = .1 \sum_{j \in \mathcal{U}_D: h_j > 0} J_j G_{ji},$$

$$\tilde{h}_i = \max(0, h_i - L_i^{\text{atk}} - L_i^{\text{dis}}).$$

Set $h'_i = 0$ if $r_i = A, I_i = 1$, otherwise $h'_i = \tilde{h}_i$. Disturb damages either side, not assets. Scouts use $d < 2000\lambda$, $\theta < \pi/3$; this sector does not restrict native actor rows. Collision victims do not move. Combat victims complete their motion; inactive velocity is zeroed next step.

G. Survival payoff and completion

Assets obey $h'_k = \max(0, h_k - L_k)$, $D'_k = D_k + L_k$.

$$z(s') = \begin{cases} -1 & \exists k : h'_k < 10^{-3}, \\ +1 & \forall b \in \mathcal{B}_A : h'_b < 10^{-3}, \\ 0 & \text{otherwise.} \end{cases}$$

Target loss has priority. Natural completion is $d = \mathbf{1}[z \neq 0]$; default Red reward $R_t = 10z$, Blue $-R_t$. Red elimination alone does not end the world. Individual $h'_i \leq 0$ retires a Survival actor after its final transition. Horizon H truncates without a win bonus; use global bootstrap $b_t = 1 - d$, including individual death and truncation. Actor action-loss masks use pre-step participation. Explicit reward weights replace the default payoff as defined on page 1.

F. Exact observations and CTDE information

For each living entity $j \neq i$, the actor row is

$$e_{ij} = [(p_j - p_i)/L, (v_j - v_i)/(2V_{\max}), h_j, 1, \mathbf{1}_{c_j=R}, \mathbf{1}_{c_j=B}, \mathbf{1}_{j \in \mathcal{P}}],$$

$$L = \sqrt{\sum_a (b_a^+ - b_a^-)^2}.$$

Here $V_{\max}$ is the preset maximum speed, not the reference speed. Rows have 11 values, in fixed roster order: Red Scout/Disturb/Attack, Blue Scout/Disturb/Attack, then targets, excluding self. Dead rows are zero. Every living entity is included; Attack, Scout and Disturb share this packing. Side flags do not encode role. Own health is absent from the self block.

The actor receives the $(M-1) \times 11$ entity table ($V_{\max} = 20$) plus unnormalized $[p_i, v_i, q_i, \omega_i] \in \mathbb{R}^{13}$. With deterministic encoding, $Z_i(o | s) = \delta_{\text{encode}_i(s)}(o)$. Other UAV attitudes/rates, opponent memory and time are omitted; hence local information is partial despite noiseless sensing.

Privileged $g(s) = \text{state}()$ packs 11 values per entity,

$$[(p - b^-)/(b^+ - b^-), v/(2V_{\max}), h, \ell, R, B, P],$$

with dead blocks zero. Rigid models append 13 raw state values per UAV (dead zero), then $[t, \sum_k D_k, \mathbf{1}_{\text{Damage}}]$: size $11M + 13N + 3$. Particle size is $11M$, omitting time and damage. g is a CTDE feature, not the complete Markov state.

H. Defender Dec-POMDP and MARL objective

For a specified frozen opposing policy π_B , define

$$\mathcal{M}_R = (\mathcal{R}, \mathcal{S}, \{\mathcal{A}_i\}, T^{\pi_B}, \{\mathcal{O}_i\}, Z, R, \gamma, \rho_0, H),$$

$$T^{\pi_B}(s' | s, a_R) = \int \delta_{F_C(s, a_R, a_B)}(s') \times \pi_B(da_B | o_B, \zeta_t).$$

F_C is the stated collision, simultaneous interaction and controlled-motion map, including any memory evolution; ρ_0 is the reset law. Native execution accepts both teams' actions. Freezing a rule or learned opponent is an experiment choice; with both teams learning, the model is a POSG.

$$\tau_i = (o_{i,0}, a_{i,0}, \dots, o_{i,t}),$$

$$\pi_R(a_R | \tau_R) = \prod_{i \in \mathcal{R}} \pi_i(a_i | \tau_i),$$

$$\pi_R^* \in \arg \max_{\pi_R} \mathbb{E} \left[\sum_{t=0}^{T-1} \gamma^t R_t \right].$$

Expectation is over ρ_0, T^{π_B}, π_R . Here T is natural completion (possibly infinite); H limits sampled rollouts. A truncated return is partial, so its successor value is retained. For $\gamma = 1$, this is respectively the native terminal Survival payoff or negative total asset damage. For $\gamma < 1$, it is the discounted task selected by the caller. Actors use local histories only. CTDE critics can use $g(s)$, joint actions and recorded task/memory features:

$$y_t = R_t + \gamma b_t Q^-(g(s'), a'_R).$$

Here Q^- is a target critic and successor actions follow the target actor policies. Use global completion, not individual death. A recurrent critic or augmented state may be needed because g omits history. This defines an objective; it does not prescribe an implemented optimizer.

Theme 16 / Quadrotor / Position / Damage

Rigid-body thrust dynamics, reference feedback and feasible actuation

A. Scenario and complete state

Red defenders $\mathcal{R}$ protect stationary assets $\mathcal{P}$ against Blue $\mathcal{B}$. Set $\mathcal{U} = \mathcal{R} \cup \mathcal{B}$, $\mathcal{E} = \mathcal{U} \cup \mathcal{P}$, $N = |\mathcal{U}|$, $K = |\mathcal{P}|$, $M = N + K$. Episode-fixed roles are Attack, Scout or Disturb.

$$s_t = (\{x_i, h_i, r_i, c_i, \text{id}_i\}_{i \in \mathcal{U}}, \{p_k, h_k, D_k\}_{k \in \mathcal{P}}, t, \zeta_t).$$

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. Participation is $\ell_i = \mathbf{1}[h_i > 0]$. Reset UAV health is 1, target health $h_k^0 = 2$ by default; $v_k = 0$, $D_k = 0$ initially. D_k accumulates raw asset damage.

Configuration C fixes roster, bounds, controls and task. ζ_t contains persistent assignments and opponent memory, if used. Reactive controllers have no hidden integrators. Last command, functional bits, pre-step positions and step damage are overwritten transition records; retain them for exact reward/event replay. RNG state is required for exact replay, not actor input.

B. Six-degree-of-freedom dynamics

World coordinates are east-north-up (ENU); body coordinates are forward-left-up (FLU). The scalar-first quaternion $q = (q_w, q_x, q_y, q_z)$ maps body to world through $R(q) \in SO(3)$, with $q \in S^3$, $\|q\| = 1$ and $q \sim -q$. Thus 13 stored values represent 12 physical degrees of freedom.

$$\begin{aligned} \dot{p} &= v, & \dot{v} &= R(q)F_B/m - ge_3, \\ \dot{q} &= \frac{1}{2}q \otimes (0, \omega), & \dot{\omega} &= J^{-1}(M_B - \omega \times J\omega). \end{aligned}$$

Here $g = 9.81 \text{ m s}^{-2}$, $\omega \in \mathbb{R}^3$ is the body angular velocity in rad/s; F_B excludes gravity. No wind, sensor noise or actuator lag is present.

$$R(q) = I + 2q_w[q_v]_{\times} + 2[q_v]_{\times}^2, \quad q_v = (q_x, q_y, q_z).$$

Here $[z]_{\times}y = z \times y$.

C. Four-rotor wrench and parameters

The rotor order is front, right, rear, left, and normalized thrust fractions are $u_j \in [0, 1]$. With $m = 4.34 \text{ kg}$, $J = \text{diag}(.0820, .0845, .1377) \text{ kg m}^2$, arm $l = .315 \text{ m}$ and yaw ratio $c_\tau = .008004 \text{ m}$ [2],

$$f_{\max} = 2.5mg/4, \quad f_j = f_{\max}u_j,$$

$$\begin{aligned} \begin{bmatrix} f \\ M_x \\ M_y \\ M_z \end{bmatrix} &= A \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}, \\ A &= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -l & 0 & l \\ -l & 0 & l & 0 \\ c_\tau & -c_\tau & c_\tau & -c_\tau \end{bmatrix}. \end{aligned}$$

The rigid-body force is $F_B = (0, 0, f)$ and moment is (M_x, M_y, M_z) . Ideal level hover has $u_j = .4$. The maximum thrust-to-weight ratio 2.5 is a simulation assumption. No aerodynamic/rotor drag, gyroscopic rotor torque or motor time constant is included [3].

Episode geometry and reset law

Default $\lambda = .16$, $\mathcal{W} = [-400, 400]^2 \times [0, 400] \text{ m}$, $d_c = 3.2 \text{ m}$, $(r_0, r_1) = (48, 96) \text{ m}$ and $r_f = r_0$ (configurable $0 < r_f \leq r_0$). Assets are independently uniform in $[-368, -304] \times [-192, 192] \times [80, 240] \text{ m}$ or use a fixed layout. Red samples an area-uniform XY annulus $[128, 352] \text{ m}$ around a uniformly chosen asset, clipped to scene bounds; Blue is uniform over $[0, 400] \times [-400, 400] \text{ m}$. UAV altitude is uniform in $[32, 240] \text{ m}$.

Reset UAVs at $v = 0$, $q = (1, 0, 0, 0)$, $\omega = 0$. Reject UAV openings within $\max(4.8, d_c + 10^{-6}) \text{ m}$ of any previously placed entity; later collisions are allowed. Only 3D is supported.

Absolute position intent

Each living agent selects $p_d \in \mathcal{W}$ in metres. At every RK4 stage compute

$$v_d = \text{sat}_{12}(.5(p_d - p)), \quad \text{sat}_L(z) = z \min(1, L/\|z\|).$$

For $z = 0$, define $\text{sat}_L(0) = 0$. The position reference is held over the external step; feedback updates using stage position and velocity. Unlike the fixed wing and minimum-speed particle, this ideal quadrotor can command zero velocity near its target. Reaching the point is a feedback outcome, not an instantaneous assignment.

D. Geometric feedback and feasible allocation

At a stage state, clip a held velocity reference to 20 m/s, then compute

$$\begin{aligned} a_d &= \text{sat}_6(2(v_d - v)), & F_d &= m(a_d + ge_3), \\ b_{3d} &= F_d/\|F_d\|, \\ b_{2d} &= (b_{3d} \times e_1)/\|b_{3d} \times e_1\|, \\ R_d &= [b_{2d} \times b_{3d}, b_{2d}, b_{3d}], \\ e_R &= \frac{1}{2}(R_d^T R - R^T R_d)^\vee, \\ M_d &= -8.81e_R - 2.54\omega + \omega \times J\omega, \\ f_d &= F_d^T R e_3. \end{aligned}$$

Fixed world-east heading avoids hidden yaw memory. The 6 m/s² bound is below g , so F_d and its heading construction are nonsingular. This uses attitude PD and gyro compensation, without the source paper's trajectory derivative or desired angular-rate feedforward.

To preserve feasible collective thrust, set $\bar{f} = \text{clip}(f_d, 0, 4f_{\max})/4$ and $d = A^{-1}(0, M_d)^T$. Define

$$\begin{aligned} \kappa_+ &= \min_{d_j > 0} (f_{\max} - \bar{f})/d_j, \\ \kappa_- &= \min_{d_j < 0} (-\bar{f})/d_j, \quad \kappa = \min(1, \kappa_+, \kappa_-). \end{aligned}$$

$$u_j = \text{clip}\left(\frac{\bar{f} + \kappa d_j}{f_{\max}}, 0, 1\right).$$

Empty minima are ignored. Uniform differential scaling sacrifices moment while retaining the clipped collective; independent rotor clipping is not assumed to preserve it.

Numerical transition and boundary

Write $f(x, u)$ for the above derivative, evaluated with normalized q . An external step is $\Delta = 1 \text{ s}$, $n = \lceil \Delta/.01 \rceil = 100$, $h = \Delta/n$. With stage feedback U ,

$$\begin{aligned} k_1 &= f(x, U(x)), \\ k_2 &= f(x + hk_1/2, U(x + hk_1/2)), \\ k_3 &= f(x + hk_2/2, U(x + hk_2/2)), \\ k_4 &= f(x + hk_3, U(x + hk_3)), \\ x^+ &= x + h(k_1 + 2k_2 + 2k_3 + k_4)/6, \\ q^+ &\leftarrow q^+/\|q^+\|. \end{aligned}$$

Repeat n times. A raw actuator is held constant; an intent is held but feedback is recomputed at every RK4 stage. Controllers have no hidden integrators.

For a candidate endpoint p^* , set $p'_a = \text{clip}(p_a^*, b_a^-, b_a^+)$. If $p_a^* > b_a^+$, $v'_a = \min(v_a^*, 0)$; if $p_a^* < b_a^-$, $v'_a = \max(v_a^*, 0)$; otherwise $v'_a = v_a^*$. Collision paths use clipped substep positions. Boundaries are task constraints; physical airspeed and attitude are not clamped to controller reference limits.

Theme 16 / Quadrotor / Position / Damage

Task interaction, exact information structure and defender learning objective

E. Collision and automatic interaction

Predict living UAV paths: particles use $p_i + \tau \Delta v_i$, rigid models use synchronized RK4 endpoints clipped to $\mathcal{W}$. On each matched segment, let $r = p_i - p_j$, $d = (q_i - p_i) - (q_j - p_j)$:

$$\tau_* = \begin{cases} 0 & \|d\|^2 \leq 10^{-12}, \\ \text{clip}(-r^T d / \|d\|^2, 0, 1) & \text{otherwise,} \end{cases} \quad d_* = \|r + \tau_* d\|.$$

Collision sets both UAV HP to zero when $\min d_* \leq d_c$; targets are excluded. Pairs use i -major, $j < i$ order. Dead outer agents are skipped at entry and dead inner agents are skipped. An outer agent killed during its inner sweep continues that sweep. After collision, surviving Attack agents automatically fire at step-start positions:

$$I_i = \begin{cases} \mathbf{1}[\min_{k \in \mathcal{P}: h_k > 0} \|p_i - p_k\| < r_f], & i \in \mathcal{B}, \\ \mathbf{1}[\min_{j \in \mathcal{B}_A: h_j > 0} \|p_i - p_j\| < r_f], & i \in \mathcal{R}. \end{cases}$$

An empty minimum is $+\infty$; dead sources have $I_i = 0$. A denotes Attack. Fire is a separate rule, not a navigation action. With unit attack intensity,

$$\eta(d) = \begin{cases} 1 & d < r_0, \\ (r_1 - d)/(r_1 - r_0) & r_0 \leq d < r_1, \\ 0 & d \geq r_1, \end{cases}$$

$$L_k = \sum_{b \in \mathcal{B}_A: h_b > 0} I_b \eta(\|p_b - p_k\|),$$

$$L_i^{\text{atk}} = \sum_{j \in \mathcal{U}_A: c_j \neq c_i, h_j > 0} I_j \eta(\|p_j - p_i\|).$$

One source can affect multiple enemies; friendly explosions are excluded. All losses use a common post-collision, pre-motion snapshot and add before HP clipping.

Disturb and Scout

For source j , $d_{ji} = \|p_i - p_j\|$ and $\theta_{ji} = \angle(v_j, p_i - p_j)$; angle is zero if either norm is below 10^{-3} . Let

$$G_{ji} = \mathbf{1}[10^{-3} < d_{ji} < 600\lambda, \theta_{ji} < \pi/6] \times e^{-(d_{ji}/\lambda)^2/2} / \sqrt{2\pi},$$

$$S_c(j) = \sum_{i \in \mathcal{U}: c_i = c, h_i > 0} G_{ji}.$$

Living Red Disturb uses $J_j = \mathbf{1}[S_R(j) < S_B(j)]$, Blue uses $J_j = \mathbf{1}[S_R(j) \geq S_B(j)]$. Dead sources contribute zero.

$$L_i^{\text{dis}} = .1 \sum_{j \in \mathcal{U}_D: h_j > 0} J_j G_{ji},$$

$$\tilde{h}_i = \max(0, h_i - L_i^{\text{atk}} - L_i^{\text{dis}}).$$

Set $h'_i = 0$ if $r_i = A, I_i = 1$, otherwise $h'_i = \tilde{h}_i$. Disturb damages either side, not assets. Scouts use $d < 2000\lambda$, $\theta < \pi/3$; this sector does not restrict native actor rows. Collision victims do not move. Combat victims complete their motion; inactive velocity is zeroed next step.

G. Damage payoff and completion

Assets keep finite reset health $h'_k = h_k^0$. Count raw contributions before HP clipping, including simultaneous overkill:

$$D'_k = D_k + L_k, \quad d_t = \sum_k L_k, \quad R_t = -d_t.$$

Blue reward is d_t ; there is no team-size, asset-count or HP normalization and no terminal Survival bonus.

$$d = \mathbf{1}[\forall b \in \mathcal{B}_A : h'_b \leq 0].$$

All agent slots remain until global completion/truncation. Dead slots have zero observations, ignored actions and still receive full team R_t , including later damage after Red extinction. Do not sum duplicate teammate rewards. Undiscounted returns are $(-\sum_k D_k, +\sum_k D_k)$. H only truncates; global bootstrap $b_t = 1 - d$ remains one at truncation. Actor action-loss masks use pre-step participation.

F. Exact observations and CTDE information

For each living entity $j \neq i$, the actor row is

$$e_{ij} = \begin{bmatrix} (p_j - p_i)/L, (v_j - v_i)/(2V_{\max}), \\ h_j, 1, \mathbf{1}_{c_j=R}, \mathbf{1}_{c_j=B}, \mathbf{1}_{j \in \mathcal{P}} \end{bmatrix},$$

$$L = \sqrt{\sum_a (b_a^+ - b_a^-)^2}.$$

Here $V_{\max}$ is the preset maximum speed, not the reference speed. Rows have 11 values, in fixed roster order: Red Scout/Disturb/Attack, Blue Scout/Disturb/Attack, then targets, excluding self. Dead rows are zero. Every living entity is included; Attack, Scout and Disturb share this packing. Side flags do not encode role. Own health is absent from the self block.

The actor receives the $(M - 1) \times 11$ entity table ($V_{\max} = 20$) plus unnormalized $[p_i, v_i, q_i, \omega_i]$. Both blocks are zero for an already-dead Damage observer. With deterministic encoding, $Z_i(o | s) = \delta_{\text{encode}_i(s)}(o)$. Other UAV attitudes/rates, opponent memory and time are omitted; hence local information is partial despite noiseless sensing.

Privileged $g(s) = \text{state}()$ packs 11 values per entity,

$$[(p - b^-)/(b^+ - b^-), v/(2V_{\max}), h, \ell, R, B, P],$$

with dead blocks zero. Rigid models append 13 raw state values per UAV (dead zero), then $[t, \sum_k D_k, \mathbf{1}_{\text{Damage}}]$: size $11M + 13N + 3$. Particle size is $11M$, omitting time and damage. g is a CTDE feature, not the complete Markov state.

H. Defender Dec-POMDP and MARL objective

For a specified frozen opposing policy π_B , define

$$\mathcal{M}_R = (\mathcal{R}, \mathcal{S}, \{\mathcal{A}_i\}, T^{\pi_B}, \{\mathcal{O}_i\}, Z, R, \gamma, \rho_0, H),$$

$$T^{\pi_B}(s' | s, a_R) = \int \delta_{F_C(s, a_R, a_B)}(s') \times \pi_B(da_B | o_B, \zeta_t).$$

F_C is the stated collision, simultaneous interaction and controlled-motion map, including any memory evolution; ρ_0 is the reset law. Native execution accepts both teams' actions. Freezing a rule or learned opponent is an experiment choice; with both teams learning, the model is a POSG.

$$\tau_i = (o_{i,0}, a_{i,0}, \dots, o_{i,t}),$$

$$\pi_R(a_R | \tau_R) = \prod_{i \in \mathcal{R}} \pi_i(a_i | \tau_i),$$

$$\pi_R^* \in \arg \max_{\pi_R} \mathbb{E} \left[\sum_{t=0}^{T-1} \gamma^t R_t \right].$$

Expectation is over ρ_0, T^{π_B}, π_R . Here T is natural completion (possibly infinite); H limits sampled rollouts. A truncated return is partial, so its successor value is retained. For $\gamma = 1$, this is respectively the native terminal Survival payoff or negative total asset damage. For $\gamma < 1$, it is the discounted task selected by the caller. Actors use local histories only. CTDE critics can use $g(s)$, joint actions and recorded task/memory features:

$$y_t = R_t + \gamma b_t Q^-(g(s'), a'_R).$$

Here Q^- is a target critic and successor actions follow the target actor policies. Use global completion, not individual death. A recurrent critic or augmented state may be needed because g omits history. This defines an objective; it does not prescribe an implemented optimizer.